

Politically Induced Internal Trade Barriers in China ^{*}

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Abstract

Competition among subnational governments generates internal non-tariff trade barriers, fragmenting markets within nominally integrated economies. In China's decentralized system, where provincial authorities evaluate city leaders on relative economic performance, politicians face incentives to maximize locally appropriable gains while minimizing spillovers to rivals. This yardstick competition creates systematic distortions in the allocation of public resources. Exploiting exogenous variation in promotion incentives driven by the age of city party secretaries (1996–2018), I show that intensified within-province competition significantly reduces inter-city trade flows, dampens international exports, and restricts firm access to intermediate inputs. Under competitive pressure, politicians reallocate resources toward non-tradable sectors, particularly real estate, while underinvesting in connective infrastructure near jurisdictional borders. Individually rational responses to career incentives thus generate collectively suboptimal market fragmentation, showing that internal trade costs are endogenous to political institutions and that yardstick competition carries dynamic costs.

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1 Introduction

Trade drives prosperity, yet it is fundamentally constrained by political institutions. While standard benchmarks predict that falling transport costs should compel convergence (McCallum, 1995), evidence reveals persistent fragmentation even within national borders (Young, 2000; Poncet, 2005). A burgeoning literature demonstrates that, internationally, trade flows are endogenous to geopolitical alignment; strategic divergence and political risk act as implicit barriers that segment global markets (Fajgelbaum et al., 2020; Caldara and Iacoviello, 2022; Kleinman et al., 2024). Whether analogous political forces generate ‘internal geopolitical distance’ between subnational governments remains an open question.

This paper establishes that bureaucratic promotion tournaments function as a distinct source of such distance, driving domestic fragmentation through strategic resource misallocation. I examine how yardstick competition (Shleifer, 1985; Besley and Case, 1995)—originally conceptualized for democratic settings—operates in hierarchical authoritarian systems, where it generates internal geopolitical distance by transforming neighboring jurisdictions into rivals rather than partners in integration.¹ Under intense competition and budget constraints, this relative evaluation induces politicians to strategically reallocate public resources toward non-tradable sectors, particularly real estate, that generate rapid and locally appropriable GDP gains.² This reallocation away from tradable industries and trade-facilitating infrastructure creates non-tariff barriers to inter-jurisdictional trade, sacrificing aggregate economic efficiency to maximize relative performance. The findings reveal a yet understudied drawback to certain forms of decentralized governance, contributing to our understanding of how institutional frameworks affect

¹The mechanism differs from canonical yardstick competition in one important respect: while Shleifer (1985) emphasized voters comparing government performance, here upper-level governments conduct the comparison. Crucially, the distortions I document do not stem from “not correctly set incentives” but from a fundamental information constraint—upper-level governments cannot easily monitor cross-jurisdictional trade flows and spillovers, leading them to rationally evaluate officials on locally observable outcomes. This creates a systemic principal-agent problem where yardstick competition based on local performance metrics produces trade barriers as an equilibrium response.

²Politicians explicitly acknowledge this preference: “I sell land, I do real estate, today I build a park, tomorrow I build a center—it’s all government investment. Once these investments go in, the KPI is completed and the GDP looks very pretty” (Voice of America Chinese, July 13, 2022, <https://www.voachinese.com/a/6655247.html>). Documented cases illustrate the pattern: Xiangtan’s city party secretary launched 345 construction projects totaling 617 billion yuan (nearly five times annual fiscal revenue), explicitly stating his goal to “quickly show achievements” and secure vice-provincial promotion, resulting in 435 billion yuan debt and 33 unfinished projects (February 1, 2024, <https://baijiahao.baidu.com/s?id=1789672752005565436&wfr=spider&for=pc>). Similarly, Zhangzhou’s city party secretary spent 2.11 billion yuan on six replica tourist stations and designed a 200-million-yuan opera house to create visible achievements “that attract attention” before superiors’ visits; both projects were later abandoned (Central Commission for Discipline Inspection investigation, January 2024, <https://news.sina.com.cn/c/2024-01-09/doc-inaawrkp5341897.shtml>).

economic outcomes.³

China’s institutional framework, which combines regional autonomy with centralized personnel control, provides an ideal setting for studying political economy and market integration. As documented by [Maskin et al. \(2000\)](#), [Li and Zhou \(2005\)](#), and [Xu \(2011\)](#), local politicians have autonomy in shaping economic policies similar to that in a democratic system, but their career prospects are not determined by voters. The promotion system operates within a strictly hierarchical structure, where politicians at each level are promoted by their immediate superiors.⁴ In this promotion system, two key characteristics significantly influence evaluation: a politician’s age and economic performance.⁵⁶ Consequently, politicians with similar characteristics face intense competition among their peers. When facing such competition under budget constraints, politicians strategically optimize their policy portfolios by reallocating scarce resources to maximize locally visible and appropriable economic gains within their evaluation period.

This evaluation system design reflects inherent information constraints in multi-tiered governance rather than policy mistakes. Upper-level governments cannot easily monitor or attribute bilateral trade flows between jurisdictions, while local GDP is readily observable through standardized reporting. Given limited administrative capacity, evaluation criteria rationally prioritize measurable local performance. When officials compete on such metrics, trade barriers emerge as equilibrium responses rather than policy failures. Section 2 details why this represents a second-best institutional solution.

I use plausibly exogenous variation in the intensity of inter-jurisdictional yardstick competition, measured through a novel proxy: the relative age differential between politi-

³The literature has identified other negative aspects of decentralized policy-making in multileveled government: suboptimal fiscal decisions ([Oates and Schwab, 1988](#)); tax policy distortions ([Keen and Kotsogiannis, 2002](#)); increased corruption ([Fan et al., 2009](#)); and populist tendencies ([Acemoglu et al., 2013](#); [Kishishita and Yamagishi, 2021](#)).

⁴It is called the ‘one-level-up’ policy ([Chen and Kung, 2019](#)). For example, provincial-level organization departments, including the provincial governor, determine the appointments and promotions of city-level leaders.

⁵Several studies, e.g., [Wang et al. \(2020\)](#); [Zeng and Zhou \(2024\)](#), demonstrate that age serves as a crucial indicator for upper-level politicians when assessing local politicians’ promotion potential. In the absence of electoral accountability, this age-based promotion system provides a standardized mechanism for screening and selecting politicians.

⁶Economic performance is measured by a jurisdiction’s economic growth during a politician’s tenure. As documented in [Li and Zhou \(2005\)](#), higher growth rates correlate with increased promotion probability.

cians and their peers.⁷ First, age is one of the two important predictors for a politician's promotion as it may suggest the politician's potential in the context of China.⁸ Second, using relative age differential rather than the absolute value of age can further represent the nature of yardstick political competition in the tournament-style promotion system, as politicians have to "win" promotion from peers.

Using plausibly exogenous competitive intensity and controlling for politician and city fixed effects, I find that a one standard deviation increase in this age differential (indicating reduced yardstick competition) corresponds to a 2.6% increase in total inter-city trade volumes, demonstrating measurable reduction in trade barriers arising from strategic resource allocation. The effect varies systematically across transportation modes: smallest for motorway trade (2.9%) and largest for waterway trade (6.2%), reflecting varying degrees of political control over different transport channels. Waterway transportation, subject to extensive government control over port infrastructure, proves most susceptible to career-driven manipulation, while railway and airway trade show non-significant effects as these primarily serve long-distance connectivity rather than regional spillovers central to intra-provincial competition.⁹ Remarkably, political fragmentation extends beyond domestic markets, with international export values increasing by 3.1% when competition intensity decreases, suggesting strategic resource reallocation inadvertently undermines export performance. Furthermore, bilateral provincial trade flows confirm this mechanism, showing origin province competition effects significantly outweigh destination effects, demonstrating that politicians primarily control outbound trade policies through their resource allocation decisions.

I further examine how yardstick competition affects firms through inter-firm linkages. The analysis reveals that politically induced frictions propagate along potential produc-

⁷For example, the relative age of a city politician is calculated as the difference between the politician's age and the average age of all politicians in the same province at the same time. Politicians whose age is closer to the provincial average face more intense competition because they fall within the optimal age range for promotion, competing directly with numerous similarly positioned peers. Politicians who are significantly older or younger than the average face reduced competitive pressure: older politicians because they approach mandatory retirement with limited advancement opportunities, while younger politicians because they have extended time horizons before facing peak competitive pressure. This age-based variation is plausibly exogenous because politicians cannot manipulate their birth year, age cutoffs for promotion are institutionally determined rather than subject to individual choice, and changes in the age differential mainly stem from political turnover in other cities. Section 3.1 shows how I construct this measure.

⁸Chinese leadership has historically emphasized age considerations in political appointments. A notable example is Deng Xiaoping's 干部队伍年轻化 (cadre rejuvenation) policy in the 1980s, which called for the promotion of cadres who were revolutionary, younger, more educated, and more technically specialized. This focus on age in political appointments continued under subsequent leaders, with Hu Jintao advocating for the promotion of experienced older politicians, and Xi Jinping further emphasizing age as a criterion in politician selection. https://news.ifeng.com/mainland/200702/0210_17_75079_1.shtml https://www.gov.cn/xinwen/2018-07/08/content_5304620.htm Retrieved on 31 January, 2025.

⁹Winston (1993) documents how different modes of transportation are subject to different regulations.

tion and information networks, influencing connected enterprises beyond the politicians' immediate jurisdictions. I identify these linkages using two complementary approaches: firm-level similarity derived from large language model (LLM) embeddings and historical input-output (IO) relationships from provincial tables. The results show that intensified political competition in one city leads to adverse outcomes—such as reduced access to intermediate inputs, lower revenues, and decreased labor productivity—not only among firms within the city but also among economically connected firms elsewhere in the province. This micro-level evidence suggests that internal market fragmentation arises partly from negative externalities transmitted through existing economic networks.

I identify how yardstick competition fragments markets through strategic resource allocation that paradoxically enhances short-term local performance. The mechanism operates in two stages. First, I establish that competition drives superior economic outcomes: politicians facing intense competition outperform provincial peers by 0.6 percentage points in GDP growth (1 SD increase in competition), validating that tournament incentives generate genuine effort effects. Second, I demonstrate that politicians achieve these performance advantages by redirecting public resources toward real estate and local infrastructure that generate rapid, locally appropriable GDP gains, while reducing support for tradable manufacturing that creates spillovers benefiting rivals. This allocation pattern distorts production composition, reducing tradable capacity and lowering trade through supply-side production distortions.

I document this resource allocation mechanism across four policy margins. First, competition shifts industrial subsidies from tradable manufacturing toward real estate, enabling politicians to maintain superior GDP performance while constraining sectors that would enhance rivals' supply chains; this provides the first empirical evidence for the strategic substitution framework proposed by [Becko and O'Connor \(2024\)](#). Second, competition reduces investment in inter-city motorway networks that lower bilateral trade costs ([Allen and Arkolakis, 2022](#)); spatial analysis reveals this underinvestment concentrates precisely in border regions where spillovers to rival cities are largest, with no effects in interior areas, confirming politicians strategically target high-spillover locations. Third, competition reallocates land from industrial and logistics uses toward residential development, supporting real estate-driven GDP growth while constraining tradable sectors' capacity to expand through agglomeration economies. Fourth, competition shifts city governments' stated policy priorities themselves: text analysis of annual government work reports shows that intense competition reduces the attention devoted to internal trade and raises the attention devoted to real estate, providing direct textual evidence that the reallocation is deliberate.

This paper relates to four strands of literature. First, it contributes to the literature on

China's decentralized political economy. A large literature argues that China's M-form organization and decentralized governance strengthened local incentives and facilitated market-oriented reform ([Montinola et al. 1995](#); [Qian and Weingast 1997](#); [Qian and Roland 1998](#); [Maskin et al. 2000](#); [Jin et al. 2005](#); [Xu 2011](#)). In particular, [Maskin et al. \(2000\)](#) emphasize that the comparability of regional governments under the M-form structure facilitates yardstick competition and relative performance evaluation, while [Xu \(2011\)](#) characterizes China as a regionally decentralized authoritarian system combining centralized personnel control with substantial local economic authority. A related literature stresses the broad discretion of Chinese local governments in allocating land, infrastructure, regulation, and other resources. [Bai et al. \(2020\)](#) argue that local governments use this discretion to favor firms, while competition across jurisdictions can constrain some forms of local predation.

I identify a distinct consequence of the same institutional structure. When local politicians are rewarded for performance relative to political competitors, they internalize the local returns to public policies but not benefits that spill over to rival jurisdictions. Promotion competition therefore reallocates resources away from policies with cross-jurisdictional benefits and toward activities whose returns are more locally appropriable. This creates implicit non-tariff barriers to internal trade through the underprovision of connective infrastructure and other policies that facilitate interregional exchange. Thus, while the existing literature emphasizes how decentralized competition strengthens local incentives and supports market development, I show that the same competition can fragment the national market when policy benefits cross jurisdictional boundaries.

Second, this paper contributes to the literature on political competition, yardstick competition, and economic growth. Classic models of yardstick competition show how comparison across jurisdictions can discipline politicians and improve policy performance ([Besley and Case 1995](#)), while [Besley et al. \(2010\)](#) show that stronger political competition induces more growth-oriented policies and raises economic growth. In China, where political competition operates through promotion rather than elections, a large literature similarly documents that career concerns and relative performance evaluation induce local officials to pursue economic growth ([Li and Zhou 2005](#); [Zhou 2007](#); [Liu 2005](#); [Lu et al. 2022](#); [Zeng and Zhou 2024](#)) and shape policy choices in other domains ([He et al. 2020](#); [Jia 2024](#)).

My findings provide a dynamic qualification to this growth-enhancing view of political competition. Consistent with the existing literature, stronger promotion competition stimulates local economic growth in the short run. However, it also changes the composition of the policies used to generate that growth. Because political rewards depend on relative local performance, politicians become less willing to undertake investments whose benefits spill over to competing jurisdictions. I show that this strategic reallocation creates non-tariff barriers to internal trade, reduces market integration, and ultimately

dampens subsequent economic growth. Political competition can therefore be locally growth-enhancing in the short run yet dynamically inefficient: it strengthens incentives to generate local growth while distorting public investment away from activities whose returns depend on an integrated national market.

Third, the paper speaks to the literature on local protectionism and internal market fragmentation. A large literature documents substantial barriers to interregional trade in China and their economic consequences (Young 2000; Bai et al. 2004; Poncet 2005; Bai and Liu 2019; Barwick et al. 2021), while related work quantifies the gains from reducing domestic trade costs (Donaldson 2015). Most closely related, Fang et al. (2025) provide a political economy explanation for local protectionism under tournament-style political competition, showing that career-concerned officials discriminate against firms from politically competing regions.

My paper identifies a distinct but complementary form of protectionism. Rather than focusing on discriminatory treatment of firms according to their geographic origin, I show that political competition changes the composition and spatial allocation of public resources themselves. Stronger promotion competition shifts government support away from tradable activities and toward non-tradable activities with geographically concentrated benefits, reduces industrial land provision, and discourages connective infrastructure investment, particularly near jurisdictional borders. These distortions raise inter-city trade costs even without formal restrictions on non-local goods or explicit discrimination against outside firms. The paper therefore broadens the political economy of local protectionism from discriminatory treatment of non-local firms to the endogenous creation of non-tariff internal trade barriers through public resource allocation. I further trace these distortions through production networks, showing consequences for intermediate-input access, firm productivity, and longer-run local development.

Finally, this paper contributes to the political economy of trade. Seminal work emphasizes how political interests and institutions shape international trade policy and economic integration (Grossman and Helpman 1995; Dollar and Kraay 2003; Dutt and Mitra 2005; Levchenko 2007; Dutt and Traca 2010; Head et al. 2010; Antràs and i Miquel 2011). Yet comparatively less is known about how political incentives operating below the national level generate trade frictions within countries, particularly when subnational governments control infrastructure, land, industrial policy, and other policies that affect the cost of exchange without taking the form of tariffs or formal trade restrictions. Existing studies of intra-national trade, including Jiang and Mei (2020) for China, show that political institutions can affect domestic integration, but do not directly identify how competition among subnational politicians endogenously shapes the policy determinants of internal trade costs.

I address this gap by showing that non-tariff internal trade barriers are endogenous to the career incentives of subnational politicians. Promotion competition changes infrastructure provision, land allocation, and industrial support in ways that systematically raise the cost of trade across jurisdictions. The paper therefore provides a political micro-foundation for domestic trade frictions: internal market fragmentation can emerge not only from explicit protectionist restrictions, but also from strategic resource-allocation decisions by local politicians who fail to internalize the benefits their policies confer on competing jurisdictions.

The paper is organized as follows: Section 2 describes detailed background information on yardstick political competition and internal trade in China; Section 3 and Section 4 describe the data and identification strategy; Section 5 presents the baseline results; Section 6 presents additional results that further shed light on the mechanism; Section 7 concludes.

2 Background

2.1 Politician's Age and Career Prospects

The institutional setup for political advancement in China's multi-level administrative system is key to understanding local policy-making. Promotions follow a strict, top-down evaluation system under the 'one-level-up' policy, where higher-level politicians evaluate subordinates primarily on age and economic performance. Age is particularly dominant in promotion decisions (Wang et al., 2020), while economic performance includes GDP growth (Jia et al., 2015) and infrastructure development (Wang et al., 2020).¹⁰

Formally, the appointment and promotion of city party secretaries run through the provincial Party Committee and its Organization Department under the 'one-level-up' (下管一级) system (Chen and Kung, 2019). The provincial Organization Department maintains cadre files, conducts evaluations, vets candidates, and proposes appointments of prefecture-level leaders, which the provincial Party Standing Committee then approves. Evaluation is anchored in the annual cadre assessment (*kaohe*, 考核), which scores leaders against target-based criteria dominated by measurable economic indicators (GDP growth, investment, and fiscal revenue), alongside social stability and party-building responsibilities (Li and Zhou, 2005; Jia et al., 2015; Wang et al., 2020). Because these criteria are standardized within a province and applied by a single principal, secretaries of the same rank in the same province are directly comparable, which is precisely the environment in which yardstick evaluation operates. Terms are nominally five years, but rotations driven

¹⁰While Chinese politicians may manipulate GDP growth for promotion, internal trade flows lack institutional salience and are not systematically monitored in cadre evaluations, making manipulation unlikely.

by promotions, transfers, and retirements produce average tenures of two to three years. Rank matters for which tournament a secretary competes in: most prefecture-level cities carry director (*zhengtingji*) rank and their secretaries are appointed by the province, while provincial capitals and sub-provincial (*fushengji*) cities carry vice-provincial rank, and their secretaries, who typically sit on the provincial Party Standing Committee, compete in a different, higher-level pool. The estimation sample accordingly excludes vice-provincial-rank cities (Section 3).

Age serves as a crucial promotion factor due to institutional features. First, the combination of local decision-making power and central control creates a tournament-like promotion system where age becomes an easily observable measure of future potential (Li and Zhou, 2005; Xu, 2011; Wang et al., 2020; Fang et al., 2025; Zeng and Zhou, 2024).¹¹ Second, mandatory retirement ages—60 years for prefecture-level cities and 65 years for province-level cities—impose strict career limits.¹² Without electoral accountability, age-based promotion offers a standardized method for comparing politicians across regions.

2.2 City Party Secretary and Within-province Competition

This study focuses on city party secretaries, who hold the highest political authority in city governance within the dual leadership structure alongside mayors. Party secretaries determine strategic priorities while mayors handle operations, highlighting the secretary’s influential role in shaping local economic policies. City party secretaries face distinctive within-province competition. Their promotion opportunities are largely confined to provincial-level positions, creating vertical mobility constraints that differ from mayors, who enjoy more flexible career paths including potential promotion to secretary roles. This restriction intensifies competition among secretaries for limited higher-level positions—while provinces may contain over ten cities, available provincial positions are far fewer.¹³

Promotion data from 2004 to 2014 confirms this dynamic: only 2% of promoted secretaries moved to central government or other provinces, while 18% advanced to provincial-level positions. In contrast, mayors enjoyed promotion rates exceeding 50% (Appendix Table I.1). Given that economic performance factors into evaluation, party secretaries

¹¹Younger politicians may demonstrate higher ability by achieving the same rank as older peers in shorter time, while having more years for potential advancement.

¹²Politicians nearing these thresholds face reduced promotion chances and are often moved to ceremonial roles.

¹³While city administrative rank may correlate with promotion likelihood—with secretaries from provincial capitals or sub-provincial cities enjoying greater visibility—the inclusion of city fixed effects absorbs time-invariant differences in baseline promotion prospects across city tiers. The age differential measure captures relative competitive pressure among secretaries within the same province at the same time, conditional on this baseline heterogeneity.

provide valuable insights into how promotion incentives shape behavior through yardstick competition.

The institutional structure of China's promotion system generates competitive rather than cooperative incentives among similarly aged secretaries. Provincial-level positions are strictly limited—typically fewer than a dozen slots for provinces containing 10–20 cities—creating a zero-sum environment where one secretary's promotion reduces peers' chances. While factional ties, personal background, and career trajectories also shape appointments at every level of the hierarchy (Shih et al., 2012), relative economic performance benchmarked against provincial peers is one of the central criteria in the evaluation of city party secretaries (Li and Zhou, 2005; Wang et al., 2020). The “one-level-up” policy means secretaries compete for the attention of the same provincial organization department, which uses standardized age and performance criteria to rank candidates. These features ensure that age similarity intensifies rivalry rather than fostering collaboration. This study hypothesizes that secretaries adjust behavior based on age relative to provincial peers—those significantly younger or older face less intense competition, while those with similar ages compete most intensely for limited positions without age-based advantages. Figure H.1 shows the hypothesized non-linear relationship between politician age and trade volumes.

2.3 City Trade, Spillover, and Local Protectionism

In China's institutional context, trade volumes between cities via motorway and waterway transportation serve as important indicators of local protectionism, particularly across internal administrative boundaries. These modes, used primarily for shorter distances unlike airway and railway transportation, are particularly useful for studying protectionist behaviors from within-province competition.

City party secretaries can implement protectionism through direct and indirect methods. Direct actions are most visible in waterway and railway sectors, where local politicians exercise significant regulatory control. Even in road transportation, secretaries can impose local tolls and administrative fees.¹⁴

Indirect methods are equally critical, driven by strong yardstick competition incentives. Competition motivates secretaries to strategically reduce subsidies to tradable sectors, preferring non-tradable sectors like real estate that generate locally-confined benefits without enhancing rival jurisdictions' performance. This strategic focus on minimizing positive spillovers is supported by evidence that trade activity correlates with beneficial

¹⁴Truck tolls in China differ widely across provinces and by charging method: some use weight, others axle or distance. Despite the nationwide ETC rollout in 2020, provincial autonomy means a loaded truck can still pay far more than an empty one, and similar routes can carry different costs across regions.

economic outcomes in nearby cities (see Table 1.2). Politicians possess additional policy instruments including land provision manipulation, road infrastructure development, and trade policy emphasis. These tools, explored in Section 6.2, collectively enable strategic management of inter-jurisdictional economic interactions for political advantage. Figure H.2 illustrates the proposed causal pathway.

3 Data

In this section, I present the data construction, sources, and key trends in the data. Other data descriptions can be found in Appendix A.

3.1 Yardstick Competition Measured by Politician Age Differential

This study utilizes biographical data for city party secretaries, including start age, education, and promotion history.¹⁵ The key measure to capture yardstick political competition intensity is the ‘Politician Age Differential.’

I calculate this measure using Equation 1:

$$\text{Politician Age Differential}_{i,c,t} = \left| \text{Start Age}_{i,c} - \text{Average Start Age}_{p,t} \right| \quad (1)$$

where i , c , p , and t represent the politician, city, province, and year, respectively. $\text{Start Age}_{i,c}$ is the age of politician i when they began their term in city c . $\text{Average Start Age}_{p,t}$ is the average starting age of other city party secretaries within the same province during year t . Figure H.3 illustrates how the replacement of a peer secretary mechanically shifts this measure for secretaries on either side of the peer norm.

The effectiveness of this measure depends on two characteristics. First, empirical evidence (see Figure 1) demonstrates that this age differential successfully predicts promotion likelihood. Second, the variation in age differentials must be independent of other factors that could influence protectionist policies.

The measure is designed to mitigate endogeneity concerns through three mechanisms. First, it relies on politicians’ ages at the *beginning* of their terms rather than during tenure, ensuring the measure remains uncorrelated with short-term economic performance (Wang et al., 2020). Second, the primary variation source stems from routine political turnover, representing an external shock to competitive pressure.¹⁶ Third, the analysis incorporates

¹⁵I do not use mayors as the main research subjects because mayors do not face within-province yardstick political competition as aforementioned. I use city mayors for the falsification test.

¹⁶Political turnover among rivals (rivals’ age) is unlikely to correlate with unobservable local characteristics affecting trade volumes. Jiang and Mei (2020) argue that political turnover in China is plausibly exogenous.

politician fixed effects to control for strategic assignment possibilities.

3.2 City Trade Volumes and Export Value

The primary outcome variables are city-level trade volumes and export values sourced from the Chinese City Statistical Yearbook, spanning 1996 to 2018. Trade volumes encompass transportation across four modes: motorway, waterway, railway, and airway (measured in tons), while export values capture international trade flows (measured in USD). These indicators primarily reflect domestic trade patterns and regional commerce within China's borders. The analysis includes only cities with complete and consistent data coverage throughout the entire study period. Reflecting the rank distinction discussed in Section 2, the baseline estimation sample further excludes cities whose party secretary holds vice-provincial rank (provincial capitals and sub-provincial cities), since these secretaries compete in a different, higher-level tournament rather than the within-province tournament studied here; they enter the analysis only through the separate placebo test in Column (5) of Table 1. Figure H.4 maps average inter-province trade flows over the sample period.

Freight tonnage is a proxy for inter-regional trade, not a direct measure of it: it records the physical weight of goods moved, embedding transshipment, return trips, and intra-city logistics, a limitation long recognized in work on China's domestic integration (Poncet, 2003, 2005). Two considerations make it suitable nonetheless. First, since tonnage is the dependent variable, any wedge between measured volume and true trade acts as classical measurement error, widening standard errors without biasing the competition coefficient unless the error correlates with competition, which the fixed effects and the exogeneity of peer turnover rule out. Second, the freight measure closely tracks an independent, input-output-based measure of inter-provincial trade (Appendix Figure H.5). By mode, road transport accounts for roughly three-quarters of domestic freight tonnage and is the mode over which city governments have the most discretion, while railway (about 9 percent) and airway are centrally administered and serve mainly long-distance connectivity. I therefore treat motorway and waterway volumes as the policy-relevant outcomes and railway and airway, over which city governments exercise little discretion, as comparison modes.

3.3 Summary Statistics

Table I.3 presents summary statistics for the primary variables employed in this analysis, spanning the period from 1996 to 2018, covering 289 cities. The data are organized across six panels, encompassing trade, politician characteristics, economic indicators, firm-level variables, city-sector aggregates, and firm registration data.

Panel A reports statistics for trade and transportation variables based on 4,819 city-year observations. The logarithm of total trade volume exhibits a mean of 8.628 with a standard deviation of 1.032. Among transportation modes, motorway trade demonstrates the highest average volume at 8.441, followed by railway (6.157) and waterway (5.977) transportation. Airway trade volume shows a mean of -1.095, consistent with its more specialized nature and limited geographic coverage. International export values average 6.445 in logarithmic terms across 3,638 observations.

Panel B documents politician characteristics, revealing an average age differential of 4.213 years between local officials and regional benchmarks. Politicians commence their tenure at an average age of 50.46 years and maintain an average tenure duration of 2.892 years. Panel C indicates that regional GDP averages 15.56 in logarithmic terms across all observations.¹⁷

Panels D and E present firm-level characteristics and city-sector aggregates respectively, with detailed statistics provided in the table. Panel F encompasses firm registration statistics from 5,967,624 observations, indicating average logarithmic firm entries of 1.705, with 48.3% classified as tradable enterprises and 4.2% as real estate firms.

4 Empirical Strategy

4.1 Specification

To test whether age differential affects trade volumes, I estimate the following equation:

$$Y_{c,t} = \alpha_0 + \alpha_1 \text{Politician Age Differential}_{i,c,t} + X'_{i,c,t} + \delta_c + \sigma_i + \tau_t + \epsilon_{c,t} \quad (2)$$

where subscripts i , c , and t denote politician, city, and year, respectively. $Y_{c,t}$ is the logarithm of trade volumes (via road, waterway, train, airway, total, and export value) in city c and year t . The key independent variable, $\text{Politician Age Differential}_{i,c,t}$, measures the absolute difference between a city leader's age and the average age of other city leaders in the same province, as defined in Equation 1. The vector $X'_{i,c,t}$ includes time-varying controls for politician characteristics (age, rivals' age) and city attributes (GDP). δ_c denotes city fixed effects;¹⁸ σ_i denotes politician fixed effects;¹⁹ τ_t denotes year fixed effects. $\epsilon_{c,t}$ is the error term. Standard errors are clustered at the city level.

¹⁷Figure H.6 shows the distribution of the signed and absolute Politician Age Differential, and Figure H.7 shows distributions for the main trade outcome variables.

¹⁸I control for city fixed effects to account for unobserved city-specific factors that may influence both age differential and trade.

¹⁹I control for politician fixed effects to account for unobserved politician-specific characteristics that might be correlated with both age differentials and trade outcomes.

The coefficient of interest is α_1 , which captures the effect of Politician Age Differential on trade volumes and export value. A positive coefficient would indicate that larger age differentials are associated with higher trade volumes, supporting the hypothesis that intense yardstick political competition (smaller age differentials) leads politicians to engage in local protectionism. Conversely, when politicians face less direct competition from age-similar peers, they may pursue policies that facilitate inter-regional trade.

4.2 Identification Assumptions

The identification strategy assumes that, conditional on city, politician, and year fixed effects and time-varying controls, variation in Politician Age Differential is exogenous to unobserved determinants of city-level trade volumes.

From Trade Volumes to Trade Barriers Trade volume and trade barriers are conceptually distinct: higher barriers reduce volume, but low volume need not reflect high barriers. I address this by splitting the determinants of a city’s trade volume into a persistent component (preferences, technology, geographic trade costs) and a variable component (local policy, business-cycle conditions, input prices). City, politician, and year fixed effects absorb the persistent component, so identification comes only from the variable component. The key assumption is that yardstick competition, driven by peer secretary turnover in other cities, shifts the policy part of this variable component while remaining orthogonal to the rest. Under this assumption, the competition-induced change in trade volume identifies the policy channel, even though volume levels do not themselves measure barrier levels; Section 6.2 documents the underlying policy instruments directly.

Plausibly Exogenous Variation Variation in age differential comes mainly from the staggered rotation of city party secretaries across the province and from politicians’ start ages. When a peer politician elsewhere is replaced, the focal politician’s relative age position shifts for reasons unrelated to conditions in the focal city, namely bureaucratic turnover elsewhere. These rotations follow institutional rules (terms up to five years, averaging 2-3 years) and are set by the Central Organization Department based on province-wide personnel considerations largely orthogonal to local trade conditions, making the resulting variation plausibly exogenous.

Fixed Effects as Additional Safeguard Politician and city fixed effects absorb time-invariant differences that might otherwise correlate with both age differential and trade outcomes, guarding against strategic assignment or selection. Identification then relies solely on within-politician variation in relative age position over time.

Balance Tests Figure H.8 shows no significant correlation between age differential and observable characteristics, including politician education, STEM background, and tenure, or city attributes such as population, GDP, and firm counts, once fixed effects are included. This supports the assumption that age differential does not proxy for other trade-related factors.

5 Results

5.1 Validating Age Differential as a Competition Measure

To validate that age differential correlates with career prospects and within-province competition intensity, I analyze how age differential affects politician career outcomes using the following specification:

$$1(\text{Politician Career Prospect})_{i,c} = \sum_j \alpha_j \text{Signed Age Differential Bin}_{j,i,c} + X'_{i,c} + \delta_c + \sigma_t + \epsilon_{i,c} \quad (3)$$

The dependent variable captures two career outcomes: (1) Promotion, an indicator for whether politician i in city c is promoted following their tenure; and (2) Transfer/Retirement, an indicator for whether the politician is transferred or retired. Demotion serves as the reference category. The key explanatory variable represents discrete bins of the signed age gap (politician's age minus provincial average), with the $[-1, 1]$ year range as the reference category; negative values indicate younger than peers, positive values indicate older. I use binned age differentials in Equation 3 to visualize the non-linear relationship across the age distribution in Figure 1; Tables I.4 and I.5 present regressions using a continuous measure. Controls include individual characteristics, city-level variables, connections to provincial politicians, city fixed effects, and tenure start year fixed effects. Standard errors are clustered at the city level.

Figure 1 presents the relationship between signed age differential bins and career prospects. The youngest politicians (relative to peers) exhibit the highest promotion probability, while the oldest have the lowest. Tables I.4 and I.5 present corresponding regressions: a one-year increase in signed age differential reduces promotion probability by 3.3 percentage points (Table I.4, column 4) and increases transfer/retirement probability by 3.9 percentage points (Table I.5, column 4). These effects are attenuated for a higher city tier, e.g., deputy-level cities, consistent with Wang et al. (2020).

The signed estimates establish that career stakes operate on both sides of the peer-age distribution. Section 5.2.4 provides complementary evidence based on the density of close-in-age competitors and tests the symmetry restriction underlying the absolute-value

specification used in the trade regressions.

5.2 Yardstick Competition and City Trade Volumes

In this subsection, I estimate Equation 2 to examine the age differential effect on city trade volumes.

5.2.1 Politician Age Differential and City Trade Volumes

Table 1 reports the baseline regression results examining the impact of Politician Age Differential on various measures of city trade volumes. Panel A presents estimates for total city trade volumes across four progressively stringent specifications, followed by two placebo columns. Column (1) employs city fixed effects and year fixed effects as the baseline specification. Column (2) augments this with politician and city characteristics. Column (3) further adds politician fixed effects and replaces year fixed effects with province-year fixed effects to absorb province-specific time trends. Column (4) instead implements city-politician fixed effects together with year fixed effects to account for within-politician variation across different postings; I treat this as the paper’s preferred specification throughout. The coefficient on Politician Age Differential remains positive and statistically significant across all four specifications. Under the preferred specification in Column (4), the estimated coefficient of 0.012 indicates that a one standard deviation increase in Politician Age Differential corresponds to a 2.6 percent increase in total trade volumes.²⁰

The magnitude of this effect is economically substantial and comparable to findings in [Li and Zhou \(2005\)](#), and is consistent with evidence that promotion incentives near age thresholds raise reported GDP growth by about 3.4 percentage points ([Zeng and Zhou, 2024](#)), suggesting that political competition represents a significant factor in economic policy implementation. My results further show a yardstick competition nature of the promotion system.

Panel B analyzes motorway trade volumes, where coefficients remain positive and significant across specifications, with a magnitude comparable to the total-trade effect (2.9 percent increase under Column 4). This finding aligns with evidence that highway-driven market access fosters trade integration ([Faber, 2014](#)). Consistent with the underinvestment mechanism discussed in Section 6.2, politicians facing stronger competition invest less in trade infrastructure connecting to their rivals.

²⁰The standard deviation of Politician Age Differential is 2.208 years, yielding the calculation: $0.012 \times 2.208 = 0.026$.

Panel C examines waterway trade, revealing the largest effect with a 6.2 percent increase. This substantial impact reflects the high degree of government control over waterway and port infrastructure, as documented by the institutional evolution and planning of China's port/IWT system (Notteboom and Yang, 2017), making waterway systems particularly responsive to local officials' strategic decisions.

Remarkably, Panel D shows that international export value increases by 3.1 percent when domestic competition intensity decreases. While politicians may not be incentivized to restrict exports directly, their fragmentation of internal markets creates unintended spillovers that damage export competitiveness. This suggests that local protectionism's costs extend far beyond domestic efficiency: it inadvertently undermines the very export performance that politicians seek to boost. I explore multiple channels through which this occurs in the later sections.

The differential effects across transportation modes reveal that politicians strategically target policy instruments where they exercise greater control and where impacts on competitors are more direct. This selective intervention is consistent with theoretical and empirical accounts of China's political tournament system.

Column (5) provides a complementary placebo by restricting the sample to secretaries of vice-provincial-rank cities (provincial capitals, sub-provincial cities, and cities whose secretary sits on the provincial Party Standing Committee) who compete in a different, higher-level tournament and do not face the same within-province promotion competition as director-level secretaries (Section 2). Consistent with this institutional difference, the estimated coefficient for this subsample is small and statistically indistinguishable from zero across all four trade outcomes.

Column (6) provides a falsification test by analyzing the behavior of city mayors, who operate under different institutional constraints and career incentives than party secretaries. While city party secretaries face intense, geographically-constrained competition within their province, mayors have more flexible career paths. For instance, mayors can be promoted to party secretary positions, meaning they do not compete in the same closed tournament as secretaries.²¹ This institutional difference implies that within-province yardstick competition, as measured by relative age, should not significantly influence mayors' policy decisions regarding trade.

The results confirm this prediction. When the same specification is run using the age differential among city mayors, the estimated coefficients are statistically indistinguishable from zero and economically negligible across all trade categories. This null result for mayors provides strong evidence for the proposed mechanism. It demonstrates that the link between relative age and trade policy is not a general phenomenon related to

²¹The promotion rate of mayors is around 50%, while that of party secretaries is approximately 18%.

age itself, but is specific to the institutional context of the party secretaries' promotion tournament. The absence of an effect for mayors, who face different career incentives, helps to isolate the paper's central mechanism. A horse-race specification that includes both the secretary's and the mayor's age differential jointly confirms this pattern: the secretary's coefficient remains positive across all outcomes and statistically significant for total, motorway, and waterway trade, while the mayor's coefficient is small and statistically insignificant throughout (Appendix Table I.7).

5.2.2 U-Shaped Relationship

To trace out the shape of this relationship rather than impose linearity, I re-estimate the trade regression using lagged discrete bins of the signed age differential in place of the continuous absolute-value measure:

$$\ln(Y_{c,t}) = \sum_{\substack{j=1 \\ j \neq 4}}^7 \beta_j \text{Signed Age Differential Bin}_{j,c,t} + \delta_c + \sigma_i + \eta_{c \times i} + \tau_t + \epsilon_{c,t} \quad (4)$$

where $Y_{c,t}$ is the trade outcome for city c in year t , and the seven bins are defined as in Equation 3, with the Peer Average bin ($j = 4$, the $[-1, 1]$ year range) omitted as the reference category. δ_c , σ_i , $\eta_{c \times i}$, and τ_t denote city, politician, city-politician, and year fixed effects, respectively. Standard errors are clustered at the city level.

The results are consistent with Figure H.1, which shows the hypothesized non-linear relationship between politician age and trade volumes. This non-parametric specification confirms the pattern across the full distribution of the age differential (Figure 2): trade volumes are lowest when a politician's age is close to the provincial average and rise as the differential widens in either direction. The pattern holds on both sides of the distribution, ruling out the possibility that the baseline result is driven by seniority alone; politicians who are much younger than their peers, not only those who are much older, also generate higher trade volumes. The U-shape is most pronounced for waterway trade and comparatively muted for motorway trade, mirroring the panel-by-panel ranking in Table 1.

This shape is consistent with career incentives that vary non-monotonically with age differential rather than linearly with age itself. Politicians far from the peak competitive window, whether much younger with a longer remaining horizon or much older with diminishing promotion prospects, face weaker incentives to erect local trade barriers than those in the middle of the distribution, where promotion is most immediately contested.

5.2.3 Robustness Checks

Multiple robustness tests confirm the baseline findings. First, a placebo test shows no effects using cross-provincial age differentials (Table I.9), which should not respond to within-province competition. Second, the trade response lines up with the transport modes that city governments actually control. Railway and airway volumes, which are centrally administered, show no significant response individually (Table I.10). To compare modes formally, I stack the four mode-level outcomes into a single regression with mode-specific coefficients and mode-interacted fixed effects (Appendix Table I.11). The mode-specific estimates reproduce the pattern, but equality across modes cannot be rejected, so I read the contrast as directionally consistent with the mechanism rather than as an independent placebo test. Third, results are robust to alternative specifications: different sets of fixed effects and alternative clustering (Table I.12), and an instrumental variable strategy (Table I.8; first-stage relationship in Appendix Figure H.9). Fourth, alternative competition measures based on education or tenure show similar but weaker effects, confirming that age-based competition drives the results (Appendix G.9). Finally, provincial-level analysis using inter-provincial railway trade (Appendix G.1) yields consistent findings, suggesting that competition-driven protectionism is not unique to city-level politicians but operates uniformly across the political hierarchy. Notably, trade-reducing effects originate primarily from competition intensity in the exporting region, indicating that politicians strategically erect barriers to boost local performance relative to rivals.

Additional checks reported in the Appendix further support the baseline specification. Because leadership turnover may itself induce changes in local policy (Shen et al., 2022), I first exclude city-years in which the focal city's own secretary turns over. The estimates remain similar, thereby isolating variation in competitive pressure generated by turnovers of peer secretaries in other cities while holding the focal city's leadership constant (Appendix Table I.18). The estimates are also robust to controlling for lagged trade volume and historical economic conditions (Appendix Table I.13). Railway infrastructure, which is centrally planned and outside local politicians' control, shows no corresponding response to competition, as expected under a valid identification strategy (Appendix Table I.15). The relationship is also robust to a battery of heterogeneity and placebo checks: it holds across the secretary's tenure year and does not vary systematically with proximity to retirement (Appendix Figures H.10 and H.11), it is not concentrated in cities with particular initial economic conditions (Appendix G.6), it shows no pre-existing trend prior to a peer turnover (Appendix G.8), and it is not an artifact of mechanical tenure effects (Appendix G.7).

5.2.4 Similar-Age Competitors

The age differential is a continuous, cutoff-free measure of a politician's distance from the provincial peer norm, but two politicians with the same differential may face different numbers of close-in-age rivals. I therefore complement it with a measure of competitive density: the number of other city party secretaries in the same province who fall within an approximately three-year age band of the focal secretary's start age. This bandwidth is motivated by Figure 1, which shows that career pressures concentrate near the provincial peer-age norm.

For career outcomes, the measure's "first stage," I use the term-average count of peers whose start ages fall within three years of the focal secretary's age. For the annual trade analysis, I use the count of peers whose ages fall within three years of the focal secretary's age.

Appendix Table I.6 relates similar-age peer density to career outcomes. One additional similar-age peer lowers the probability of promotion and raises the probability of transfer or retirement, with similar estimates when deputy-province cities, provincial capitals, or both are excluded. These associations are consistent with tournament congestion: a denser set of close-in-age rivals worsens a secretary's career prospects.

An analogous annual count reproduces the main trade result. Appendix Table I.16 shows that one additional similar-age peer is associated with lower total, motorway, and waterway trade; the export estimate is also negative but imprecise. This sign mirrors the positive coefficient on Politician Age Differential: greater close-in-age rival density indicates stronger competition and predicts lower trade.²²

The absolute-value construction also embeds a symmetry restriction: a given distance from the peer norm is assumed to matter equally for secretaries above and below it. Appendix Table I.17 tests this restriction directly by decomposing the differential into years younger and years older than the peer average. The coefficients are positive on both sides for every trade outcome, and tests of equality yield p -values of 0.240, 0.212, 0.491, and 0.869 for total, motorway, waterway, and export trade, respectively, so I do not detect meaningful asymmetry in how the two sides of the peer-age distribution affect trade. Appendix B complements this evidence by formalizing how the measure responds to the replacement of a peer secretary and showing that the case in which an extreme old-for-young replacement leaves the measure unchanged is confined to a small share of the observed data.

²²The pattern extends to complementary outcomes. More similar-age peers predict stronger short-run GDP performance relative to provincial peers (Table 3), although the corresponding nighttime-light estimates are noisier (Appendix Table I.19). In provincial bilateral trade, both origin and destination counts are negative (Appendix Table I.29).

5.2.5 Event-Study: Similar- Versus Dissimilar-Age Peer Arrivals

To provide dynamic evidence and distinguish the age-similarity channel from a generic turnover shock, I compare the arrival of a similar-age peer secretary (within three years of the focal secretary's start age) with the arrival of a dissimilar-age peer in another city of the same province. For each group $g \in \{S, D\}$, I estimate the following event-study specification separately:

$$Y_{c,t} = \sum_{\substack{k=-4 \\ k \neq 0}}^4 \beta_k^g D_{i,c,t}^g(k) + X'_{i,c,t} \gamma + \delta_c + \sigma_i + \eta_{c \times i} + \tau_t + \epsilon_{c,p,t}, \quad g \in \{S, D\}, \quad (5)$$

where $D_{i,c,t}^g(k)$ indicates that city c is observed k years relative to the arrival of a peer of type g : S denotes an incoming secretary whose age is within three years of the focal secretary's age, and D denotes an incoming secretary outside that band. The specification includes four leads and four lags, omits the arrival year ($k = 0$), and includes the same controls and city-politician and year fixed effects used in the preferred specification. Standard errors are clustered at the city level.

Interpreting the dynamics requires that current shocks to a focal city's trade do not determine the timing and age composition of assignments in other cities. This is plausible because bureaucratic rotations are governed by the centralized personnel system and reflect term limits, promotions, and retirement rather than trade conditions in any one city. The lead coefficients provide a diagnostic for pre-trends: a competition response should follow, rather than precede, the arrival of a similar-age rival.

Figure 3 reports the results. The pre-arrival estimates do not display a systematic decline in trade for the similar-age series. After arrival, the similar-age series shifts downward, whereas the dissimilar-age series shows no comparable decline. Because dissimilar-age arrivals are also turnover events, this contrast is consistent with an age-similarity channel rather than a generic province-wide turnover effect.

The effect weakens when I use an alternative, wider similar-age band of five years (Appendix Figure H.12), as the wider band admits less comparable peers and weakens rivalry intensity, confirming that the three-year result reflects genuine age-based competition rather than the bandwidth choice.

6 Additional Empirical Evidence

Having documented that yardstick competition reduces inter-city trade, I now examine the mechanisms through which political tournaments fragment markets. The analysis proceeds in four steps. First, I establish that competition drives politicians to outperform their provincial peers in GDP growth, validating that yardstick competition operates as theorized. Second, I demonstrate that politicians achieve these performance gains not through explicit trade barriers or price manipulation, but through strategic resource allocation that systematically favors non-tradable sectors over tradables. This allocation pattern distorts production composition, reducing tradable capacity and thereby lowering trade flows through supply-side distortions. Third, I turn to firm-level data to show that these effects propagate through inter-city business networks, directly linking the city-level trade results to disruptions in firm-level input sourcing, revenue, and productivity. Fourth, I show that while this strategy generates short-term GDP gains, the cumulative costs of market fragmentation substantially undermine long-term economic development.

6.1 Mechanism: Competition and Relative Economic Performance

I examine how competition intensity affects short-term economic performance. Understanding this relationship is crucial for identifying the mechanism through which political tournaments shape policy choices: do politicians restrict trade because they are economically weak, or do trade restrictions serve as strategic instruments to enhance competitive position?

Table 3 presents results on the relationship between Politician Age Differential and contemporaneous GDP performance relative to provincial peers. The analysis employs city-politician fixed effects and year fixed effects to isolate within-politician variation across postings and control for common temporal shocks. Column (1) measures the absolute GDP gap in hundreds of millions of RMB between a city and the provincial average, while Column (2) expresses this gap as a percentage deviation from the provincial mean.

The results reveal a significant negative relationship between competition intensity and relative economic performance. A one standard deviation increase in Politician Age Differential is associated with a 15.4 hundred million RMB decrease in GDP relative to the provincial average (Column 1), and a 0.6 percentage point reduction in relative GDP performance (Column 3).²³ A parallel specification using nighttime lighting as an alternative proxy for short-term local economic performance yields a consistent, though noisier, pattern (Appendix Table I.19).

²³Calculations: $6.953 \times 2.208 = 15.4$ hundred million RMB; $0.270 \times 2.208 = 0.6$ percentage points, where 2.208 years is the standard deviation of Politician Age Differential.

The negative coefficients indicate that politicians facing stronger competitive pressure—those with smaller age differentials from their provincial peers—achieve better short-term GDP performance relative to competitors. This finding is consistent with tournament theory predictions ([Lazear and Rosen, 1981](#); [Li and Zhou, 2005](#)).

Integrating these findings with the trade restriction patterns documented in Table 1 reveals the strategic logic underlying local protectionism. Politicians facing intense competition (small age differential) simultaneously (1) achieve better short-term GDP growth relative to provincial peers, yet (2) impose greater trade barriers against those same competitors. This apparent paradox dissolves when recognizing that trade restrictions serve as a complementary policy instrument rather than a substitute for growth-enhancing efforts. By fragmenting markets, politicians can capture a larger share of economic activity within their jurisdiction, protect local industries from inter-regional competition, and boost short-term GDP metrics that are crucial for promotion evaluation—all while constraining competitors’ access to their local markets and resources.

This strategic use of protectionism is further rationalized by China’s official rotation system. With average party secretary tenures of 3-4 years, politicians face strong incentives to prioritize policies that generate immediate, observable performance improvements over those that maximize long-term welfare. Trade restrictions offer such short-term benefits: they can quickly boost local GDP by redirecting economic activity inward, increase fiscal revenues through heightened administrative discretion, and demonstrate responsiveness to local interest groups—all measurable within a single term. The longer-term costs of market fragmentation, including reduced specialization, diminished economies of scale, and foregone agglomeration benefits, typically materialize beyond the politician’s tenure and are difficult to attribute to specific policy choices.

6.2 Four Non-tariff Barriers that Distort Trade

The reduced-form results show that stronger yardstick competition raises internal trade barriers, but not through which instruments. Identifying these channels matters for three reasons: it clarifies whether the main result reflects specific, manipulable policy margins rather than diffuse protectionism; it has direct welfare implications, since infrastructure, subsidies, and land allocation carry different efficiency costs than border tariffs; and it matters for measurement, since these barriers are invisible to conventional checkpoint-based trade surveys despite appearing to be the dominant channel linking political incentives to market fragmentation. I identify four channels: intercity motorway investment, the sectoral composition of industrial subsidies, land allocation across tradable and non-tradable uses, and policy mentions. Across all four, politicians facing intense competition do not

cut spending indiscriminately; instead, they strategically withhold investments most likely to spill over to rival jurisdictions while sustaining spillover-free alternatives.

6.2.1 Endogenous Trade Costs: Competition and Intercity Motorway Network Development

This subsection examines how competition shapes investment in inter-city road infrastructure, focusing on controlled-access motorways. Motorway construction represents the canonical high-spillover policy: it connects multiple jurisdictions, lowers bilateral trade costs, and re-routes economic activity beyond investing city boundaries. When competitive pressure is acute, incumbents tilt spending toward low-spillover, place-based amenities with geographically contained benefits. When competitive pressure weakens, the political return to high-spillover projects rises, making cross-jurisdictional roads more attractive (Bougheas et al., 1999; Limao and Venables, 2001).

Motorway networks are a key determinant of within-country trade frictions and market integration. Spatial trade models show that improving transport links lowers generalized trade costs across many origin-destination pairs, triggering general-equilibrium adjustments in production, consumption, and routing, with benefits diffusing well beyond the directly treated links.²⁴ In China, expressways lower ad-valorem transport costs by roughly 20 percent relative to regular roads and raise aggregate exports, confirming motorways as a trade-cost-reducing, high-spillover technology (Fan et al., 2023).

I test whether competition depresses connective infrastructure investment using a GIS panel that combines political data with geospatial transport networks (Davis et al., 2025).²⁵ Rather than estimate a uniform average effect, I exploit spatial variation in spillover exposure across counties, since the theory predicts where underinvestment occurs, not that it occurs everywhere. I classify each county as *border* or *interior* depending on whether it adjoins a rival city's counties within the same province. Motorway investment in border counties mechanically lowers bilateral trade costs to rivals and generates substantial cross-jurisdictional spillovers through general equilibrium adjustments (Allen and Arkolakis, 2014); investment in interior counties, buffered from rival boundaries, yields mostly local returns with little leakage. If politicians concentrate underinvestment where the spillover-to-local-cost ratio is highest, the effect of Politician Age Differential on motorway provision should be concentrated in border counties and weak or absent in interior ones.

Table 6 reports interaction-term regressions across five motorway network measures.

²⁴Allen and Arkolakis (2014) estimate the U.S. Interstate Highway System raised welfare by about 3.5 percent; Allen and Arkolakis (2022) similarly show network improvements reshape flows and welfare far beyond treated links.

²⁵The Davis et al. (2025) dataset provides annual GIS records for highways, motorways, and rail, with routes and access points detailed enough to construct consistent connectivity measures over time.

The interaction between Politician Age Differential and the border-county indicator is positive and statistically significant across all five outcomes, while the uninteracted (interior-county) effect is small in magnitude and statistically indistinguishable from zero. This pattern indicates that competitive underinvestment in motorway infrastructure is not a uniform cut to spending but concentrates precisely in border locations, where each foregone infrastructure dollar imposes the greatest relative damage on rival jurisdictions' access, exactly where the spillover-based mechanism predicts it should.

Spatial trade theory clarifies why this location-specific pattern, rather than a uniform average effect, is the relevant test. In models with gravity-consistent trade and endogenous routing, social returns to motorway links depend on bilateral trade cost reductions across the network, not centrality alone. Investments increasing link density on secondary corridors can compress shipping times and improve reliability for many city pairs, raising market access in neighboring jurisdictions. [Allen and Arkolakis \(2014, 2022\)](#) demonstrate such network-wide spillovers are first-order for welfare and activity distribution, implying even "local" roadwork has non-local consequences. Chinese expressway evidence supports this: lower per-kilometer costs translate into higher aggregate trade, even for off-trunk expansions. Table [I.20](#) shows the effect of motorway development on trade volume and exports.

This spatial targeting provides direct evidence that politicians strategically differentiate infrastructure investments based on spillover incidence. Rather than uniformly cutting all motorway spending, which would impose indiscriminate costs on local economic performance, politicians concentrate underinvestment in border counties where cross-jurisdictional spillovers are most salient, while maintaining relatively stronger investment in geographically insulated interior counties. This pattern reveals sophisticated cost-benefit optimization: politicians accept local economic costs precisely where those costs generate maximum relative harm to competitors through reduced trade spillovers, consistent with politicians understanding and exploiting the geographic structure of infrastructure spillovers within promotion tournaments ([Besley and Case, 1995](#)).

A natural question is how an instrument that accumulates over multi-year horizons can be reconciled with annual variation in trade flows. The answer is that while the motorway stock itself moves slowly, the margins through which it affects trade costs are adjusted year by year. Construction is incremental: local governments decide annually which segments to fund, accelerate, or delay, so even a persistent infrastructure strategy translates into yearly changes in effective network capacity ([Donaldson, 2018](#); [Allen and Arkolakis, 2014](#)). Moreover, trade responds to effective rather than physical trade costs, and local governments operate the existing network at high frequency, adjusting tolls, weight limits, inspection intensity, and maintenance on routes serving rival cities ([Young,](#)

2000; Bai et al., 2004; Barwick et al., 2021). The stock of motorway capacity thus pins down the medium-run level of trade costs a city carries, while annual choices over construction pace and network operation generate the year-to-year responses documented in Table 1.

6.2.2 Production Costs and Productivity: Industrial Policy Spending

This subsection studies how yardstick competition shapes the allocation of government subsidies across sectors. Industrial policy, particularly subsidies, is a central tool of Chinese economic governance (Aghion et al., 2015). Recent work on geoeconomics argues that peacetime industrial policy substitutes for credible conflict-time trade punishment: governments unable to threaten future trade taxes rely more heavily on industrial policy to build strategic capacity today (Becko and O'Connor, 2024). Applied to the subgovernmental context, where local politicians lack direct trade instruments such as tariffs, this logic implies that political incentives should instead show up in the composition of subsidies across tradable and non-tradable sectors. Politicians facing intense yardstick competition may shift subsidies away from tradable sectors, whose productivity gains could spill over to rival jurisdictions, and toward non-tradable sectors, whose benefits stay local. If yardstick competition discourages policies that generate positive spillovers to rivals, it should produce systematic differences in subsidy allocation across these two sector types.

To test this mechanism, I construct a city-sector panel and estimate:

$$\begin{aligned} \text{Log(Subsidy)}_{s,c,t} = & \beta_0 + \beta_1 \text{Politician Age Differential}_{c,t} \times \text{Sector}_s \\ & + X'_{s,c,t} + \delta_{c,s} + \sigma_i + \tau_{s,t} + \gamma_{c,t} + \epsilon_{c,s,t}, \end{aligned} \quad (6)$$

where $\text{Log(Subsidy)}_{s,c,t}$ is the logarithm of subsidies received by sector s in city c during year t . Sector_s equals 1 if the sector is tradable (manufacturing and wholesale following (Brandt et al., 2017)), 0 otherwise. Controls include city-sector characteristics and politician characteristics; fixed effects are at the city-sector, politician, sector-year, and city-year levels. Standard errors are clustered at the city level.

Panel A of Table 4 shows that the interaction between Politician Age Differential and the tradable-sector indicator is positive and significant across specifications. In the full model (Column 4), the coefficient of 0.087 implies that a one-standard-deviation increase in age differential (weaker yardstick competition) is associated with a 27.3 percent increase in subsidies to tradable sectors relative to non-tradables.²⁶ Consistent with the strategic-substitution logic in Becko and O'Connor (2024), when direct trade-policy instruments are unavailable at the local level, politicians facing less competitive pressure deploy industrial

²⁶Using $(e^{0.087} - 1) \times 100 = 9.1\%$, then $2.98 \times 0.091 = 0.273$.

policy more aggressively toward tradables despite cross-jurisdiction spillovers. By contrast, the interaction with real-estate sectors is negative (-0.121) and statistically significant, suggesting that a one-standard-deviation increase in age differential (weaker yardstick competition) is associated with a 34.0 percent decrease in subsidies to real-estate sectors relative to other sectors.

Overall, these patterns indicate that yardstick competition systematically distorts public resource allocation. Under intense competition, politicians limit support for tradables to avoid strengthening rivals, redirecting resources instead toward non-tradables with geographically confined benefits, consistent with evidence on local political incentives in China (Chen and Kung, 2019; Wang et al., 2020; Lu et al., 2022). The magnitudes, on the order of 50 percent shifts in relative subsidy allocations, represent economically meaningful policy distortions and support the broader logic that industrial policy composition adjusts when other policy instruments are constrained.

6.2.3 Agglomeration and Capacity: Land Allocation

Land allocation is a first-order policy lever for local politicians, shaping both urban form and the sectoral composition of economic activity. In decentralized settings, allocation choices respond to yardstick competition. Land-use planning and the timing, quantity, and composition of land released to the market thus become instruments for influencing observable outcomes (GDP growth, investment, fiscal take) (Brueckner, 2000). Given the post-1994 fiscal architecture, land-lease revenues (“land finance”) provide a large, discretionary funding source for local governments, further heightening the political salience of land supply decisions (Cai et al., 2013; Fang et al., 2019).

To examine these mechanisms, I re-estimate Equation 2 at the city level, rotating the outcome variable across distinct measures of land provision and value. Specifically, I compile city-year aggregates for (i) manufacturing/tradable industrial land, (ii) real estate land (residential/commercial), and (iii) other non-tradable sector land. Industrial land transactions are drawn from the Ministry of Natural Resources’ public registry,²⁷ which reports universe-level transactions from 2000 to 2018; I classify parcels by designated end use following standard administrative categories used in urban land markets (see, e.g., Cai et al., 2013). For each category, I study both physical provision (area released) and contemporaneous market value (total transaction value), recognizing that quantities and prices capture different margins of political intervention.

Table 7 shows how the intensity of yardstick competition maps into land allocation across sectors. Consistent with the mechanism, weaker competition (larger age differential) induces a reallocation toward tradables and away from non-tradables.

²⁷China’s Ministry of National Land and Resources website <https://www.landchina.com>.

From Columns (1) to (2), I find that a one-standard deviation increase in Politician Age Differential raises industrial land provision by 12.4% and the total value of industrial land sold by 20.1%. This pattern is consistent with politicians facing less immediate benchmarking pressure being more willing to invest in slower-maturing, high-spillover industrial capacity ([Duranton and Puga, 2004](#)).

In Columns (3) to (4), I find the same increase in age differential is associated with a 13.9% decline in real estate land provision and a 32.5% decline in the total value of real estate land sold. This contrasts sharply with the industrial margin and aligns with the notion that intense yardstick competition pushes politicians toward projects with rapid, locally appropriable gains in measured GDP and fiscal revenues. From Columns (5) to (6), for land provisions for other non-tradable land, one standard deviation increase in age differential falls by 14.2% and value by 23.8%.

The allocation responses fit a political economy logic. Industrial land and related infrastructure generate broader spatial spillovers, through supply chain linkages, labor pooling, and knowledge externalities, that extend beyond jurisdictional borders ([Duranton and Puga, 2004](#); [Greenstone et al., 2010](#); [Combes et al., 2011](#)). When peer rivalry eases, the political cost of conferring benefits that partially accrue to neighbors falls, making such investments more attractive. Real estate and other non-tradables, by contrast, deliver benefits that are highly visible, quickly realized, and geographically confined. These incentives are reinforced by governance shocks that reduce rent-seeking in land markets, altering both the composition and pricing of land transactions ([Chen and Kung, 2019](#)). Consistent with this reallocation, weaker competition is also associated with more new firm registrations in tradable sectors and fewer in real estate (Appendix Table I.21); see Appendix G.2 for details.

6.2.4 Stated Policy Priorities: Trade and Real Estate Attention in Government Work Reports

The three instruments above operate through budgets, infrastructure, and land quotas. A complementary question is whether the underlying reallocation is visible in politicians' own stated priorities. Annual government work reports are the canonical statement of a city government's policy agenda: delivered at the start of each year, they set out the leadership's priorities for the year ahead. Because the reports are drafted under the direct control of the city leadership and revised every year, stated priorities can respond to competitive pressure at the same annual frequency as the competition measure itself, making this the fastest-moving of the four channels, in contrast to infrastructure and land

allocation, which adjust over multi-year horizons.²⁸

To measure policy attention, I construct length-weighted indices of trade-related and real-estate-related policy mentions from the full text of each city's annual report, weighting the frequency of topic-related mentions by the length of the corresponding policy text. Appendix Figure H.13 reports the keyword composition underlying the two indices. I then estimate specifications analogous to Equation 2, with the attention indices as outcomes.

Table 5 reports the results. Columns (1) to (3) examine trade-policy mentions and Columns (4) to (6) real-estate mentions, progressively adding politician characteristics and city-politician fixed effects. In the most demanding specification, a larger Politician Age Differential (weaker competition) significantly raises attention to internal trade (0.302, Column 3) and significantly lowers attention to real estate (−0.181, Column 6). Equivalently, politicians facing intense yardstick competition shift their stated agenda away from internal trade and toward real estate. This textual evidence complements the budgetary and allocative margins: the reallocation documented in subsidies, infrastructure, and land shows up in what politicians say they will do, indicating a deliberate shift in policy priorities rather than an incidental by-product of other decisions.

6.3 Competition and Firm Supply Chain Disruption

In this section, I examine the effect of competition on firm outcomes, especially those connected via business networks.

6.3.1 Local and Spillover Effects of Politician Age Differential on Firms

Yardstick competition can affect economic activity not only through a city's own trade flows but also through the inter-city linkages in which firms are embedded: intense competition may lead politicians to underinvest in policies that generate beneficial network spillovers. I therefore hypothesize that stronger competition (a smaller age differential) worsens the performance of firms reliant on inter-regional network ties. To test this, I construct a firm-level panel from the Annual Survey of Industrial Firms (ASIF), spanning 2000–2007 following Brandt et al. (2017), combined with inter-firm business network connections.

Business networks are identified in two complementary ways. The first approach follows Breitung and Müller (2025): firm pairs are linked when the text-based similar-

²⁸Government work reports are widely used in the literature on Chinese local governance to measure the targets and priorities of city leaderships. Li et al. (2019) extract GDP growth targets from provincial and prefectural work reports to study target setting in promotion tournaments, and Campante et al. (2023) track the frequency of stability-related language in work reports to show that local governments reallocate attention toward public security. I follow this approach and treat the allocation of text across policy topics as a revealed measure of the leadership's stated priorities.

ity of their industry classification, business activities, and main products exceeds 95%; Appendix C details the construction, embedding model, and threshold choice, Table I.22 provides example matched pairs, and Appendix Figure H.14 visualizes the resulting networks. The second approach uses province-specific Input-Output (IO) tables to measure sectoral connectedness within each province (Appendix D).

The empirical strategy estimates the impact of yardstick political competition faced by the politician in firm i 's city on the economic outcomes of the connected firm j located in a different city. I estimate the following equation:

$$Y_{j,s,c_j,t} = \beta_0 + \beta_1 \text{Politician Age Differential}_{i,c_i,t} + \mathbf{X}'_{j,s,c_j,t} + \delta_{pair} + \sigma_i + \tau_{s,t} + \gamma_{c_j,t} + \epsilon_{j,s,c_j,t} \quad (7)$$

where $Y_{j,s,c_j,t}$ represents firm outcomes (e.g., log intermediate inputs, log revenue) for the connected firm j in sector s and city c_j at year t . The key independent variable is Politician Age Differential $_{i,c_i,t}$ for the politician governing firm i 's city c_i at year t . Note that c_i and c_j denote distinct cities: c_i is the focal city where political competition is measured, and c_j is the connected city where firm j 's outcomes are observed. $\mathbf{X}'$ includes time-varying controls for firm j . The specification incorporates firm-pair fixed effects (δ_{pair}) to control for time-invariant characteristics of each specific firm linkage, politician fixed effects (σ_i) for the politician in city c_i , sector-year fixed effects ($\tau_{s,t}$), and connected-city-year fixed effects ($\gamma_{c_j,t}$). Standard errors are clustered at the city level. The coefficient of interest, β_1 , captures the effect of yardstick political competition in city c_i on its network partners in city c_j .

Table 2 presents the results. Columns (1)-(4) report direct effects on the focal firm i .²⁹ a larger Politician Age Differential (weaker competition) is associated with significantly better performance across all four outcomes. A one standard deviation increase raises intermediate good inputs by 5.0% and revenue by 3.5%, with profits and revenue per worker of similar magnitude (Table 2).³⁰ Columns (5)-(8) show that the spillover effects on connected firms j , estimated from Equation (7), are qualitatively identical and slightly attenuated, as expected for effects transmitted through the network. The estimates are robust to replacing the text-based network with pre-sample (1997) provincial input-output linkages (Table I.23). Together, the direct and spillover estimates indicate that intense competition generates negative externalities (trade frictions and underinvestment in shared infrastructure) that harm firms within the jurisdiction and propagate to their network

²⁹For direct effects, I estimate an analogous specification in which firm i is both the focal and outcome unit; the firm-pair fixed effect then collapses to a firm fixed effect. Outcomes are intermediate inputs, profit, revenue, and revenue per worker, all in logs.

³⁰The standard deviation of Politician Age Differential is 2.192 in the focal-firm sample and 1.986 in the connected-firm sample.

partners elsewhere in the province.

A natural objection is that tournament competition should *help* local firms: career-concerned officials favor them through preferential procurement and implicit barriers against outsiders (Fang et al., 2025; Barwick et al., 2021). Because favoritism accrues to politically connected and state-linked firms while private, trade-reliant firms bear the cost of fragmented markets, the two channels can be separated by ownership. The split (Appendix Table I.24) confirms this: the adverse competition effect is borne almost entirely by private firms, while for state-owned enterprises the SOE interaction offsets the main effect to a near-zero net, for both focal and connected firms. Both channels are therefore present: favoritism insulates state firms, and the trade-barrier channel falls on unprotected private firms. The average effect is negative because private, trade-reliant firms dominate the sample.

This decomposition also resolves the apparent tension between the firm-level losses and the short-run GDP gains in Section 6. The reconciliation is sectoral: under competitive pressure, leaders channel resources toward non-tradables, above all real estate, which accounts for roughly a quarter of Chinese GDP and is the lever most directly under a leader's control through land conveyance (Rogoff and Yang, 2026). Non-tradable value added therefore delivers the GDP boom while trade barriers depress tradable value added. Because the ASIF panel covers industrial (tradable) firms only, the firm-level losses measure the tradable side of this composition effect by construction. A direct decomposition of city GDP points the same way, with a higher tradable share and lower non-tradable share under weaker competition, though the estimates are imprecise and serve as directional corroboration only (Appendix Table I.25). Measured GDP can therefore rise even as tradable firms contract, a composition effect whose land-financed gains are short-lived, consistent with the long-run reversals documented below.

A placebo test using connected firms located in bordering cities across provincial lines shows no effect (Table I.26), consistent with the within-province tournament mechanism rather than broader regional shocks or unobserved confounders.

I further examine firm heterogeneity in competition effects and find that the effect is concentrated among downstream firms, consistent with these firms being more embedded in and reliant upon upstream network connections for inputs. The effect is also concentrated among supplier/partner relationships rather than potential competitors, consistent with the mechanism operating through disruption of input-sourcing linkages. Section G.3 in the Appendix reports these heterogeneity results in detail, distinguishing effects by counterparty type (potential competitor versus supplier/partner) and by supply-chain position (upstream versus downstream).

6.3.2 Economic Costs of Yardstick Competition on Firms

To gauge magnitudes, I translate the firm-level estimates into an illustrative productivity cost. The logic chain runs from inputs to aggregate output: the 5.0% intermediate-input reduction estimated above implies a total factor productivity decline of roughly 1.2%, using the imported-inputs elasticities of Halpern et al. (2015); sector-specific versions of this calculation (Table I.27) are then aggregated with sector GDP shares from China's 2000 Input-Output table. The result, in Table I.28, is that a one standard deviation increase in yardstick competition costs 0.54% of GDP, or approximately \$6.48 billion in 2000 USD. Two scope qualifications apply. First, this is a tradable-sector productivity cost, not an overall GDP effect: the calculation covers only the tradable sectors matched to input-output shares (roughly one-third of GDP) and deliberately excludes the non-tradable side.³¹ Second, the short-run effect on measured total GDP runs in the opposite direction, because competition simultaneously expands non-tradable activity (Section 6.2): a real productivity loss initially masked by the non-tradable boom, precisely the configuration that generates the long-run reversals documented below. Appendix F details the production-function assumptions, elasticities, and aggregation underlying the calculation.

6.4 Persistent Consequences of Yardstick Competition

Having established that yardstick political competition correlates with both protectionist policies and better short-term GDP performance, this section examines whether these effects persist over extended horizons. While competitive pressure generates immediate economic gains, the trade barriers it induces may create institutional distortions that accumulate over time. I employ a long-run perspective to assess whether sustained yardstick competition yields persistent growth advantages or ultimately undermines development.

To examine long-term consequences, I estimate the following cross-sectional specification:

$$\Delta Y_{c,1996-2018} = \alpha_0 + \alpha_1 \overline{\text{Politician Age Differential}}_{c,1996-2018} + X'_c + \delta_p + \epsilon_c \quad (8)$$

where $\Delta Y_{c,1996-2018}$ represents the change in economic outcomes for city c between 1996 and 2018. The key variable, $\overline{\text{Politician Age Differential}}_{c,1996-2018}$, captures a city's cumulative exposure to competitive pressure over more than two decades by averaging the age differential between each party secretary and their provincial peers. The vector X'_c

³¹The four sectors entering Table I.28 (Processing, Products, Textiles, and Natural Resources) account for 33.5% of GDP in the 2000 input-output table. Sectors outside this set are omitted rather than assigned a zero effect.

includes baseline city characteristics, and δ_p denotes province fixed effects. Standard errors are bootstrapped (500 times) at the province level.

Table 8 reveals a reversal of the short-term patterns: cities with higher average age differentials, indicating lower competitive intensity and less protectionism, exhibit significantly better long-term outcomes. A one standard deviation increase in the average age differential is associated with 2.25% higher annual GDP growth, 2.14% higher lighting growth, 2.82% higher population growth, and 2.89% higher trade growth, with particularly strong effects on population and trade. These patterns suggest that cities whose party secretaries faced less intense competition, and consequently erected fewer trade barriers, achieved stronger market integration, which attracted labor and capital, fostered agglomeration economies, and enabled sustained development (Donaldson, 2018; Faber, 2014).

6.4.1 Reconciling Short-Term Gains with Long-Term Costs

The contrast between Tables 3 and 8 reveals a fundamental tension: competitive pressure enhances short-term GDP performance, yet sustained competition undermines long-term development. The resolution lies in the distinction between effort and instruments. Competitive pressure induces greater effort, consistent with tournament theory (Lazear and Rosen, 1981; Li and Zhou, 2005), but the protectionist instruments deployed create persistent institutional distortions. Trade barriers become entrenched through vested interests and bureaucratic inertia (Acemoglu et al., 2001), and the accumulated stock of protectionist policies erected by successive administrations ultimately outweighs the transient benefits of enhanced effort. The multitask agency problem reinforces this mechanism (Holmstrom and Milgrom, 1991): competitive politicians rationally prioritize policies that boost short-term, measurable GDP even when market fragmentation undermines long-term growth.

Trade fragmentation generates dynamic efficiency losses that compound over time through obstructed technology diffusion, foregone scale economies, and resource misallocation (Frankel and Romer, 1999). Local protectionism thus represents not merely a distributional inefficiency but a dynamic one that reduces aggregate growth. Even well-designed tournament systems generate substantial costs when performance metrics emphasize short-term outcomes and agents possess policy instruments with persistent negative externalities.

These findings illuminate a tension in China's political tournament distinct from democratic contexts. While Besley et al. (2010) show that a lack of yardstick competition in the U.S. may harm growth through reduced electoral discipline, China's promotion system, based on relative performance rather than electoral accountability, generates the opposite

problem: intense peer competition incentivizes inefficient protectionism as politicians prioritize demonstrable short-term gains over long-term regional development. The desirable level of yardstick competition thus depends critically on the institutional arrangements governing political selection.

7 Conclusion

This paper establishes that internal trade barriers are endogenous to the design of subnational political institutions. I demonstrate that yardstick competition – often praised for disciplining local politicians – can generate perverse incentives that systematically impede economic integration. My theoretical framework shows that when career advancement is tied to relative performance and economic activities generate spillovers, rational politicians face a dominant strategy to fragment markets.

To test this framework, I exploit exogenous variation in yardstick competition intensity in China, proxied by relative age differentials among city-level party secretaries within provinces. I find that a one standard deviation increase in this age differential (indicating lower competition) is associated with a 2.6% increase in inter-regional trade volume.

My analysis uncovers specific policy levers used by politicians under competitive pressure: misallocation of industrial policy spending, biased land provision, and underinvestment in trade-enhancing infrastructure. These distortionary effects are not transitory; regions subject to more intense yardstick competition exhibit lower long-term economic development, evidenced by reduced nighttime lights and population growth.

The policy implications extend beyond China's context. Any system combining subnational autonomy with relative performance evaluation risks creating incentives for local protectionism. Policy reforms should address underlying political economy mechanisms by redesigning personnel evaluation systems to incorporate inter-jurisdictional cooperation measures, establishing institutions that reward collaborative competition, and creating oversight structures to counter politically motivated trade restrictions.

This research demonstrates that the political architecture of decentralization, not just its degree, fundamentally shapes economic outcomes. As countries devolve authority to subnational governments, addressing these institutional design challenges becomes critical for realizing the benefits of both political decentralization and economic integration.

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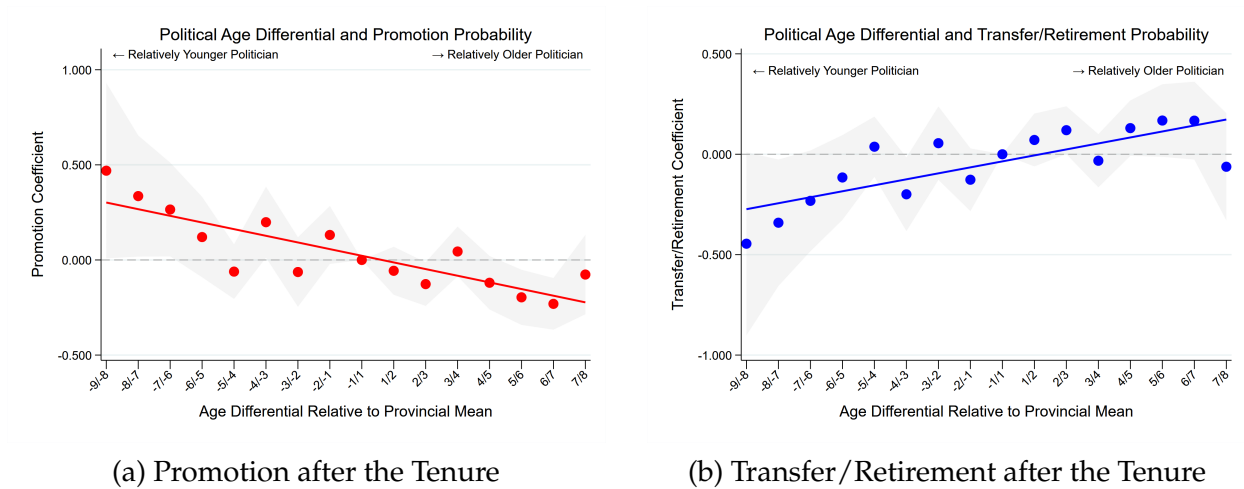
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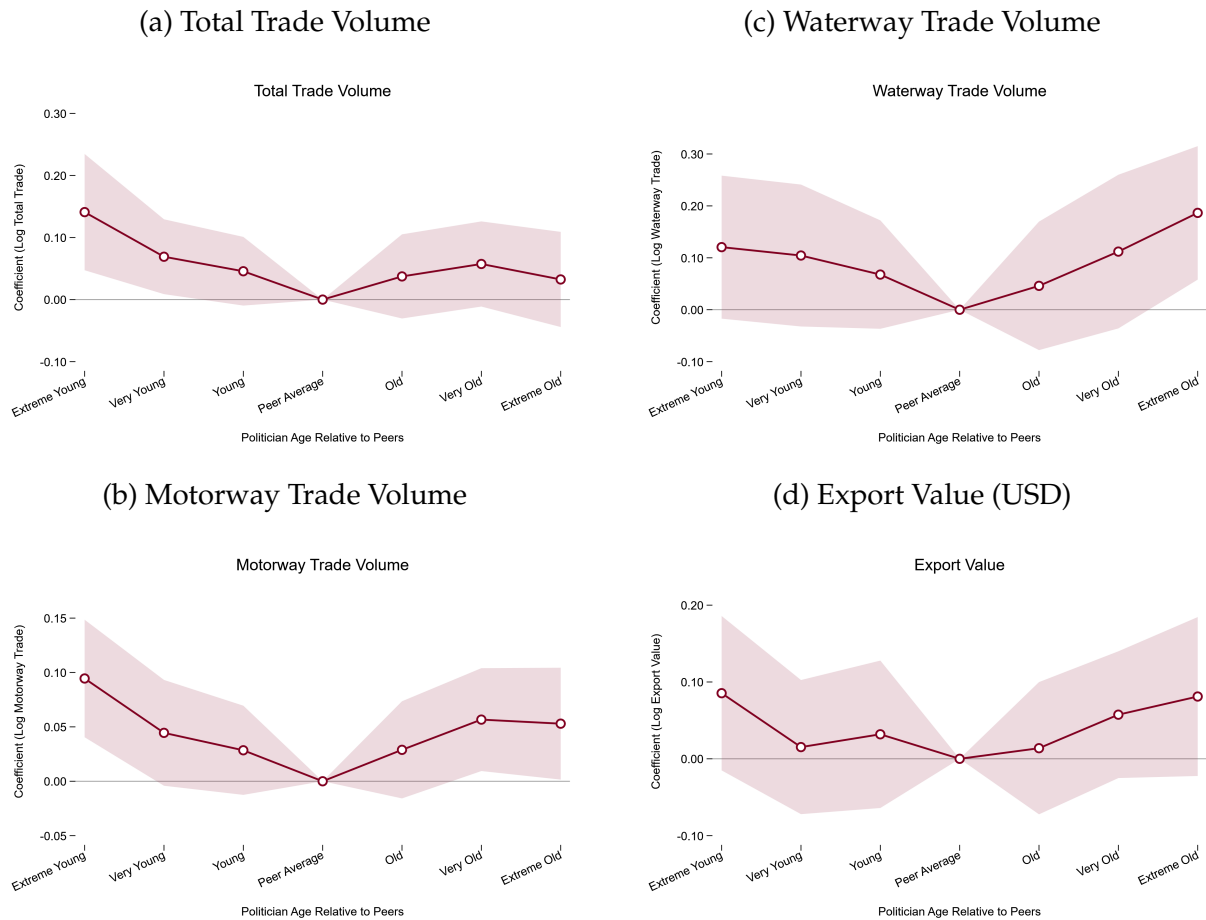
Figures

Figure 1: Signed Politician Age Differential and Career Prospects (Controlling Politician's Age)



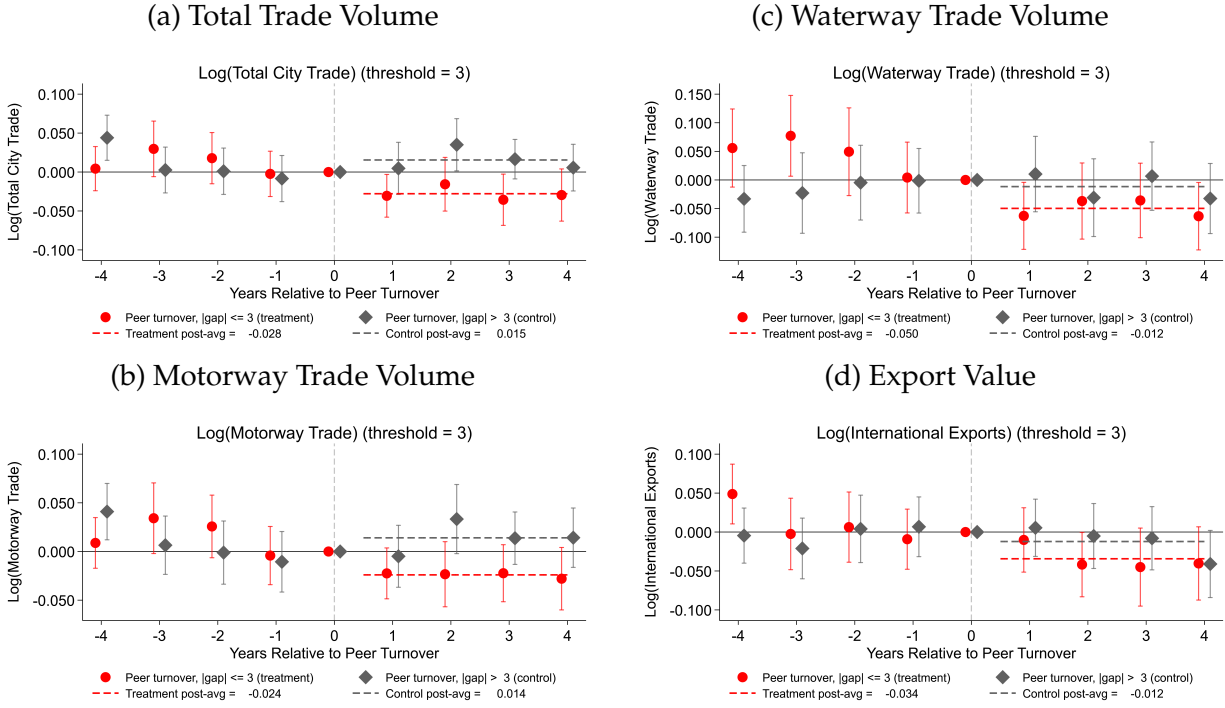
Notes: This figure shows how the signed age differential affects career prospects. The signed age differential is calculated as a politician's age minus the mean age of other party secretaries in the same province during the term; negative values indicate younger than peers, positive values indicate older. This signed measure is used here to separately examine career incentives for younger versus older politicians; the main trade regressions instead use the absolute value of this gap, which captures competition intensity regardless of direction. The x-axis shows binned age differentials, with $[-1,1]$ as the reference category. The analysis uses politician term-level data from 2000 to 2014. Controls include city characteristics, politician characteristics, politician's provincial connections, city fixed effects, and tenure start year fixed effects. Standard errors clustered at the city level. 95% confidence intervals shown.

Figure 2: The U-shaped Relationship between Politician Age Differential and Trade Volumes



Notes: These figures show the estimated effects of Politician Age Differential on different trade volumes across different age categories. The analysis uses city-level panel data from 1996 to 2018. The x-axis shows seven categories of age differential relative to the provincial peer average: 'Extreme Young', 'Very Young', 'Young', 'Average' (reference category), 'Old', 'Very Old', and 'Extreme Old'. The y-axis shows the estimated coefficients from a regression of log trade volume on age differential category indicators. I control for city and politician characteristics, city fixed effects, politician fixed effects, year fixed effects, and city-politician fixed effects. Standard errors are clustered at the city level. 95% confidence intervals are shown.

Figure 3: Event Study: Similar- versus Dissimilar-Age Peer Arrivals



Notes: These figures present event-study estimates of the effect of a within-province peer secretary's arrival on city trade outcomes. The similar-age series (red circles) consists of arrivals for which the incoming peer secretary's age is within three years of the focal secretary's start age; the dissimilar-age series (gray diamonds) consists of arrivals outside that band. The x-axis shows years relative to the peer arrival, and the y-axis shows the estimated coefficients. The horizontal dashed lines report the average post-arrival coefficient for each group. All specifications include city, politician, city-politician, and year fixed effects and the baseline controls. Standard errors are clustered at the city level. Vertical bars show 90% confidence intervals.

Tables

Table 1: Effects of Politician Age Differential on City Trade

Variables	City Party Secretary (Within-province Competition)				Placebo: Vice-Provincial City Secretary	Placebo: City Mayor
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Total Trade Volume</i>						
Politician Age Differential	0.018*** (0.007)	0.017** (0.007)	0.012** (0.005)	0.012** (0.005)	0.005 (0.008)	-0.000 (0.006)
Observations	4,624	4,624	4,624	4,624	657	5,107
R-squared	0.656	0.657	0.871	0.871	0.971	0.937
<i>Panel B: Motorway Trade Volume</i>						
Politician Age Differential	0.023*** (0.006)	0.023*** (0.006)	0.013*** (0.004)	0.013*** (0.004)	0.007 (0.008)	0.004 (0.004)
Observations	4,480	4,480	4,480	4,480	621	4,858
R-squared	0.801	0.801	0.927	0.927	0.958	0.954
<i>Panel C: Waterway Trade Volume</i>						
Politician Age Differential	0.015 (0.015)	0.016 (0.015)	0.028*** (0.010)	0.028*** (0.010)	0.000 (0.016)	-0.003 (0.008)
Observations	2,530	2,530	2,530	2,530	328	2,708
R-squared	0.864	0.864	0.964	0.964	0.976	0.976
<i>Panel D: Export Value (USD)</i>						
Politician Age Differential	0.016* (0.008)	0.014* (0.008)	0.014* (0.008)	0.014* (0.008)	0.006 (0.006)	-0.007 (0.004)
Observations	3,404	3,404	3,404	3,404	500	3,754
R-squared	0.924	0.924	0.971	0.971	0.986	0.988
City Characteristics	–	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	–	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	–	–	–
Politician FE	–	–	Yes	–	–	–
Year FE	Yes	Yes	–	Yes	Yes	Yes
Province-Year FE	–	–	Yes	–	–	–
City-Politician FE	–	–	–	Yes	Yes	Yes

Notes: This table examines how politician age differentials between party secretaries (within the same province) affect city-level trade volumes. The analysis uses city-level panel data from 1996 to 2018. The key independent variable Politician Age Differential measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Columns (5) and (6) report placebo tests: column (5) restricts the sample to vice-provincial cities, and column (6) analyzes the city mayor's age differential, neither of which features within-province competition of the same kind. Panel A shows effects on total trade volume, Panel B on road trade, Panel C on waterway trade, and Panel D on the value of international exports (USD). All trade volumes are logged and measured at 10,000 tons. I control for politician characteristics, city fixed effects, politician fixed effects, year fixed effects, and city-politician fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Effect of Politician Age Differential on Firm Network and Firm Outcomes

VARIABLES	Firm _i Located in City _i Direct Effect on Focal Firms				Firm _j Connected to Firm _i in Business Network Indirect/Spillover Effect on Firms Show a 95% Similarity Score (LLM)			
	Log(Intermediate Good Inputs) (1)	Log(Profits) (2)	Log(Revenue) (3)	Log(Revenue per Worker) (4)	Log(Intermediate Good Inputs) (5)	Log(Profits) (6)	Log(Revenue) (7)	Log(Revenue per Worker) (8)
<i>Panel A: Baseline Specification</i>								
Politician Age Differential in City _i	0.027*** (0.007)	0.030* (0.016)	0.023*** (0.007)	0.023*** (0.007)	0.020** (0.009)	0.057** (0.028)	0.022** (0.009)	0.026*** (0.009)
Observations	27,336	27,344	27,342	27,337	120,305	120,324	120,322	120,308
R-squared	0.824	0.731	0.879	0.846	0.790	0.691	0.845	0.819
<i>Panel B: With Firm Controls</i>								
Politician Age Differential in City _i	0.028*** (0.008)	0.031* (0.016)	0.024*** (0.007)	0.024*** (0.007)	0.026*** (0.009)	0.065** (0.028)	0.029*** (0.010)	0.028*** (0.009)
Observations	27,336	27,344	27,342	27,337	120,305	120,324	120,322	120,308
R-squared	0.844	0.744	0.902	0.850	0.818	0.708	0.878	0.825
<i>Panel C: With Firm Controls, Sector-Year Fixed Effects, and City_j-Year Fixed Effects</i>								
Politician Age Differential in City _i	0.023*** (0.005)	0.011 (0.013)	0.016*** (0.005)	0.016*** (0.006)	0.008*** (0.003)	0.015* (0.009)	0.005** (0.002)	0.008*** (0.003)
Observations	27,336	27,344	27,342	27,337	120,305	120,324	120,322	120,308
R-squared	0.870	0.760	0.911	0.864	0.870	0.777	0.906	0.862
Firm Pair FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how age differentials between party secretaries affect firm-level outcomes and business network connections. The analysis uses firm-level panel data from 2000 to 2007, where Columns (1)-(4) show effects on the focal firms (Firm_i), and Columns (5)-(8) show effects on firms connected to the focal firms through business networks (Firm_j). The key independent variable 'Politician Age Differential in City_i' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Panel A shows baseline results, Panel B adds firm-level controls, and Panel C further includes sector-year and connected city-year fixed effects. All specifications include firm pair fixed effects, politician fixed effects, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * p < 0.10, ** p < 0.05, *** p < 0.01.

Table 3: Competition Mechanism: Effect of Politician Age Differential on Short-Term GDP Performance Among Competitors

VARIABLES	GDP Gap Relative to Provincial Average (100M RMB)		GDP as % Deviation From Provincial Average		Next-Year GDP (100M RMB)	
	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	-6.953*		-0.270*		-6.696	
	(4.135)		(0.157)		(4.431)	
# Similar Age Peers		9.933**		0.275**		9.772**
		(4.470)		(0.120)		(4.917)
Observations	4,967	4,967	4,967	4,967	4,967	4,967
R-squared	0.972	0.972	0.989	0.989	0.980	0.980
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the relationship between within-province politician competition and short-term GDP performance relative to provincial peers. The analysis uses city-level panel data from 1996 to 2018. Politician Age Differential measures the lagged absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. # Similar Age Peers is the count of within-province peer secretaries whose age similar to the focal secretary. Columns (1) and (2) measure the GDP gap in hundreds of millions of RMB between a city and the provincial average. Columns (3) and (4) express this gap as a percentage deviation from the provincial mean. Columns (5) and (6) use next-year GDP (100M RMB) as the outcome. All specifications include city characteristics, politician characteristics, and city, politician, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects of Politician Age Differential on Industrial Policy Spending: Tradable vs. Real Estate

VARIABLES	City-Sector Log(Mean Subsidies)					
	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential $\times$ Tradable Sector	0.043* (0.024)	0.076*** (0.027)	0.087*** (0.030)			
Politician Age Differential $\times$ Real Estate Sector				-0.066 (0.057)	-0.101* (0.058)	-0.121** (0.056)
Observations	9,739	9,739	9,739	9,739	9,739	9,739
R-squared	0.644	0.707	0.743	0.644	0.707	0.743
City-Sector Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	Yes	Yes	–
Politician FE	–	Yes	Yes	–	Yes	Yes
Sector-Year FE	–	Yes	Yes	–	Yes	Yes
City-Year FE	–	–	Yes	–	–	Yes

Notes: This table examines how age differentials between party secretaries (within the same province) differently affect tradable and real estate sectors. The analysis uses city-sector panel data from 2005 to 2017. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. The dependent variable is the log of city-sector subsidies. 'Tradable' is a dummy variable for sectors producing tradable goods, while 'Real Estate' is a dummy for the real estate sector. Controls include city-sector fixed effects, city-sector characteristics, politician fixed effects, sector-year fixed effects, and city-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Effect of Politician Age Differential on Policy Attention: Government Report

VARIABLES	Trade Policy Mentions			Real Estate Policy Mentions		
	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	0.335* (0.170)	0.350** (0.173)	0.302** (0.153)	-0.180* (0.096)	-0.196** (0.097)	-0.181** (0.084)
Observations	2,679	2,636	2,574	2,679	2,636	2,574
R-squared	0.782	0.784	0.780	0.678	0.680	0.672
City Characteristics	–	Yes	Yes	–	Yes	Yes
Politician Characteristics	–	Yes	Yes	–	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	–	–	Yes	–	–	Yes

Notes: This table examines how the politician age differential affects the policy attention of city governments, measured by length-weighted policy mentions in city policy documents, where the frequency of topic-related mentions is weighted by the length of the corresponding policy text. Columns (1) to (3) report trade-related mentions, and Columns (4) to (6) report real estate mentions. The key independent variable ‘Politician Age Differential’ measures the absolute difference between a city party secretary’s age and the average age of party secretaries in other cities within the same province. Columns (1) and (4) include city, politician, and year fixed effects; Columns (2) and (5) add city and politician characteristics; Columns (3) and (6) further add city-politician fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Spatial Heterogeneity: Trade Infrastructure Under-investment under Competition

VARIABLES	Log(Motorway Sum Length) (1)	Log(Motor Exit/ Entry Count) (2)	Log(Motorway Link Count) (3)	Log(Motorway Node Count) (4)	Log(Motorway Betweenness) (5)
Politician Age Differential	-0.063 (0.055)	-0.005 (0.006)	-0.008 (0.008)	-0.005 (0.006)	-0.045 (0.031)
Politician Age Differential $\times$ Border Region	0.114** (0.057)	0.013* (0.007)	0.016* (0.008)	0.013* (0.007)	0.106*** (0.039)
Observations	38,094	38,094	38,094	38,094	38,094
R-squared	0.925	0.949	0.947	0.949	0.924
County Characteristics	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes
County-Politician FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the differential effects of politician age differential on motorway infrastructure investment between border and interior counties through interaction terms. Border counties are defined as counties that share geographic boundaries with counties belonging to rival cities within the same province, where cross-jurisdictional spillovers from infrastructure investment are most salient. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. The interaction term 'Politician Age Differential $\times$ Border Region' captures the additional effect of yardstick competition in border counties of the politician's jurisdiction. Dependent variables include log-transformed measures of motorway sum length, motor exit/entry count, motorway link count, motorway node count, and motorway betweenness centrality. All specifications include county characteristics, politician characteristics, county-politician fixed effects and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Effects of Politician Age Differential on Land Provision and Value: Tradable vs. Real Estate

VARIABLES	Manufacturing/Tradable Land		Real Estate Land		Other Non-tradable Land (Commercial)	
	Log(Industrial Land Provision)	Log(Total Value of Industrial Land Sold)	Log(Real Estate Land Provision)	Log(Total Value of Real Estate Land Sold)	Log(Other Sector Land Provision)	Log(Total Value of Other Sector Land Sold)
	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	0.040* (0.022)	0.065* (0.039)	-0.045* (0.026)	-0.105** (0.052)	-0.046 (0.028)	-0.077* (0.047)
Observations	4,155	4,155	4,155	4,155	4,155	4,155
R-squared	0.955	0.956	0.896	0.904	0.864	0.881
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how age differentials between party secretaries affect land provision and usage across different categories. The analysis uses city-level panel data from 2000 to 2018. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Columns (1)-(2) show effects on different aspects of industrial land allocation: total provision and total value of land sold. Columns (3)-(4) show effects on real estate land provision and value, while Columns (5)-(6) show effects on land provision of other non-tradable sectors. All dependent variables are in logarithmic form. All specifications include controls, city-politician fixed effects, and province-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Effects of Politician Age Differential on Long-Term Economic Outcomes (1996 to 2018)

VARIABLES	$\Delta\text{Log}(\text{City GDP})$ (1)	$\Delta\text{Log}(\text{Night-time Lighting})$ (2)	$\Delta\text{Log}(\text{City Population})$ (3)	$\text{Log}(\text{Average Trade Volumes})$ (4)
Avg. Politician Age Differential _{1996–2018}	0.234*** (0.061)	0.223*** (0.050)	0.294** (0.122)	0.301*** (0.072)
Observations	153	288	139	283
R-squared	0.424	0.637	0.308	0.560
Province FE	Yes	Yes	Yes	Yes
Baseline Characteristics	Yes	Yes	Yes	Yes

Notes: This table examines the long-term impacts of politician age differentials on city development. The analysis uses cross-sectional city-level data comparing outcomes between 1996 and 2018, averaging the politician age differential over the full window. The key independent variable is the average absolute age differential between a city's party secretary and those in other cities within the same province over the entire period. The dependent variables are log-differences (ΔLog) between 1996 and 2018: city GDP (Column 1), nighttime lighting intensity (Column 2), city population (Column 3), and average trade volumes over the period (Column 4). All specifications include province fixed effects and baseline city controls. Standard errors are bootstrapped at the province level by 500 times. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix

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A Data Description

A.1 Provincial Bilateral Trade Flows

To validate the baseline city-level findings and examine the underlying mechanism – namely, how local politicians strategically influence inter-regional trade – I employ provincial bilateral trade flow data spanning 2000 to 2018. These data, obtained from the Railway Statistical Yearbook, capture trade volumes between province pairs transported via railway networks.

A.2 City-sector Subsidies and Firm-level Data

City-sector government subsidies data are drawn from the China Stock Market & Accounting Research (CSMAR) Database. Firm-level data, including employment, revenue, and intermediate inputs, come from the Annual Survey of Industrial Firms (ASIF). The ASIF provides comprehensive coverage of industrial enterprises exceeding a revenue threshold (typically 5 million RMB), accounting for over 95% of China’s total industrial output.

A.3 Land Provision Data

Land provision data is obtained from China’s Ministry of National Land and Resources website.³² The dataset contains comprehensive land transaction information including city, purpose, area, and price. I track keywords to identify industrial/residential land provision from purpose descriptions. During 2000-2018, there are over 2.5 million land transactions, collapsed into city-year panel data.

A.4 Motorway Network Data

Motorway network data are sourced from the comprehensive GIS database of China’s transport network compiled by Davis et al. (2025). This database provides granular, annual GIS records of national highways, motorways, and railways with detailed spatial coordinates and network characteristics. The original data are aggregated to the city level for analysis. Key variables include motorway length, network nodes, network links, and betweenness centrality metrics that capture each city’s position within the broader transportation network.

³²The data can be obtained from www.landchina.com

A.5 Firm Entry Data

This study utilizes comprehensive firm registration data from China's State Administration for Industry and Commerce (SAIC), including all registered companies. The dataset provides registered addresses, registration dates, operational status, and industry classifications. I developed a city-industry-year panel tracking new firm entries from 1996-2018, containing approximately 6 million observations.

B The Politician Age Differential: A Bounded Symmetry

We measure the yardstick competition facing secretary i as

$$\text{AgeDiff}_i = |a_i - \bar{a}_{-i}|, \quad \bar{a}_{-i} = \frac{1}{n} \sum_{j \in \mathcal{N}_{-i}} a_j, \quad (9)$$

the absolute distance between i 's age and the mean age of her n peer secretaries in the same tenure cohort. The Politician Age Differential captures i 's distance from the local age norm rather than the direction of any single peer comparison, consistent with a yardstick-competition interpretation in which relative standing, not the sign of any one comparison, disciplines behavior.

A natural concern with this construction is that it may obscure real differences in competitive dynamics: replacing an old peer with a young one plausibly intensifies competition for i , while replacing a young peer with an old one plausibly relaxes it, yet both events could leave the Politician Age Differential similarly valued. We show this concern, while valid in principle, applies only under an extreme and empirically rare configuration of peer turnover. Let $x_i = a_i - \bar{a}_{-i}$ be i 's signed distance from the peer norm. Replacing one peer of age a_{old} with one of age a_{new} shifts the peer mean by $\Delta = (a_{old} - a_{new})/n$, so that

$$\text{AgeDiff}'_i - \text{AgeDiff}_i = |x_i + \Delta| - |x_i| = \begin{cases} +\Delta, & x_i \geq 0 \\ 2x_i + \Delta, & -\Delta < x_i < 0 \\ -\Delta, & x_i \leq -\Delta. \end{cases} \quad (10)$$

For politicians at or beyond the peer norm, the Politician Age Differential moves by a fixed $\pm\Delta$ regardless of the direction of turnover, correctly signing the change in competitive pressure. Only within $x_i \in (-\Delta, 0)$, an interval of width Δ , can an intensifying and a relaxing replacement produce similar changes in the measure, and only at the single point $x_i = -\Delta/2$ does it remain exactly unchanged.

To bound how often this can occur, we evaluate Δ at its theoretical maximum in our data. The sample minimum age is 36 and the sample maximum is 60; pairing these two values, the single youngest official in the sample directly succeeding the single oldest, gives $\Delta = (60 - 36)/n = 2.0$ years at the average peer group size $n = 12$. We are explicit that this is a worst-case bound rather than a documented event: because 36 and 60 are the extreme values realized anywhere in the sample, very few officials are aged at or near either endpoint, and a turnover event pairing them directly is necessarily uncommon, plausibly occurring only once or twice across the entire sample period, if at all. Any replacement actually observed in the data almost certainly induces a smaller Δ , and therefore a narrower ambiguous interval, than the one we evaluate below.

Figure H.15 plots the resulting response function against the empirical distribution of x_i . The function is flat at $\pm\Delta$ over most of its domain and bends continuously only inside the shaded interval. Even under this extreme, near-limiting case, only 19.6% of observations fall inside the ambiguous interval, and only 11.3% lie within half a standard error of the exact offsetting point. Since $\Delta = 2.0$ itself is already an upper bound rarely if ever attained in practice, these shares overstate, rather than understate, the true exposure of our identification strategy to this concern. We conclude that the symmetry embedded in equation (9) is a bounded feature of the construction, confined to a narrow and extreme case, rather than a first-order threat to identification.

C Business Network Construction

To analyze the propagation of politically induced trade barriers through inter-firm relationships, this study constructs a comprehensive business network using China’s Annual Survey of Industrial Firms (ASIF) dataset covering 2000-2007. The methodology follows [Breitung and Müller \(2025\)](#) for network identification.

The network construction proceeds in three stages. First, firm-level data is extracted for enterprises meeting the revenue threshold (typically 5 million RMB), representing over 95% of China’s industrial output. Key attributes include industry subcategory, primary business activities, geographical location, and financial metrics (output value, sales, exports, assets, liabilities, revenue, profit, and intermediate goods usage). Firms with missing values in critical fields are excluded.

Second, composite feature vectors are engineered for each firm by combining textual and quantitative characteristics. Textual descriptions are transformed into embeddings using the paraphrase-multilingual-MiniLM-L12-v2 SentenceTransformer model, while financial variables are standardized to zero mean and unit variance. These components are concatenated to form unified feature vectors.

Third, business connections are established using cosine similarity between feature vectors. Network linkages require: (1) cosine similarity ≥ 0.95 to capture only the strongest relationships, (2) firms located in the same province but different cities to identify inter-city connections relevant to internal trade barriers, and (3) unique pair recording to avoid redundancy. On the threshold choice: [Breitung and Müller \(2025\)](#) use a 99% similarity threshold to identify global firm networks; this study adopts the lower 95% threshold to capture the closer business relationships that exist among firms within the same province, where geographic proximity makes moderately similar firms far more likely to be genuinely connected.

The matched firm pairs from 2000 are extended across the 2000-2007 period, incorporating time-varying financial data when both firms remain active and satisfy geographical constraints. All industry subcategories are grouped into 45 sectors using K-Means clustering on text embeddings, with dummy variables indicating sector membership and same-sector linkages.

This methodology yields a dynamic panel dataset of inter-city firm relationships within provinces, providing the empirical foundation for analyzing how political competition and trade frictions propagate through interconnected firms across jurisdictions. Figure [H.16](#) visualizes the intensity of connections among the top 15 sectors, while Figure [H.14](#) displays network structures for selected provinces.

D Construction of Sectoral Connectedness (IO Table)

To capture structural interdependencies at the sectoral level, I construct a measure of connectedness using province-specific Input-Output (IO) tables. I specifically use the 1997 IO tables, which predate the main sample period, to mitigate potential endogeneity concerns arising from contemporaneous shocks that might jointly affect firm performance and sectoral linkages.

While the text-based measure identifies latent similarities in business scope, the IO approach explicitly maps vertical supply chain linkages between industries. Let Z_{sk}^p denote the intermediate input flows from upstream sector s to downstream sector k within province p , and let X_k^p represent the total inputs of sector k . The direct input coefficient (technical coefficient) a_{sk}^p is calculated as:

$$a_{sk}^p = \frac{Z_{sk}^p}{X_k^p}. \quad (11)$$

This coefficient represents the value of inputs from sector s required per unit of total inputs

in sector k , thereby proxying the intensity of the direct supply chain linkage.

To distinguish significant economic linkages from negligible transactions, I define sectoral connectedness based on the distribution of these coefficients within the IO table. Let M^p denote the median value of all nonzero technical coefficients in province p . The binary connectedness indicator C_{sk}^p is defined as:

$$C_{sk}^p = \mathbf{1}(a_{sk}^p > M^p), \quad (12)$$

where $\mathbf{1}(\cdot)$ is the indicator function. Sectors s and k are thus considered *connected* only if their direct input-output dependency exceeds the median linkage intensity observed in the 1997 provincial economy. This threshold approach ensures that the measure captures salient structural relationships while filtering out minor or incidental cross-sector transactions.

E Firm Pair Classification Procedure: Potential Competitors/Suppliers

This appendix details the classification of high-similarity firm pairs into “potential competitors” and “potential suppliers/partners” for the heterogeneity analysis in Section 6.3.1.

E.0.1 Conceptual Framework

Following [Chen \(1996\)](#), I identify competitors based on two dimensions: market commonality and resource similarity. Direct competitors are firms that operate in the same market segment *and* possess comparable strategic endowments. [Bergen and Peteraf \(2002\)](#) emphasize that true horizontal competitors should exhibit similarity in both product offerings and operational characteristics. In contrast, vertically related firms may have overlapping product descriptions but differ in their financial profiles due to different positions in the value chain ([Fresard et al., 2020](#)).

E.0.2 Classification Criteria

Among the firm pairs already identified as having high product similarity using text-based methods following [Hoberg and Phillips \(2016\)](#) (Section 6.3.1), I classify pairs as **potential competitors** if they satisfy two additional conditions:

1. **Same sector:** Both firms operate in the same 2-digit industry, indicating direct market overlap.
2. **Similar financial status:** Firms have comparable operating profit margins (within 5 percentage points) and asset turnover ratios (within 0.5), indicating similar business models and scale of operations.

Firm pairs that do not meet both conditions are classified as **potential suppliers/partners**. This residual category captures vertical relationships where firms have related products but occupy different positions in the supply chain, typically reflected in different profitability and asset utilization patterns.

E.0.3 Why One Similarity Metric Identifies Both Relationship Types

The text-based similarity score and the classification criteria play distinct roles, and it is the combination, not the similarity score alone, that separates horizontal from vertical relationships. High text similarity establishes only that two firms are *economically related*: firms whose industry descriptions, business activities, and products overlap are plausibly connected, but the nature of the connection is ambiguous, since both a rival producing the same good and a supplier producing its key intermediate will describe closely related products (Hoberg and Phillips, 2016; Fresard et al., 2020). The second-stage criteria then resolve this ambiguity using information that text similarity does not carry: firms in the same two-digit industry with comparable margins and asset turnover occupy the same market position and are classified as horizontal competitors, while related-product pairs with divergent financial profiles are consistent with different positions along the value chain and are classified as vertical suppliers/partners. The similarity metric is therefore a *relatedness filter* applied to all pairs, and the competitor/supplier distinction rests entirely on the market-overlap and financial-profile criteria, so no single measure is asked to proxy for two different relationship types at once. The IO-table-based connectedness measure (Appendix D), which identifies vertical linkages from input-output coefficients without any text input, provides an independent check: results using this purely vertical measure closely track the supplier/partner classification (Table I.23).

F Details of the Welfare Calculation

This appendix details the assumptions behind the tradable-sector productivity cost reported in Section 6.3.1 (Tables I.27 and I.28).

From input losses to TFP. The starting point is the firm-level effect of competition on intermediate inputs. Across the full ASIF sample, a one standard deviation decrease in

Politician Age Differential reduces intermediate input availability by 5.0%.³³ Input losses are converted to total factor productivity losses using the elasticities in Halpern et al. (2015), who find that a 1% increase in intermediate inputs raises revenue productivity by 0.22% and quantity productivity by 0.24%; I apply the quantity elasticity under imperfect substitution and constant labor and capital inputs. Sector-specific input losses and the implied TFP losses are reported in Table I.27.

Reallocation. The market fragmentation underlying these input losses also reallocates production from efficient producers toward less efficient local producers, since firms producing intermediate inputs are, on average, more productive than the local non-tradable firms that benefit from protection.³⁴ The calculation below captures only the direct input-loss channel and therefore understates the total cost.

Aggregation. Sectoral TFP losses are aggregated to a GDP impact using China’s 2000 Input-Output table:

$$\text{GDP Impact} = \sum_{s=1}^S \text{TFP Loss}_s \times \text{GDP Share}_s \quad (13)$$

where TFP losses come from Table I.27 and GDP shares from the IO table.³⁵ The four tradable sectors entering the calculation jointly account for 33.5% of GDP; the resulting total of 0.54% of GDP (Table I.28) is accordingly a tradable-sector productivity cost rather than an economy-wide effect.

G Additional Results

G.1 Bilateral Trade Flows and Provincial Yardstick Competition

To validate that the yardstick competition mechanism operates beyond the city level and through bilateral trade relationships, I examine inter-provincial trade flows using province-pair panel data. This analysis addresses two objectives: first, demonstrating that the competition effects persist when using bilateral trade flows rather than aggregate city volumes; second, showing that the mechanism operates across different levels of government where yardstick competition exists.

³³Estimated coefficient 0.023×2.192 (the standard deviation of Politician Age Differential in the firm sample) ≈ 0.050 . This is the full-sample coefficient from Column (1), Panel C of Table 2; the corresponding figure for manufacturing firms alone is 4.2%, as reported in Table I.27.

³⁴The relationship between trade participation and productivity is well established (Melitz, 2003; Bøler et al., 2015), as is the efficiency cost of misallocation (Hsieh and Klenow, 2009).

³⁵Under a production function with constant capital and labor inputs, a 1% decrease in TFP translates directly into a 1% decrease in GDP (Solow, 1957).

A note on the choice of transport mode is warranted. In the city-level analysis, railway freight served as a comparison outcome because rail primarily handles long-distance inter-provincial routes beyond the control of city-level officials, whose jurisdiction is confined to local road and waterway infrastructure. At the provincial level, however, railway freight becomes the appropriate outcome variable precisely because it captures inter-provincial trade. Provincial leaders possess greater authority over railway logistics within their jurisdictions—including freight scheduling, terminal access, and coordination with national rail networks. Thus, the shift from road/waterway freight at the city level to railway freight at the provincial level reflects the change in spatial scale: the politically salient transport mode depends on which infrastructure falls within the relevant leader's sphere of influence.

At the provincial level, party secretaries compete nationally for central government positions, creating a different but analogous tournament structure to the within-province competition among city leaders. I construct the provincial-level age differential by comparing each provincial party secretary's age to the average age of all other provincial party secretaries nationally. This measure captures the intensity of yardstick competition among provincial leaders, where those closer to the national average age face more intense competition for limited central government positions.

The empirical specification examines bilateral trade flows between province pairs:

$$\text{Log(Bilateral Trade)}_{ij,t} = \alpha_0 + \alpha_1 \text{Age Diff}_{i,t} + \alpha_2 \text{Age Diff}_{j,t} + \delta_{ij} + \tau_t + \gamma_{i,pol} + \gamma_{j,pol} + \epsilon_{ij,t} \quad (14)$$

where i and j index origin and destination provinces, δ_{ij} represents province-pair fixed effects controlling for time-invariant bilateral trade relationships, τ_t denotes year fixed effects, and $\gamma_{i,pol}$, $\gamma_{j,pol}$ are province-politician fixed effects capturing individual leader characteristics.

Table I.29 presents the results using bilateral provincial trade data spanning approximately 900 province-pairs per year. The findings demonstrate that yardstick competition affects trade flows at the provincial level, with origin province competition showing stronger effects than destination competition. A larger age differential in the origin province (indicating weaker competitive pressure) is associated with significantly higher bilateral trade volumes, with coefficients of 0.021-0.022 that are statistically significant at the 1% level.

The asymmetric effects between origin and destination provinces align with the theoretical framework. Politicians have greater control over policies affecting outbound trade flows from their jurisdiction—through industrial policies, regulatory approaches, and inter-provincial coordination—than over inbound flows, which depend primarily on destination province policies.

Notably, the destination province age differential shows weaker and statistically insignificant effects (0.012-0.014), consistent with the notion that trade-restricting policies primarily operate through origin-side mechanisms. When examining only origin effects (Column 3) or destination effects (Column 4) separately, the origin competition measure remains robust while destination effects remain insignificant.

These results provide two important validations of the main findings. First, they confirm that the competition mechanism operates through bilateral trade relationships rather than being an artifact of aggregate city-level trade measurement. Second, they demonstrate that yardstick competition effects extend beyond the specific institutional structure of city-provincial promotion tournaments to operate across different levels of government where similar competitive dynamics exist.

The consistency between city-level and provincial-level findings strengthens the interpretation that career-driven yardstick competition systematically influences politicians' willingness to implement policies that facilitate or restrict inter-jurisdictional economic integration, regardless of the specific administrative level at which this competition occurs.

G.2 New Firm Entry: Tradable vs Real Estate

Building upon established evidence that political incentives shape policy outcomes under yardstick competition, I further examine how such competition influences local economic structure through its effects on new firm entry.³⁶ I hypothesize that yardstick competition generates heterogeneous effects on firm creation contingent upon sectoral characteristics, particularly tradability and potential for generating inter-jurisdictional spillovers that may affect competing politicians' performance metrics.

Using the SAIC firm registration data previously described, this dataset covers the universe of all firms that were ever registered in China. The data contains detailed information on firms' registered address, date of registration, current state of existence, and industry (3-digit). I thus construct a county-industry-year panel data, measuring the number of firm entries. Overall, I have obtained about 5.8 million county-industry-year observations from the data period 1996 to 2018. I estimate the following equation:

$$Y_{s,c,t} = \alpha_0 + \alpha_1 \text{Politician Age Differential}_{c,t} \times \text{Sector}_s + \delta_{s,c} + \sigma_i + \gamma_{c,t} + \epsilon_{s,c,t} \quad (15)$$

where $Y_{s,c,t}$ denotes the logarithm of the number of firm registrations plus one in 3-digit

³⁶While firm entry represents a critical driver of economic dynamism, politicians can strategically manipulate entry patterns to enhance their relative performance within promotion tournaments, effectively constituting a form of industrial policy.

sector s , county c , and year t . Politician Age Differential $_{c,t} \times \text{Sector}_s$ denotes the interaction terms between Politician Age Differential and tradable/real estate sector indicators. The specification includes county-sector fixed effects ($\delta_{s,c}$), politician fixed effects (σ_i), and county-year fixed effects ($\gamma_{c,t}$). Note that the main effect of Politician Age Differential is absorbed by the county-year fixed effects, while the main effect of Sector is absorbed by the county-sector fixed effects. Standard errors are clustered at the city level.

Table I.21 reports the results. I find that weaker yardstick competition increases the number of new firm entries within tradable sectors. Specifically, the interaction term between Politician Age Differential and an indicator for tradable sectors is positive and significant across all specifications in Columns (1)-(3). A one standard deviation increase in Politician Age Differential would lead to a 1.7 percent increase in tradable firm entries.³⁷ This finding is congruent with the study's central argument that diminished competitive pressures incentivize politicians to adopt policies more conducive to activities with potential inter-jurisdictional spillovers, such as fostering tradable industries through favorable subsidy allocations or infrastructure investments, thereby creating a more attractive environment for entrepreneurial endeavors in these domains.

In contrast, the analysis reveals an inverse relationship concerning the real estate sector. The interaction term between Politician Age Differential and real estate sectors yields a negative and statistically significant coefficient across specifications in Columns (4)-(6). The results suggest that a one standard deviation decrease in Politician Age Differential (stronger yardstick competition) would lead to a 2.1 percent increase in real estate firm entries.³⁸ This observation aligns with the theoretical framework wherein politicians facing intense rivalry strategically reallocate resources towards non-tradable sectors that offer primarily localized benefits, such as real estate, which generate economic outcomes confined to their own jurisdiction without benefiting competing politicians.

These patterns of firm entry, conditional on the interplay between yardstick competition intensity and sectoral characteristics, reveal a further channel through which the career concerns of local politicians can shape the contours of local industrial structure. An environment characterized by attenuated political competition appears to foster greater dynamism in tradable sectors, whereas heightened competition may, *ceteris paribus*, make entry into non-tradable sectors like real estate relatively more prevalent as politicians prioritize geographically circumscribed economic outcomes.

³⁷Using the coefficient 0.0077 from the most complete specification in Column(3) and assuming a standard deviation of 2.2 for Politician Age Differential: $0.0077 \times 2.2 = 0.017$ or 1.7 percent.

³⁸Using the coefficient -0.0094 from Column(6): $0.0094 \times 2.2 = 0.021$ or 2.1 percent.

G.3 Firm-Level Heterogeneity: Supplier/Competitor Relationships and Supply-Chain Position

This subsection extends the firm-level network evidence in Section 6.3.1 along two additional dimensions: the type of relationship connecting firm pairs, and each firm's position in the supply chain.

G.3.1 Disruption of Supply Chain: Competitors versus Suppliers/Partners

The preceding analysis established that the intensity of yardstick political competition generates spillover effects impacting firms connected through business networks. This section examines these network effects more deeply by distinguishing between different types of inter-firm relationships. Specifically, I investigate whether the impact of competition in city c_i on a connected firm j varies depending on whether firm j is primarily a potential competitor or a potential supplier/partner to firms in city c_i . The underlying hypothesis is that the mechanisms of local protectionism induced by intense competition might disproportionately affect vertical supply chain linkages compared to horizontal competitive relationships.

To explore this heterogeneity, I categorize the firm pairs identified through network analysis (as described in Section 6.3.1) into two groups: potential competitors (characterized by high similarity in products and business activities) and potential suppliers/-partners (representing likely vertical or collaborative links).³⁹ I then re-estimate the core specification from Equation 7 separately for each group, examining the effect of the Politician Age Differential in city c_i on the performance outcomes of firm j . The results are presented in Table I.30.

Panel A of Table I.30 displays the estimation results for firm pairs classified as potential suppliers or partners. In this group, a larger Politician Age Differential in city c_i (indicating weaker yardstick competition) is associated with statistically significant improvements in the performance of the connected firm j . Specifically, the coefficients are positive and significant for intermediate good inputs, revenue, and revenue per worker. The coefficient for Log(Profits) is positive but does not reach statistical significance. These findings imply that when yardstick competition is less intense in city c_i , its potential suppliers or partners in city c_j experience tangible benefits, notably in terms of access to intermediate inputs

³⁹For potential competitors, if firms have very similar products and business activities and comparable firm scale (based on firm assets, revenue, profits and other financial status) but are located in different cities, they are classified as potential competitors operating in the same market segment. I use LLM to match their business activities and product information, which helps identify firms with highly comparable industry classifications, product offerings, and target customer bases, making them direct rivals in their respective local markets. Detailed classification can be found in Appendix E.

and overall revenue generation.

Conversely, Panel B presents the results for firm pairs identified as potential competitors. The coefficients associated with the Politician Age Differential are consistently small and statistically insignificant across all four outcome variables: Log(Intermediate Good Inputs), Log(Profits), Log(Revenue), and Log(Revenue per Worker). This lack of significance suggests that the intensity of political competition faced by the politician in city c_i does not translate into discernible positive or negative performance impacts for its potential competitor firms located in city c_j , at least through the network channels captured here.

Comparing the results across Panel A and Panel B reveals a clear distinction. The economic consequences of yardstick political competition, as transmitted through inter-city firm networks, appear to primarily manifest in relationships involving suppliers and partners, rather than direct competitors. Intense competition (smaller age differential) seems to impede economic activity within these vertical or collaborative chains more substantially than it affects parallel competitors. This suggests that the strategic trade barriers or reduced inter-regional cooperation resulting from political pressures may be particularly detrimental to the functioning of supply networks and partnerships connecting different jurisdictions. This finding refines our understanding of the market fragmentation mechanism, indicating that politically induced frictions disproportionately disrupt established inter-firm dependencies along the value chain.

G.3.2 Heterogeneity by Firm Supply-Chain Position: Upstream vs. Downstream

This subsection examines how yardstick competition affects firm performance differentially across the value chain. I estimate the direct effects specification (as described in Section 6.3.1) at the firm level, focusing on firm i 's own performance across four key metrics: intermediate inputs, revenue, revenue per worker, and profit (all in logarithmic form). A positive coefficient indicates that weaker competition (larger age differential) correlates with improved firm outcomes.

I organize firms by their position in supply chains: upstream and midstream producers of intermediate goods including Manufacturing and Processing sectors, downstream producers of final goods including Products and Textiles, and resource-based firms in Natural Resources, alongside the Full Sample and the broader Tradable sector classification. Figure H.17 presents the estimated coefficients for each category across the four outcome variables (Panels a-d).

Across the Full Sample and Tradable sector, weaker yardstick competition associates with statistically significant increases in all performance measures, confirming that firms benefit when local political incentives generate fewer inter-jurisdictional frictions. The effects are particularly pronounced for upstream and downstream producers embedded

in complex value chains.

The heterogeneity by supply chain position reveals substantial variation in responses to political competition. Manufacturing firms, representing key upstream and midstream suppliers, exhibit the largest and most precisely estimated gains in intermediate input utilization (Panel a) and revenue generation (Panel c). Processing firms demonstrate similarly robust positive effects across all outcomes. Downstream categories including final Products and Textiles also benefit significantly, showing positive and statistically significant coefficients for revenue, productivity, and profitability (Panels b-d). In contrast, Natural Resource firms display negligible and statistically insignificant responses across all measures.

The differential responses across supply chain positions reflect distinct economic channels through which political frictions affect firm performance. Political barriers that impede inter-jurisdictional coordination disrupt sequential production processes through both cost-side and demand-side mechanisms. Downstream firms face cost-side exposure as final goods producers depend critically on timely access to diverse intermediate inputs sourced from multiple jurisdictions. Political frictions increase delivered input costs, elevate supply volatility, and reduce operational efficiency. When competition intensity weakens and such frictions diminish, downstream firms expand input procurement (Panel a) and scale production, generating higher revenue, labor productivity, and profits.

Upstream firms experience demand-side exposure as intermediate goods suppliers depend on purchasing capacity from downstream customers operating across administrative boundaries. Political barriers that constrain downstream activity translate into reduced orders and lower capacity utilization for upstream producers. Relaxing these constraints expands market access and increases sales volumes, consistent with the substantial revenue gains observed for Manufacturing and Processing sectors.

The results in Figure [H.17](#) demonstrate that sectors embedded in multi-jurisdictional value chains capture the largest benefits when political competition weakens and supply chain frictions diminish. Resource extraction activities, characterized by greater location specificity and lower input-output intensity, remain relatively insulated from these political dynamics and consequently exhibit muted responses. These differential impacts across supply chain positions provide compelling evidence that value chain integration constitutes a primary channel through which local political incentives shape firm performance and economic outcomes.

G.4 Heterogeneity by Motorway Administrative Class: Evidence from the National Expressway Network

This section decomposes the border-county result in Table 6 by motorway class: national expressways (G-prefix), provincial expressways (S-prefix), and urban motorways. National expressways stand out on the two dimensions that matter for the incumbent's calculus. They are the largest projects a locality hosts, absorbing the most land, financing, and administrative effort per segment, and they form the trunk network carrying long-distance inter-city trade, so their benefits leak most heavily to rival jurisdictions. Provincial segments serve within-region connectivity, and urban motorways yield almost entirely local returns. The spillover mechanism therefore predicts underinvestment should concentrate in the national network.

Although national expressways are centrally planned, route designation and implementation are distinct stages. The center fixes the corridor map, but local governments conduct land expropriation, arrange matching funds, issue permits, and control construction pacing; the larger the segment, the more such margins for delay. Central designation thus determines whether a link exists, while local implementation determines when it is completed. This division also aids identification: corridor placement is plausibly exogenous to local conditions, so any effect of competition on G-class capacity reflects protectionist foot-dragging on centrally designated projects.

Table I.14 reports the border-county interaction by different class. The effect is concentrated in national motorways (column 2), with no significant response for provincial or urban motorways (columns 1 and 3). Underinvestment thus concentrates in the largest, most spillover-intensive projects, operating through the local implementation of central plans.

G.5 Instrumental Variable Estimates

To address potential concerns about endogeneity and reverse causality between politician age differentials and regional trade outcomes, I employ instrumental variable estimation using the one-year lagged value of Politician Age Differential. While the variation in Politician Age Differential is plausibly exogenous to unobservable characteristics affecting regional trade volume, the IV approach provides additional robustness by exploiting predetermined variation in political competition.

I instrument Politician Age Differential using its one-year lagged value. This approach exploits the predetermined nature of lagged political competition while maintaining relevance to current competitive dynamics. The lagged instrument is valid under the assumption that past realizations of politician age differentials affect current economic

outcomes only through their influence on current political competition, rather than through direct channels.

The relevance condition is satisfied because political competition exhibits persistence over time—age differentials that existed in the previous period continue to influence current competitive dynamics through ongoing career trajectories, established networks, and accumulated political capital. However, the one-year lag ensures that the instrument is predetermined with respect to current-period shocks that might simultaneously affect both political competition and economic outcomes.

The exclusion restriction requires that lagged politician age differentials do not directly affect current trade outcomes except through their impact on current political competition:

$$E[\epsilon_{i,c,t} | Z_{i,p,t-1}, X_{i,c,t}, \alpha_c, \gamma_i, \delta_t] = 0 \quad (16)$$

where $Z_{i,p,t-1}$ represents the one-year lagged Politician Age Differential, $\epsilon_{i,c,t}$ is the error term in the structural equation, $X_{i,c,t}$ denotes control variables, and α_c , γ_i , and δ_t represent city, politician, and year fixed effects, respectively.

Table I.8 presents the second-stage results from the instrumental variable estimation, which confirm the baseline findings of a positive relationship between Politician Age Differential and internal trade volumes. The IV estimates are positive and statistically significant for total trade volume (0.021, significant at the 5 percent level), motorway trade volume (0.022, significant at the 5 percent level), and waterway trade volume (0.040, significant at the 10 percent level). In contrast, the coefficient for export value is small and statistically indistinguishable from zero, indicating that political competition affects trade flows within domestic borders rather than external trade.

The first-stage F-statistics range from 277 to 363 across specifications, far above conventional thresholds for weak instrument concerns, indicating strong relevance of the lagged instrument. Figure H.9 illustrates the first-stage relationship, showing a clear positive correlation between the one-year lagged Politician Age Differential and its current value.

These IV results provide strong evidence that the positive relationship between politician age differentials and internal trade performance reflects a causal effect of political competition rather than spurious correlation. The contrast between the significant effects on internal trade volumes and the null effect on exports is consistent with the interpretation that political competition operates through domestic trade barriers, reinforcing confidence in the robustness of the main findings.

G.6 Heterogeneous Effects by Initial Economic Conditions

A key question in political economy is how a region's initial economic conditions shape the impact of distortions caused by competition. This study's framework suggests that politicians' incentives to manipulate trade under yardstick competition may vary with local market integration and industrial development. To test this implication, the baseline model is extended by interacting the Politician Age Differential with six measures of city characteristics: an indicator for the eastern region (more developed region), the initial trade share, firm tax revenue, the manufacturing value-added share, industrial employment, and the number of industrial firms. Table I.31 reports the results. The estimated coefficient on the Politician Age Differential remains positive in every specification and is statistically significant at conventional levels, ranging from 0.016 to 0.056. Thus, the baseline relationship between a larger age differential (weaker political competition) and trade is robust across these alternative dimensions of initial economic conditions.

The interaction coefficients are negative in all six specifications, indicating that the positive association between the Politician Age Differential and trade weakens in more developed or economically integrated cities. The interaction with the eastern-region indicator is -0.012 (Column (1)). Given the main coefficient of 0.016, the implied effect for eastern cities is approximately 0.004, roughly a quarter of the effect for inland cities, so the distortion associated with political competition is largely, though not entirely, eliminated in the developed coastal region. The interaction with the initial trade share is -0.048 , against a main coefficient of 0.033 (Column (2)). Taking the point estimates at face value, the effect declines with initial trade exposure and reaches zero at an initial trade share of approximately 0.69 ($0.033/0.048$); in cities above this threshold, political competition no longer distorts trade. The interaction with firm tax revenue is negative but small and imprecisely estimated (-0.003 , Column (3)), offering little indication of heterogeneity along this dimension. The interactions with the manufacturing value-added share (-0.001), industrial employment (-0.007), and the number of industrial firms (-0.006) are likewise negative (Columns (4)–(6)). The corresponding turning points are a manufacturing value-added share of approximately 43 percent ($0.043/0.001$), manufacturing employment of approximately 4.9 ($0.034/0.007$), and approximately 9.3 for the number of industrial firms ($0.056/0.006$), in the units reported in Table I.31. Beyond these levels of market integration and industrial development, the estimates imply that the politically induced trade distortion disappears entirely. Taken together, these patterns indicate that political competition suppresses trade primarily in less developed and less industrialized cities, and that this distortion attenuates, and ultimately vanishes, once market integration and industrial depth are sufficiently advanced.

From an institutional standpoint, these findings suggest that politicians' strategic re-

sponses are conditioned by their local economic environment. In eastern, trade-oriented, and industrially developed cities, firms are more deeply embedded in interregional markets and production networks. Policies that disrupt trade can therefore impose larger costs by damaging supply chains, reducing firm performance, and weakening the absolute economic outcomes used to evaluate local officials. These economic constraints counteract the incentives generated by relative political evaluation. Consequently, although weaker political competition remains positively associated with trade across the specifications, the relationship is substantially attenuated in cities with stronger initial trade linkages and deeper industrial structures.

G.7 Politician Age Differential and Tenure Years

One alternative explanation for the effect of the Politician Age Differential on trade volumes is that it merely captures a politician's time in office or accumulated experience rather than the intensity of yardstick competition. Politicians with longer tenures might mechanically exhibit particular age differentials while independently influencing trade. Two pieces of descriptive evidence and one formal test speak against this interpretation.

Figure H.18 documents how the differential behaves over a secretary's term. The measure exhibits no mechanical relationship with tenure: its average level remains flat at roughly three years of age distance from provincial peers at every tenure year, and the average *signed* year-over-year change, reported below the line, never exceeds two percent of the level. At the same time, the measure is far from static. The average *absolute* year-over-year change is 0.53 to 0.58 years, roughly one-fifth of the level, in every year of a term. Because a secretary's own start-of-term age is fixed within a term, all of this movement originates in the turnover of peer secretaries in the same province, which is precisely the variation that the city-politician fixed-effects design exploits. The measure thus generates ample identifying variation within terms, none of which tracks the focal politician's time in office.

Figure H.19 plots the same two objects by calendar year and delivers two further messages. First, the identifying variation is present throughout the sample period: the average absolute within-term change is sizable in every year from 1996 to 2018, and its peaks follow the political appointment calendar. The reshuffle waves after the Party Congresses (most visibly 1998, 2007, and 2012 to 2013) rearrange provincial age structures and generate bursts of variation in the measure. The estimates therefore do not rely on any particular episode or era. Second, the average level of the differential declines steadily, from about 3.8 years in the late 1990s to about 2.1 years by 2018: city secretaries have become increasingly similar in age to their provincial peers, consistent with the progressive

standardization of age-based appointment norms documented in Section 2. Through the lens of the tournament framework, this age compression implies that the average intensity of yardstick competition has been *rising* over the sample period. Because this aggregate trend is common to all cities, it is absorbed by the year fixed effects and does not contribute to the within-city estimates. It does, however, underscore that the tournament pressure this paper studies is a growing, not a fading, feature of the system.

To test formally whether the estimated effect reflects tenure or experience rather than competition, I decompose the age differential by the *tenure proximity* of provincial peers. For each city-year, peer secretaries are split into three disjoint bins according to the absolute gap between the focal and peer secretary's tenure year: near-tenure peers (a gap of at most two years), mid-tenure peers (a gap of three to five years), and far-tenure peers (a gap of six or more years). The lagged age differential computed within each bin then enters the baseline specification jointly. If the baseline coefficient merely proxied for the focal politician's tenure duration, accumulated experience, or incumbency status, there would be no reason for the effect to depend on *which* peers the differential is measured against. The results are presented in Table [I.32](#).

The effect is concentrated among near-tenure peers: the age differential computed over peers within a two-year tenure gap is positive and significant for total trade (0.008), motorway trade (0.009), and waterway trade (0.030), whereas the differentials computed over mid- and far-tenure peers are small and generally insignificant. This pattern is exactly what the tournament interpretation predicts, since secretaries compete most intensely with peers at a comparable stage of their terms, who will be evaluated by the provincial leadership over the same horizon. It is difficult to reconcile with tenure- or experience-based alternatives, which imply no differential sensitivity to peers' tenure proximity. Together with the descriptive evidence in Figures [H.18](#) and [H.19](#), these results indicate that the yardstick-competition effect is distinct from the influence of tenure duration, accumulated experience, or incumbency status.

G.8 Pre-trend in Trade Volume

To further support the causal interpretation of the relationship between Politician Age Differential and trade volumes, I conduct a placebo test to examine the possibility of pre-existing trends. Specifically, I test whether future levels of political competition (measured by the average age differential in a later period) are correlated with past changes in city trade volumes. If cities that were set to experience lower or higher competition intensity in the future already exhibited different trade growth trajectories before those competition levels were realized, it would cast doubt on the identification strategy, suggesting potential

reverse causality or omitted variables driving both competition and trade patterns.

For this test, I regress the change in city trade volumes (total, motorway, waterway, and export) between 2000 and 2005 on the average Politician Age Differential observed during the subsequent 2006 to 2010 period. The regression includes province fixed effects and baseline city controls from 2000.

The results are presented in Figure [H.20](#). I find no statistically significant relationship between the average Politician Age Differential during 2006-2010 and the change in trade volumes during the earlier 2000-2005 period. The estimated coefficients are close to zero and statistically insignificant across all trade categories examined.

This lack of correlation between future competition intensity and past trade dynamics suggests that the main findings are unlikely to be driven by pre-existing trends. It provides additional evidence supporting the interpretation that changes in yardstick political competition, proxied by the Politician Age Differential, have a causal impact on inter-regional trade volumes, rather than merely reflecting pre-determined city-specific trade trajectories.

G.9 Alternative Measures of Yardstick Competition

To assess the robustness of the main findings, this section re-estimates the baseline specification using alternative measures of yardstick competition. While the primary hypothesis centers on age, competition could plausibly occur along other dimensions such as tenure, education, or experience. The results of these estimations are presented in Table [I.33](#).

First, I examine alternative specifications of the age-based competition measure. Panel B restricts the sample to politicians aged 57 or younger, isolating the group with the strongest promotion incentives. The results are nearly identical to the baseline in Panel A, confirming that the findings are not driven by politicians near retirement who might have different motivations. Panel C replaces the start-of-term age with the politician's current age in constructing the differential. The estimates remain positive and statistically significant for motorway trade, waterway trade, and exports, though the coefficient for total trade volume is smaller and imprecisely estimated, consistent with the start-of-term age being the more relevant benchmark in the promotion tournament.

Next, I test whether competition based on characteristics other than age produces similar effects. Panels D, E, and F replace the age differential with differentials in tenure, education level, and total working years, respectively. Across all trade categories, these alternative measures of competition are consistently statistically insignificant, with the sole exception of a marginally significant coefficient for waterway trade in Panel F. This suggests that the observed effect is specific to the institutional importance of **age** in the promotion tournament, rather than being a general proxy for differences in experience or

human capital.

Finally, Panel G measures competition intensity by counting the number of peers within a close range (± 3 years) of a politician's working experience. A higher count signifies a more crowded competitive field, so the expected sign is negative. Consistent with this, a greater number of similarly experienced peers is associated with significantly lower waterway trade volumes, while the remaining coefficients are small and insignificant. Taken together, these robustness checks reinforce the conclusion that yardstick competition in this context is uniquely and robustly tied to politicians' relative age.

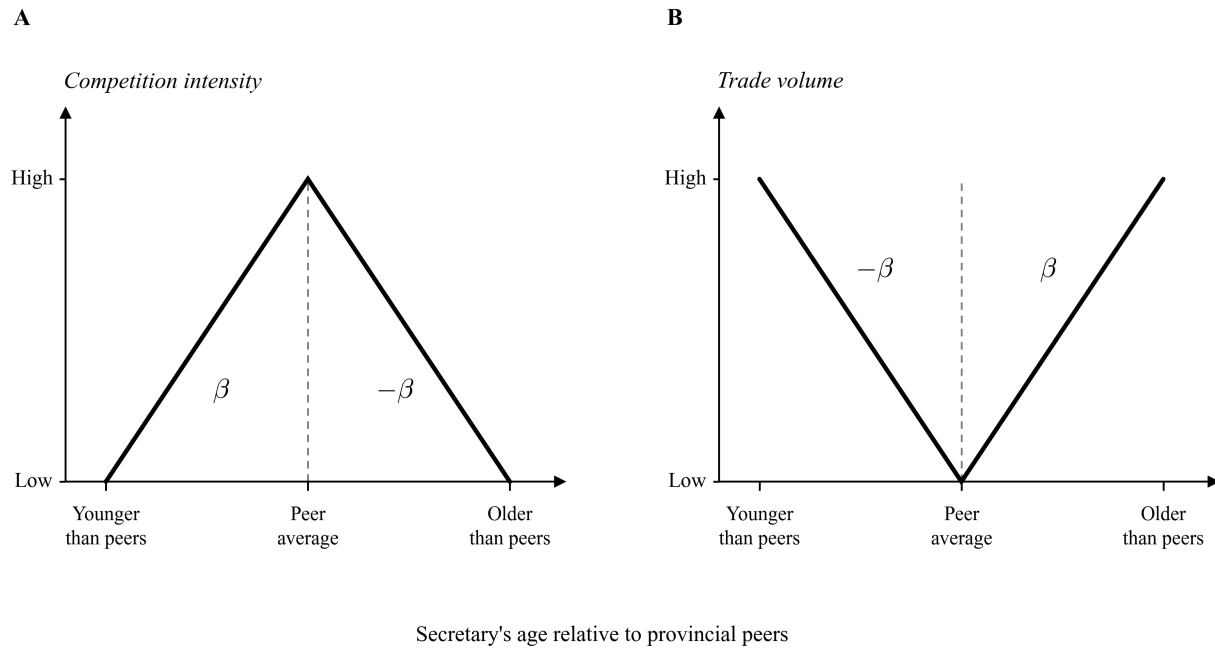
G.10 Controlling for Past Trade Volume and Economic Growth

To ensure the main findings are not driven by pre-existing economic trends that might correlate with both political competition and current trade volumes, I conduct a robustness check by controlling for lagged economic indicators. The main specification is re-estimated, including lagged values of nighttime lighting (as a proxy for historical economic conditions) and the specific trade volume being analyzed. The specifications in Table [I.13](#) use both a one-year lag and a three-year average for these controls, while maintaining the full set of fixed effects.

The results confirm the robustness of the primary finding. The coefficient for the Politician Age Differential remains positive and statistically significant across all three freight trade modes in both the lagged and three-year average specifications, ranging from 0.008 for total trade with lagged controls to 0.024 for waterway trade with three-year average controls. For exports, the coefficient is positive and significant with lagged controls (Column 7) and remains positive, though imprecisely estimated, with the three-year average controls in the smaller sample of Column (8). As expected, the lagged trade volumes themselves are strong predictors of current trade, indicating a high degree of persistence. The stability of the main coefficient after including these dynamic controls provides further confidence that the estimates are not biased by unobserved, slowly evolving city-specific trends.

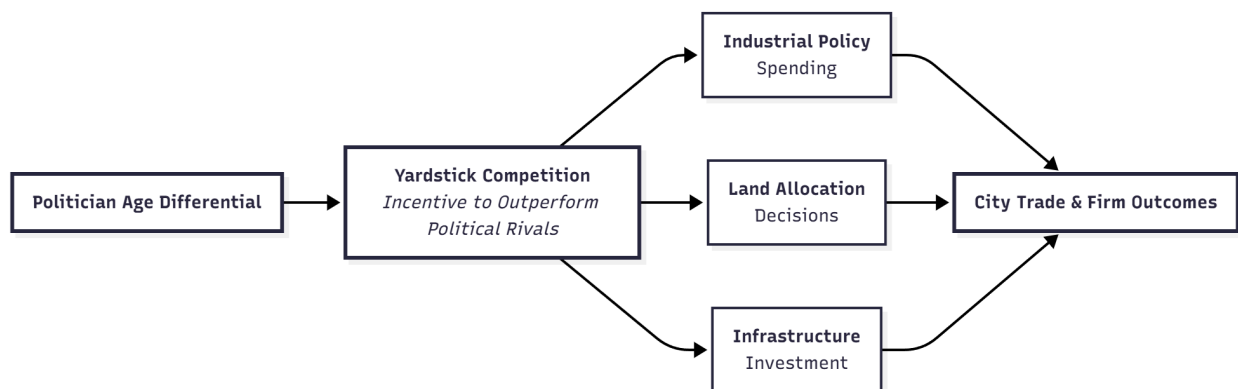
H Figures

Figure H.1: Hypothesized Relationship between Politician Age Differential and Trade Volumes



Notes: This figure illustrates the hypothesized relationship between politicians' age differential, competition intensity, and city trade volumes.

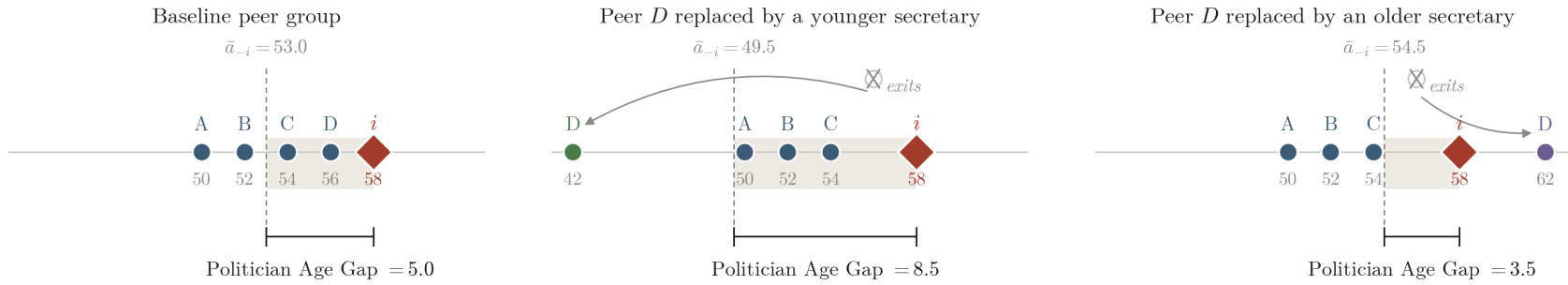
Figure H.2: Causal Chains from Politician Age Differential to Trade Outcomes



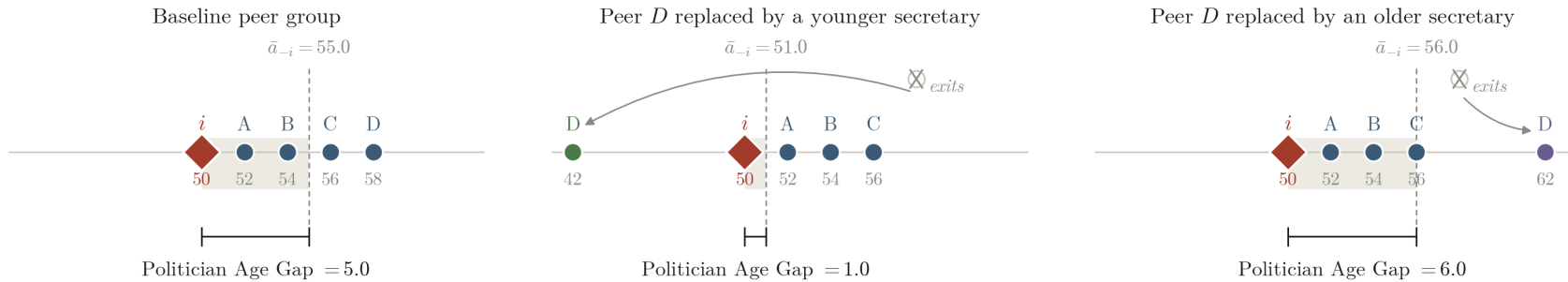
Notes: This figure illustrates the flow chart of causal chains from politician age differential to city and firm trade outcomes.

Figure H.3: Construction of the Politician Age Differential Under Peer Turnover

Panel A. Target is older than the peer norm

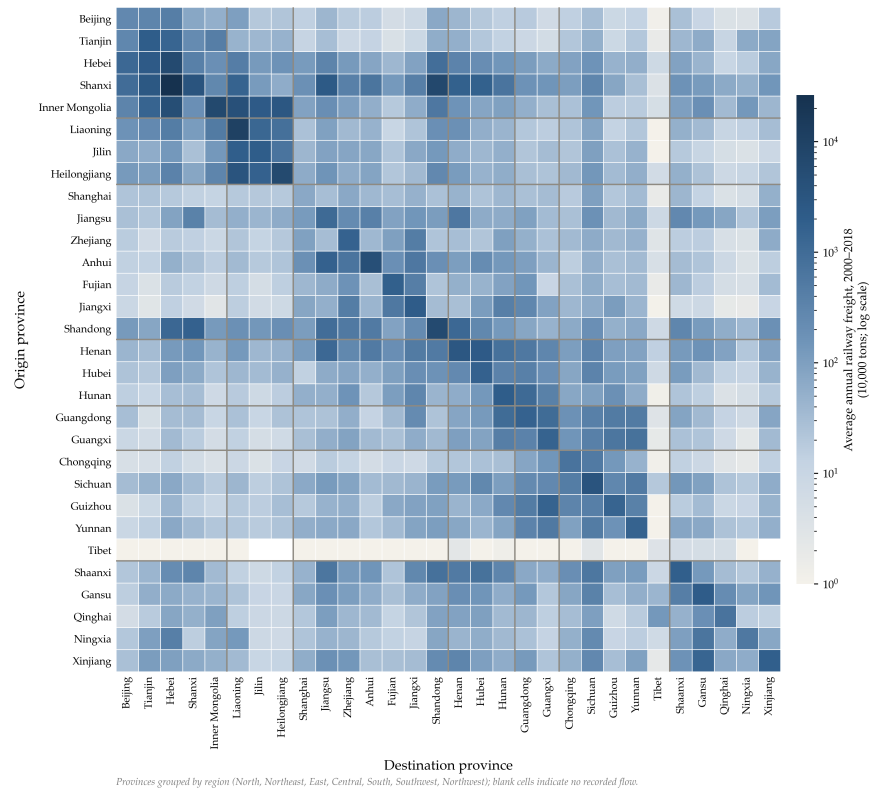


Panel B. Target is younger than the peer norm



Notes: Politicians A – D denote the peer group of secretary i within the same tenure cohort. The Politician Age Gap is defined as $|a_i - \bar{a}_{-i}|$, the absolute value of the difference between i 's own age and the mean age of i 's peers. Panel A considers a target politician who is older than the peer norm at baseline; Panel B considers a target politician who is younger than the peer norm at baseline. Within each panel, the first column shows the baseline peer group, the second column shows the effect of replacing peer D with a younger secretary, and the third column shows the effect of replacing peer D with an older secretary. Reading down each column, the same replacement event moves the Politician Age Gap in opposite directions for the two targets, since each begins on a different side of the peer norm; this illustrates that the measure is symmetric in distance from the peer norm, not in the direction of peer turnover.

Figure H.4: Inter-province Trade Flows



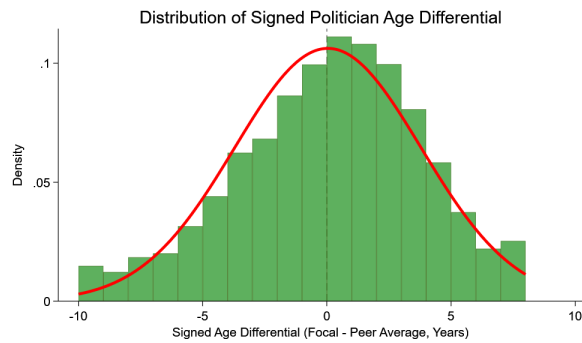
Notes: This figure illustrate the average inter-province trade flows from 2000 to 2018.

Figure H.5: Validation of the Freight-Based Trade Measure against Input-Output Inter-provincial Trade

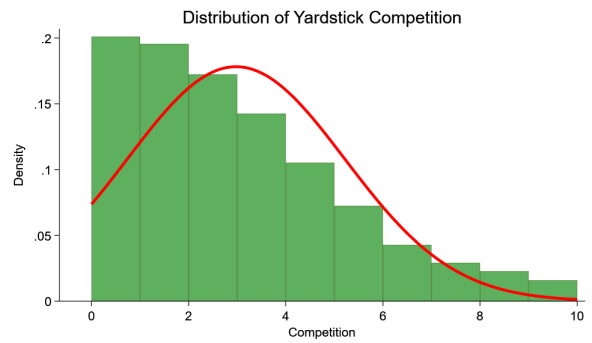


Notes: This figure validates the freight-based trade measures against an independent, accounts-based trade proxy constructed from China's 30-province multi-regional input-output (MRIO) tables. For each benchmark year (2002, 2007, 2012, and 2017), inter-provincial trade is computed by summing each province's tradable-sector shipments to and from all other provinces. City-level freight volumes from the City Statistical Yearbook are aggregated to the province level. Each panel plots the log freight measure against the log IO inter-provincial trade value, pooling the four benchmark years; the dashed line is the fitted linear relationship. The pooled log-log correlations are 0.67 for total trade, 0.65 for motorway freight, and 0.48 for waterway freight. Because the IO flows are denominated in value while freight is denominated in tonnage, the two measures are not expected to coincide in levels; the tight co-movement indicates that the freight measure is a valid volume proxy for inter-provincial trade.

Figure H.6: Distribution of Politician Age Differential



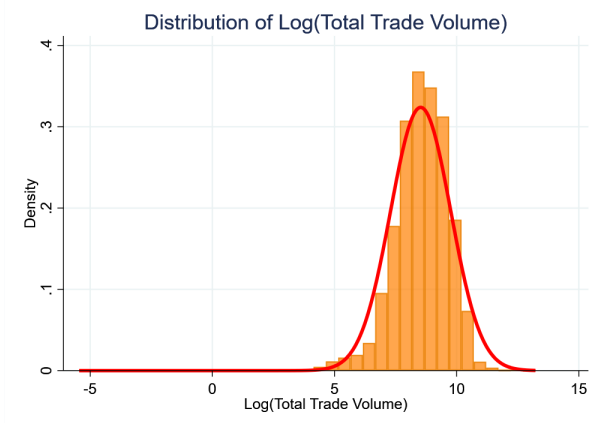
(a) Signed Politician Age Differential



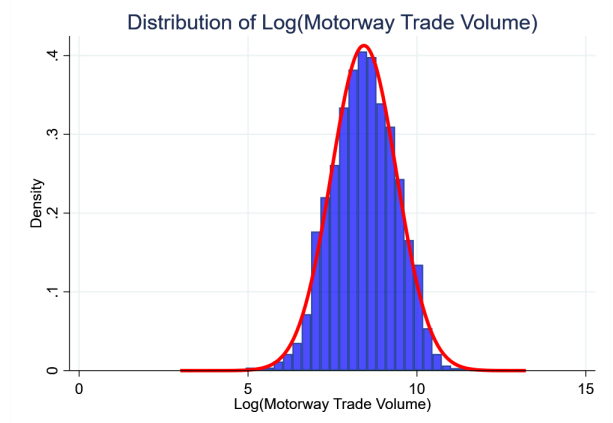
(b) Politician Age Differential

Notes: This figure presents frequency distributions of key variables in the analysis. Panel (a) displays the signed Politician Age Differential. Panel (b) illustrates Politician Age Differential. All distributions include normal curve overlays in red for reference.

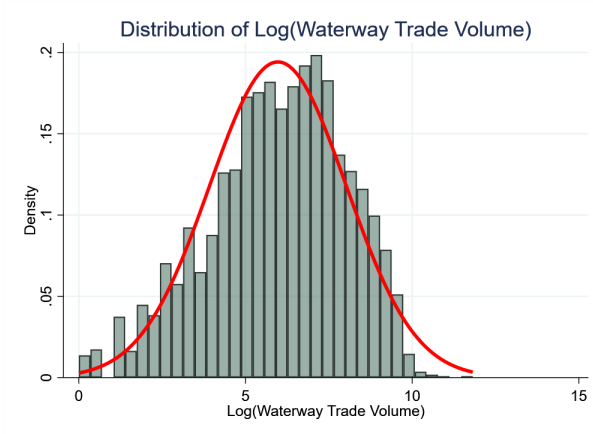
Figure H.7: Distribution of Trade Indicators



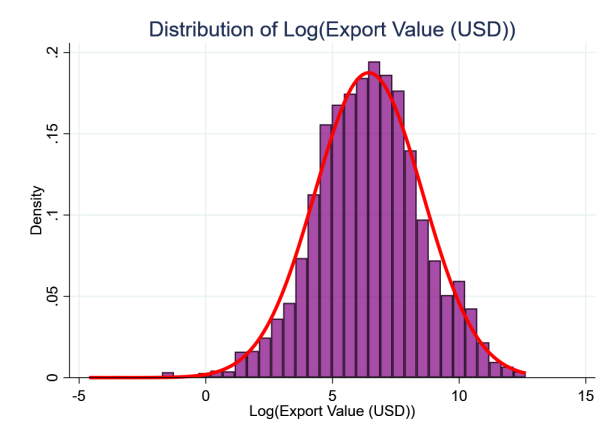
(a) Log (Total Trade Volume)



(b) Log (Motorway Trade Volume)



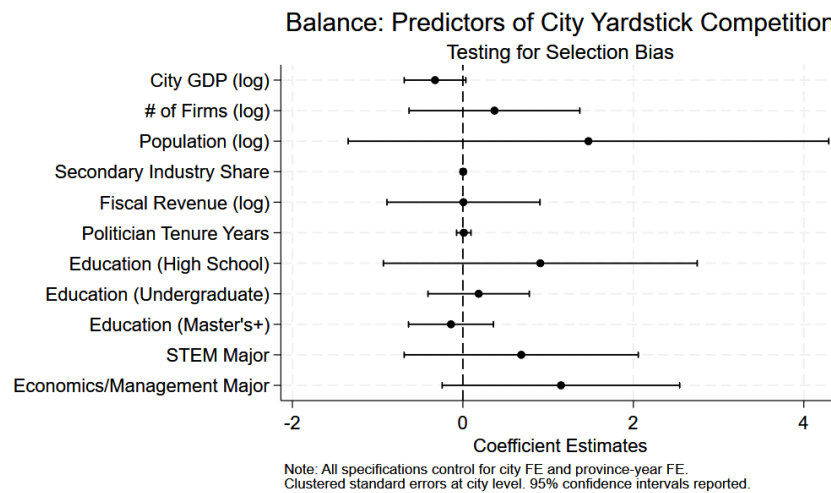
(c) Log (Waterway Trade Volume)



(d) Log (Export Value USD)

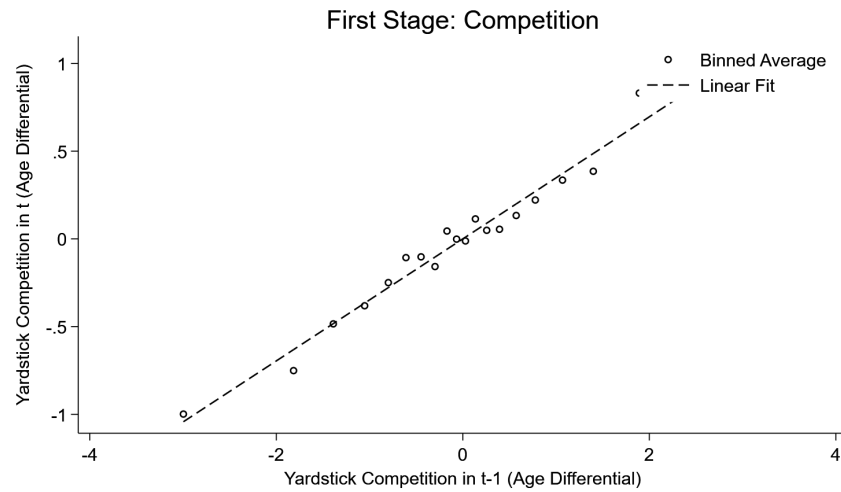
Notes: This figure presents frequency distributions of key variables in the analysis. Panel (a) displays the log-transformed total trade volume. Panels (b), (c), and (d) present infrastructure-related trade measures: log-transformed motorway trade volume, waterway trade volume, and export value in USD, respectively. All distributions include normal curve overlays in red for reference.

Figure H.8: Balance Test: Relationship between Politician Age Differential and Politician and City Characteristics



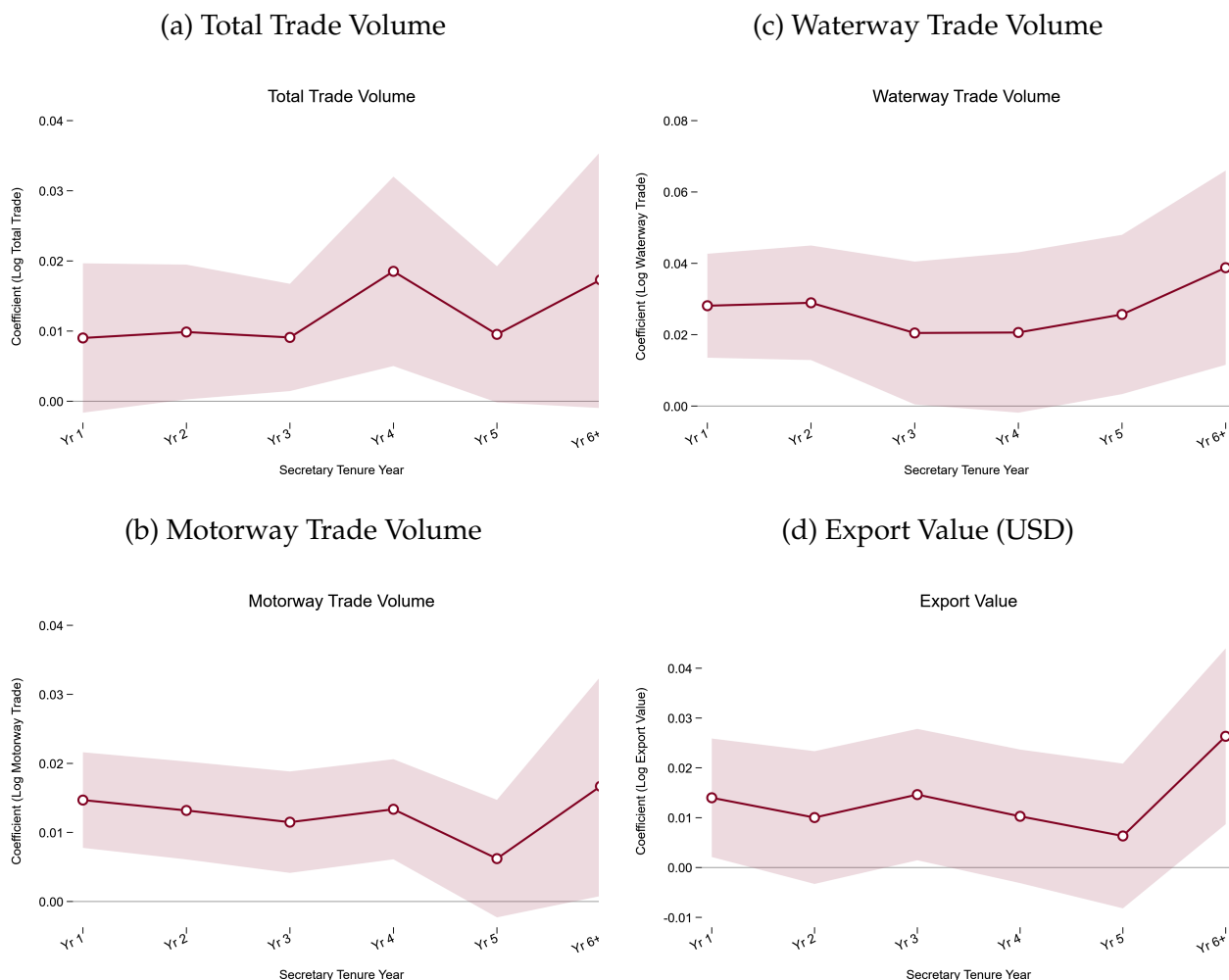
Notes: This figure presents a balance test examining the relationship between Politician Age Differential and various politician and city characteristics. The lack of statistically significant relationships suggests that age differential is plausibly exogenous to these characteristics.

Figure H.9: First Stage Relationship: One-Year Lagged Politician Age Differential



Notes: This figure displays the first-stage relationship underlying the instrumental variable estimates in Table I.8, which instrument the Politician Age Differential with its one-year lag. Both the endogenous variable and the instrument are residualized on politician characteristics and city, politician, city-politician, and year fixed effects (Frisch-Waugh-Lovell), using the first-stage estimation sample. Observations are grouped into twenty equal-sized bins of the residualized instrument; each circle plots the within-bin mean of the residualized endogenous variable against the within-bin mean of the residualized instrument, and the dashed line is a linear fit through the bin means weighted by bin size. The strong positive relationship corresponds to first-stage F-statistics between 277 and 363 across specifications, well above conventional weak-instrument thresholds.

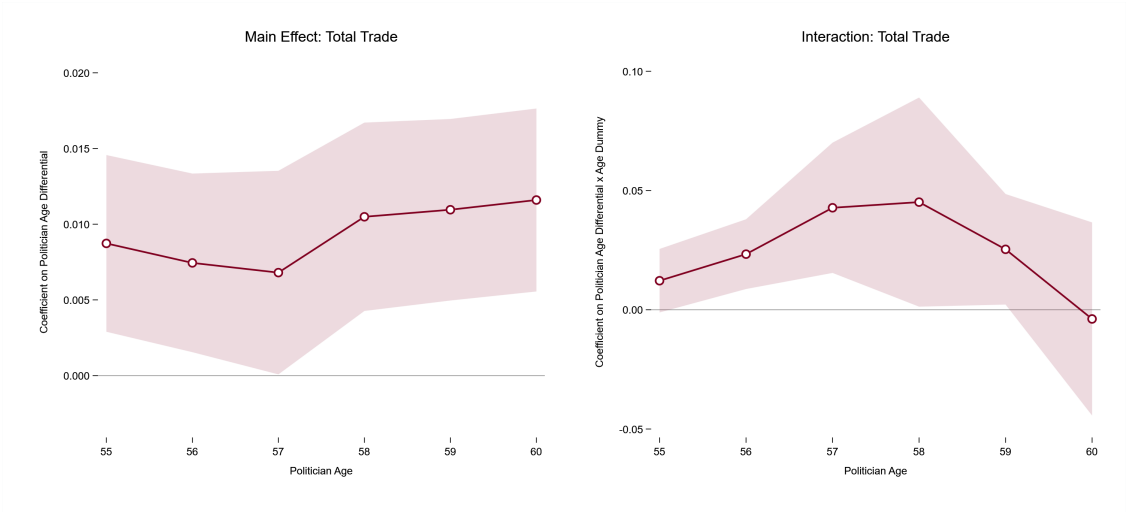
Figure H.10: Evolving Effects of Politician Age Differential on Trade Volumes by Secretary's Own Tenure Year



Notes: These figures plot the estimated effect of Politician Age Differential on different trade outcomes by the city party secretary's tenure year. The analysis uses city-level panel data from 1996 to 2018. The x-axis shows the secretary's tenure year, from Year 1 through Year 6 and above. The y-axis shows the estimated coefficients from a regression of the log trade outcome on Politician Age Differential interacted with tenure-year indicators. I control for city and politician characteristics, city fixed effects, politician fixed effects, year fixed effects, and city-politician fixed effects. Standard errors are clustered at the city level.

Figure H.11: Effect of Politician Age Differential on Trade Outcomes Closing to Retirement

(a) Total Trade Volume



(b) Motorway Trade Volume

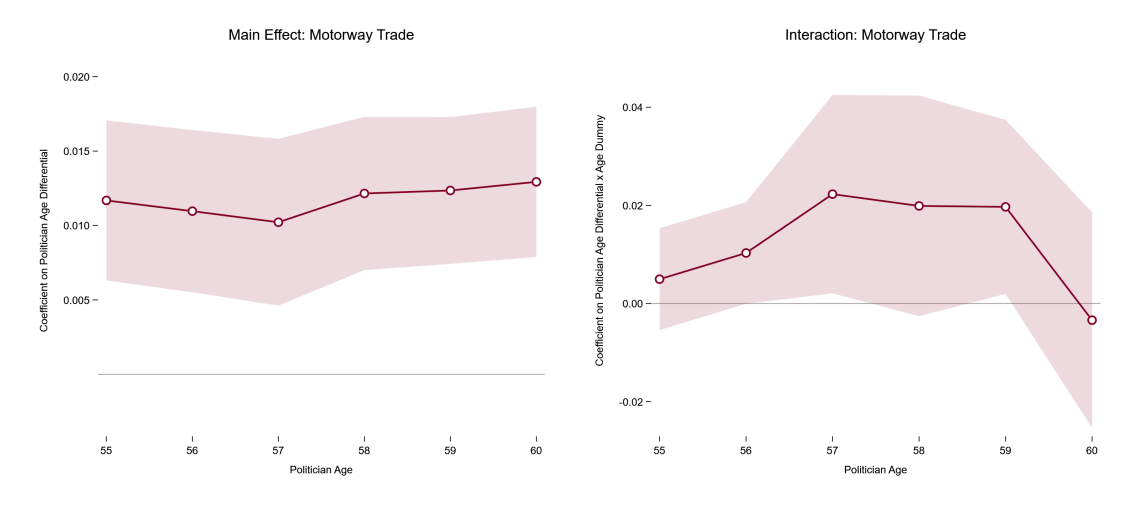
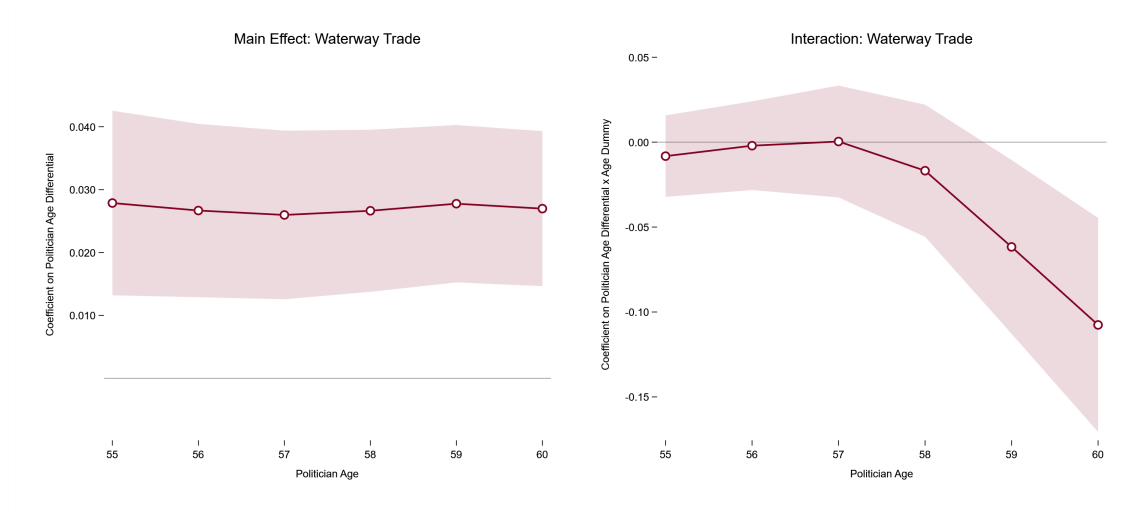
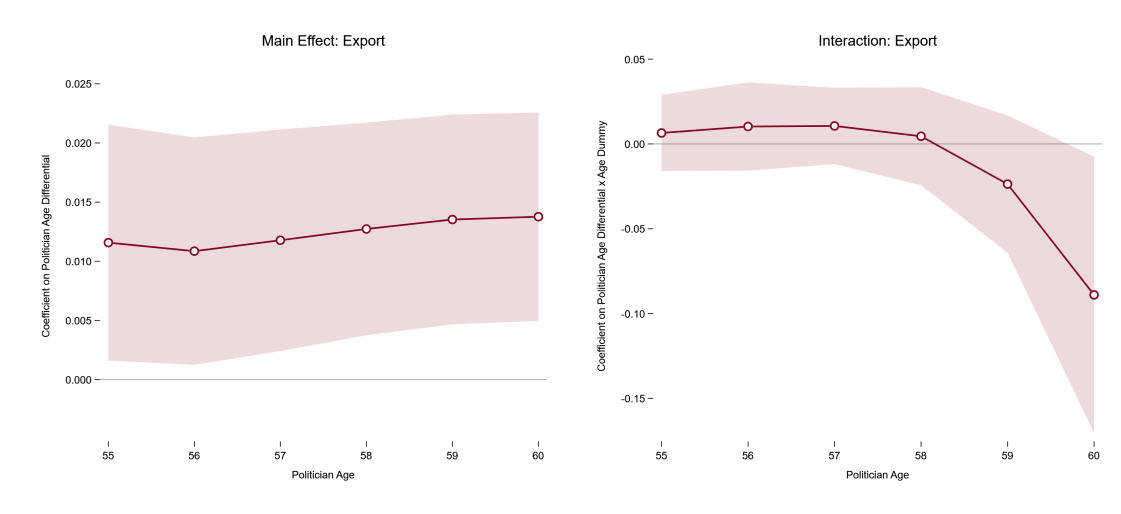


Figure H.11: Effect of Competition on Trade Outcomes Closing to Retirement (continued)

(c) Waterway Trade Volume

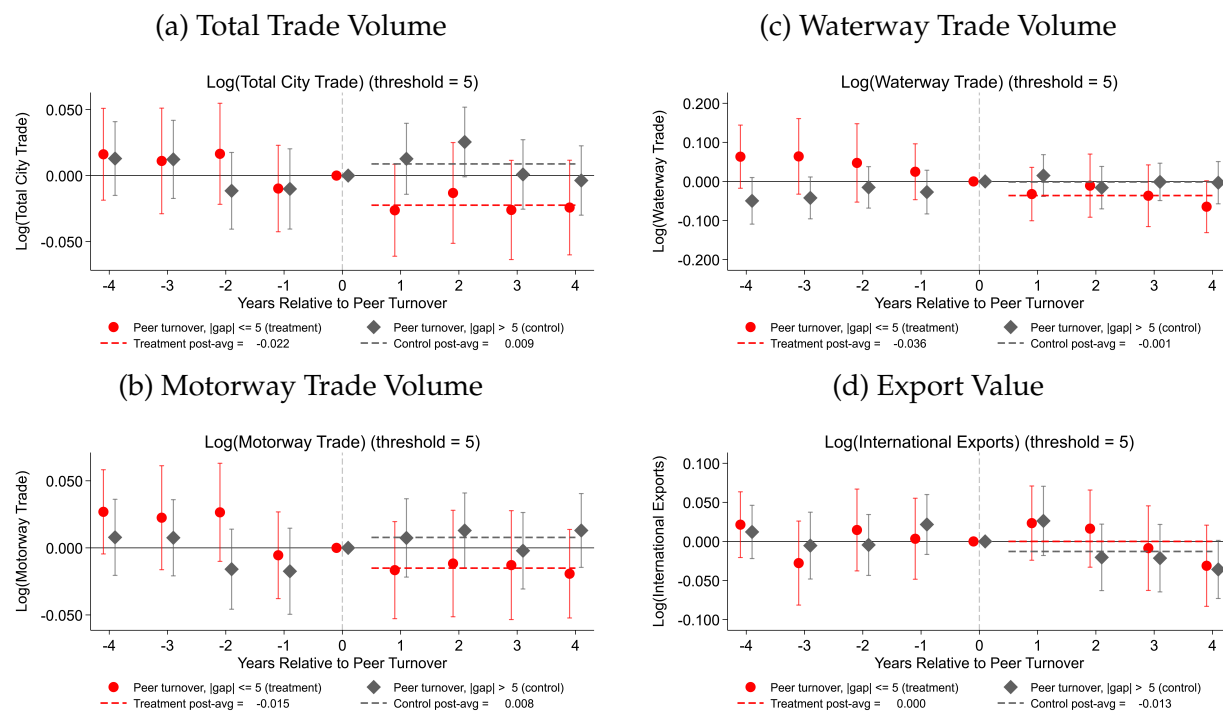


(d) Export Value (USD)



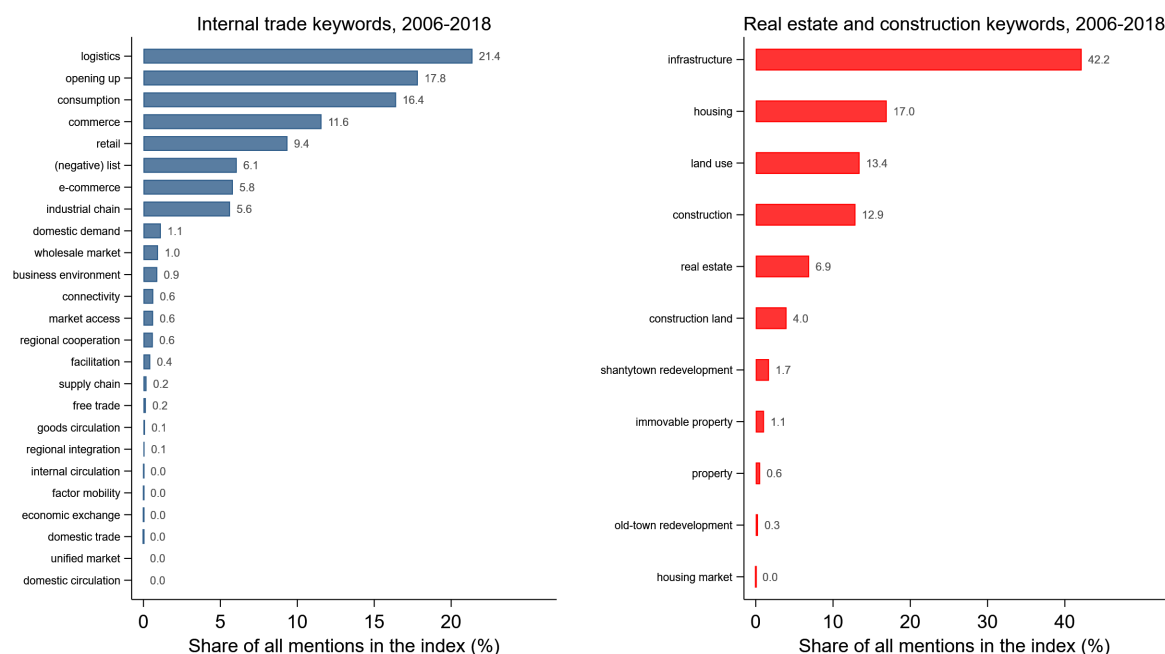
Notes: These figures plot how the coefficient on Competition varies with the city party secretary's age. For each trade outcome, the left subfigure reports the main coefficient on Politician Age Differential estimated at each politician age, and the right subfigure reports the interaction coefficient (Politician Age Differential $\times$ Age Dummy) relative to the omitted reference category. The analysis uses city-level panel data from 1996 to 2018. All trade outcomes are logged and measured at 10,000 tons. I control for city and politician characteristics, city fixed effects, politician fixed effects, year fixed effects, and city-politician fixed effects. Standard errors are clustered at the city level.

Figure H.12: Robustness: Similar- versus Dissimilar-Age Peer Arrivals (Five-Year Threshold)



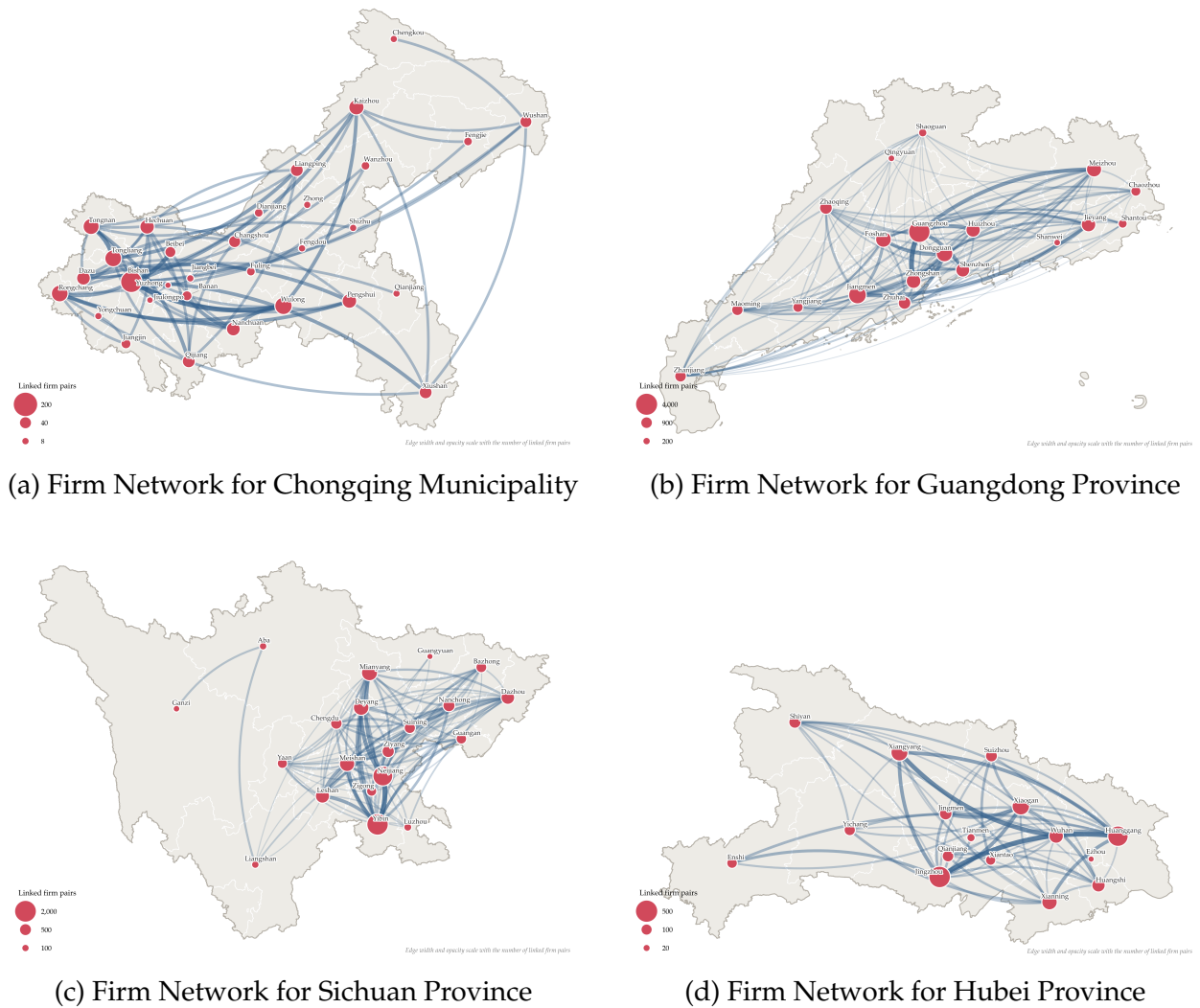
Notes: These figures report an alternative event study using a five-year age-gap threshold. The similar-age series (red circles) consists of arrivals for which the incoming peer secretary's age is within five years of the focal secretary's start age; the dissimilar-age series (gray diamonds) consists of arrivals outside that band. The x-axis shows years relative to the peer arrival, and the y-axis shows the estimated coefficients. The horizontal dashed lines report the average post-arrival coefficient for each group. All specifications include city, politician, city-politician, and year fixed effects and the baseline controls. Standard errors are clustered at the city level. Vertical bars show 90% confidence intervals.

Figure H.13: Keyword Composition of the Internal Trade and Real Estate Policy Indices



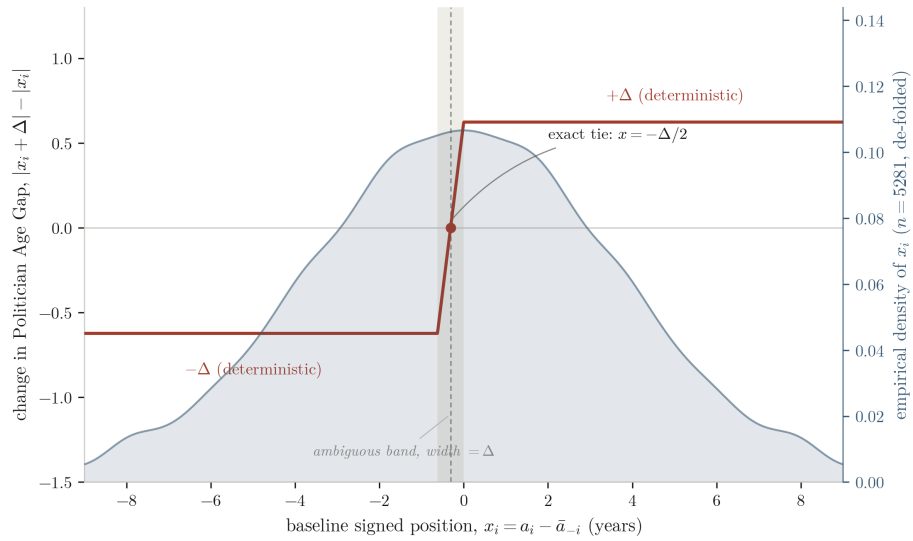
Notes: This figure reports the composition of keywords underlying the internal trade (panel a) and real estate/construction (panel b) policy attention indices. Keywords are extracted from the text of annual government work reports issued by regional governments in China, with mentions counted across the full text of each report. Each bar reports a keyword's share of total mentions within its category, so that shares sum to 100% within each panel. Keyword-to-category assignment is mutually exclusive. English labels are translations of the original Chinese keywords.

Figure H.14: Firm Networks in Four Selected Provinces and Municipality



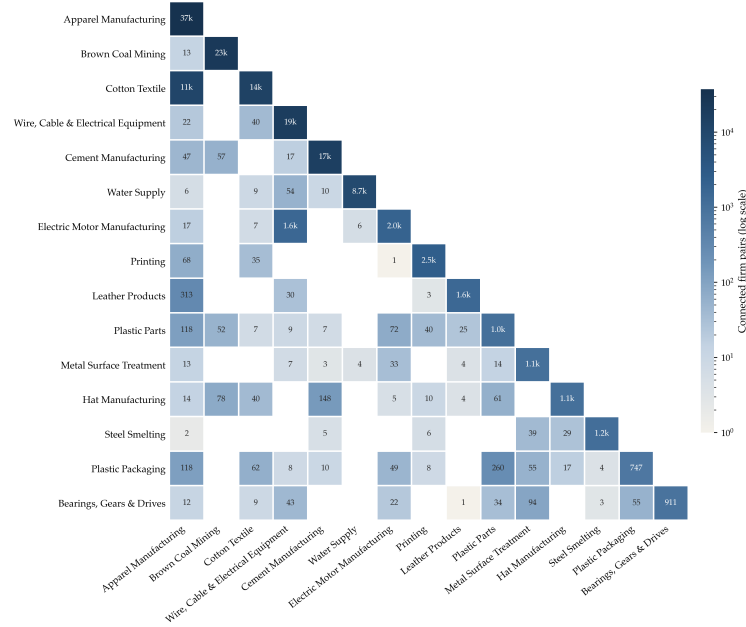
Notes: This figure illustrates business network structures in four selected Chinese provinces/municipalities. Each node represents a different industry, with colors indicating industry categories according to the legend. The connecting lines (edges) represent business relationships between industries, with thicker lines indicating stronger connections.

Figure H.15: Change in the Politician Age Differential by peer-norm distance



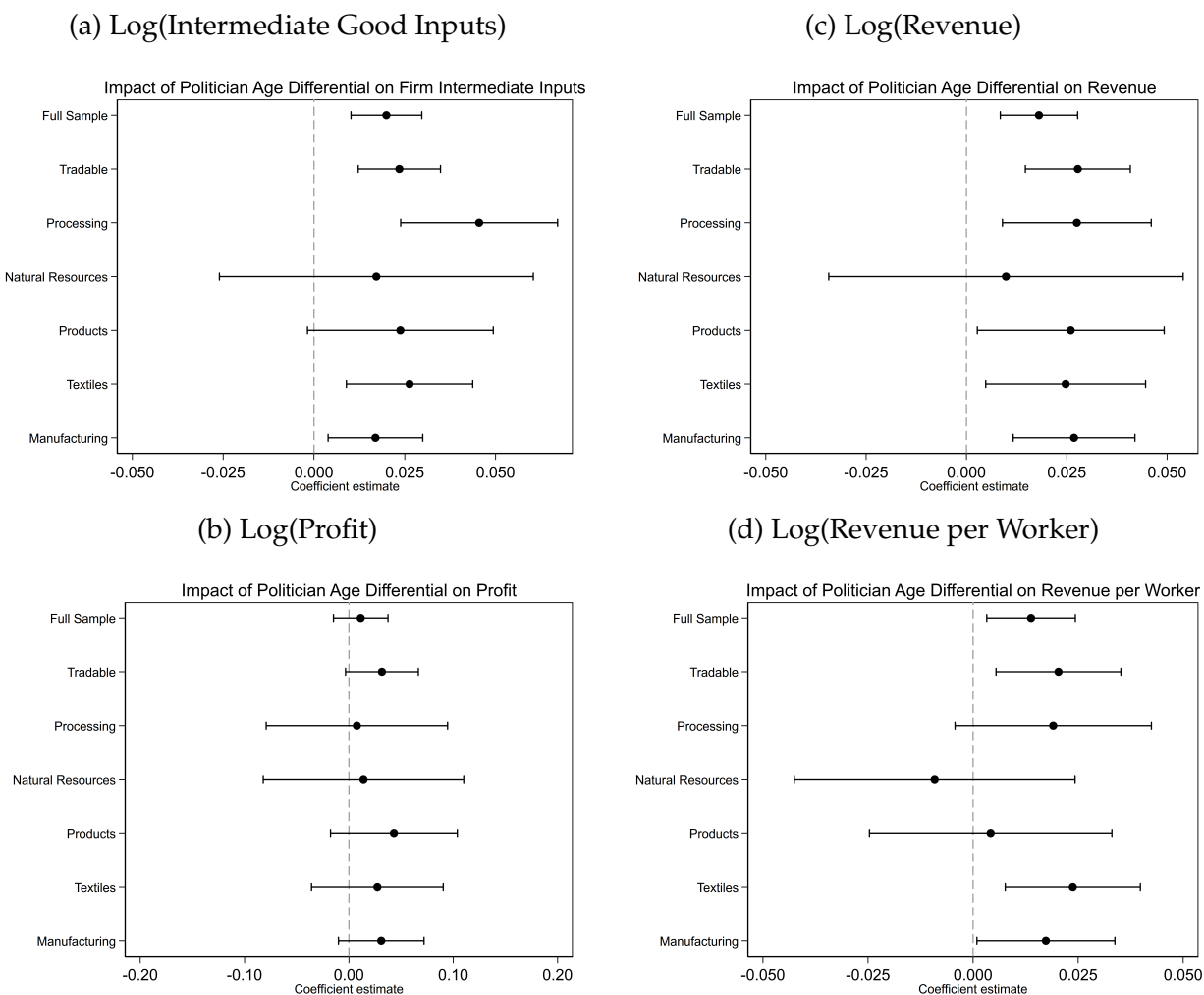
Notes: The solid line plots $|x_i + \Delta| - |x_i|$, the change in the Politician Age Differential from replacing one peer, against x_i , the target's signed distance from the peer norm before replacement, evaluated at the most extreme peer replacement mathematically possible in the data: the sample's single youngest official (age 36) replacing its single oldest (age 60), $\Delta = 2.0$ years, at the average peer group size $n = 12$. This pairing is a theoretical upper bound rather than a frequently observed event. The shaded curve is the empirical density of x_i in the estimation sample, recovered from the observed distribution of the Politician Age Differential under the assumption that x_i is symmetric around zero. The shaded vertical band marks the interval $x_i \in (-\Delta, 0)$ within which the response varies continuously; the dashed line marks $x_i = -\Delta/2$, the unique point at which the measure is unchanged by the replacement.

Figure H.16: Connected Sectors in Business Networks (Heatmap)



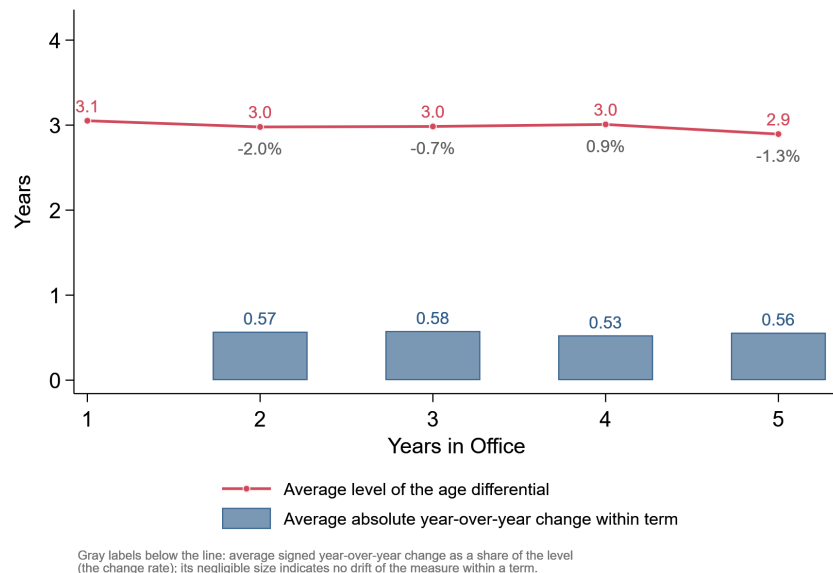
Notes: This heatmap illustrates inter-sectoral firm connections within the business network, showing connection intensity between sectors using 2000-2007 data. Lighter colors indicate higher connection strength between corresponding sectors.

Figure H.17: Effects of Politician Age Differential on Different Firms: Upstream vs. Downstream Sectors



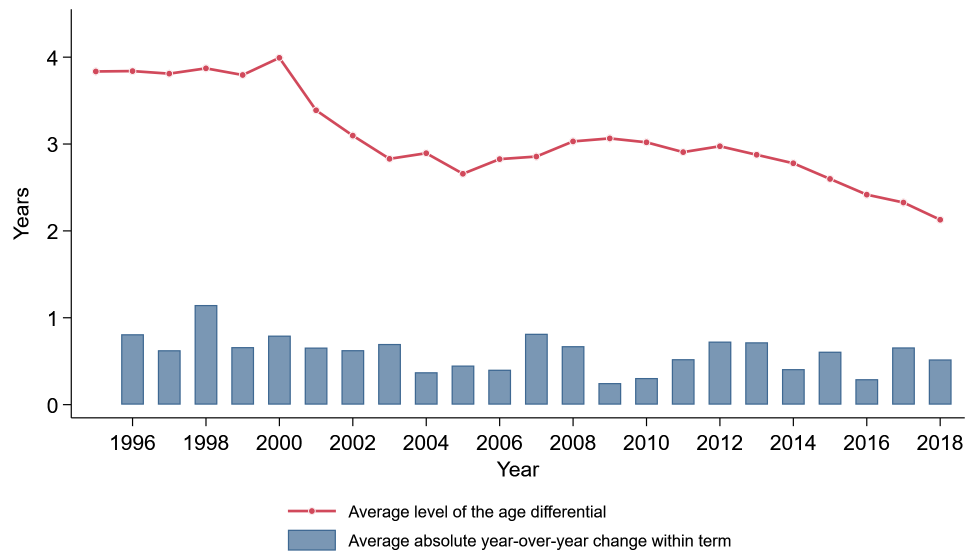
Notes: These figures illustrate the effect of Politician Age Differential on various outcomes of firms operating in different sectors. Panel (a) shows the effect on firm intermediate good inputs; panel (b) shows the effect on firm profit; panel (c) shows the effect on firm revenue; and panel (d) shows the effect on firm revenue per worker. Across all specifications, I control for politician controls, firm controls, firm fixed effect, politician fixed effect, and sector-year fixed effect. Standard errors are clustered at the city level. 95% confidence intervals are shown.

Figure H.18: Changes in Politician Age Differential



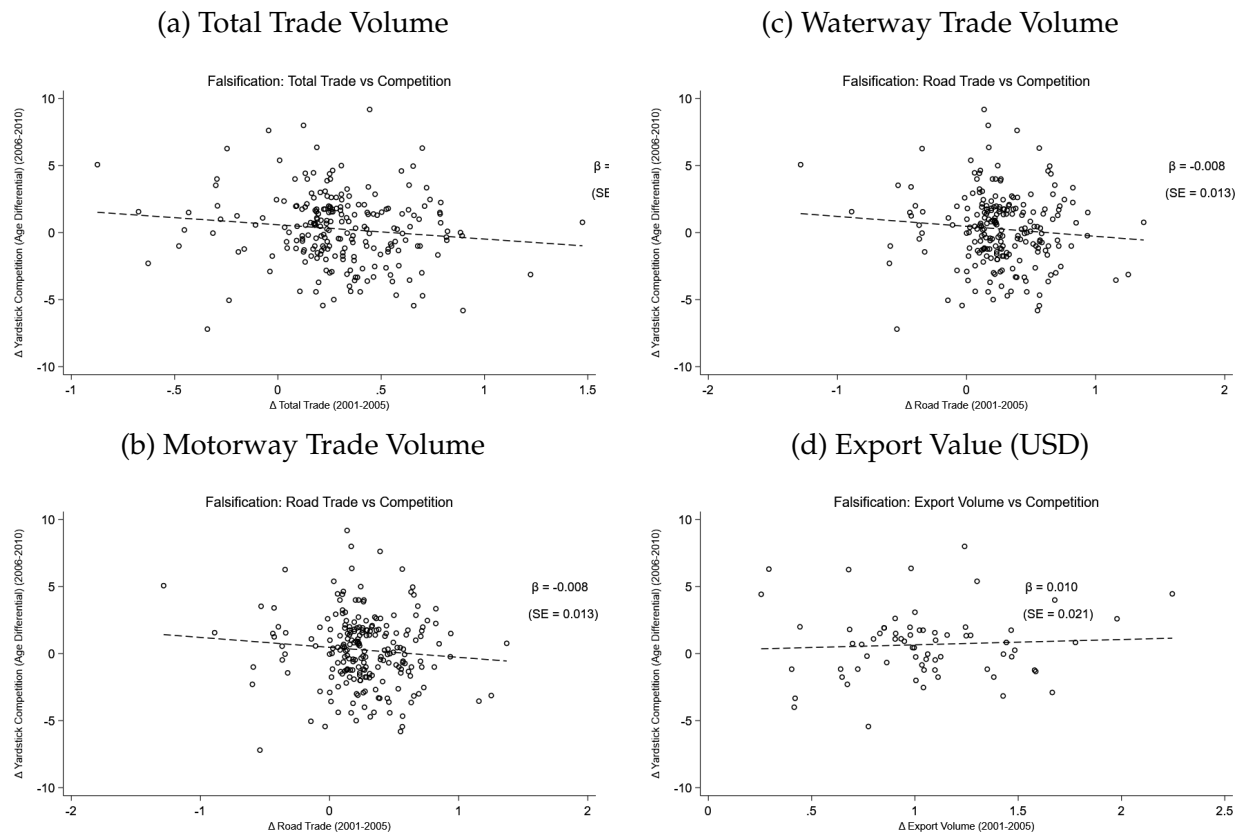
Notes: This figure documents the within-term variation of the Politician Age Differential across tenure years (horizontal axis: years in office, 1–5). The line reports the average *level* of the differential, which is stable at about three years of age distance from provincial peers; the gray labels below the line report the average *signed* year-over-year change as a share of the level (the change rate), which never exceeds two percent in absolute value, indicating no mechanical drift of the measure over a term. The bars report the average *absolute* year-over-year change within a city-secretary term, defined from the second year in office onward. Because a secretary’s own start-of-term age is fixed within a term, all of this movement – 0.53 to 0.58 years annually, about one-fifth of the level – comes from the turnover of peer secretaries in the same province, the variation that identifies the fixed-effects estimates.

Figure H.19: Changes in Politician Age Differential over Time



Notes: This figure plots the Politician Age Differential by calendar year. The line reports the average *level* of the differential across city-years, which declines from about 3.8 years in the late 1990s to about 2.1 years in 2018, reflecting the increasing compression of secretaries' age profiles as age-based appointment norms became standardized. The bars report the average *absolute* year-over-year change within city-secretary terms, which is sizable in every year of the sample – on average about one-fifth of the level annually – and peaks after Party Congress reshuffle waves (1998, 2007, 2012–2013), when the turnover of peer secretaries rearranges provincial age structures. The aggregate decline in the level is common to all cities and absorbed by the year fixed effects; identification relies only on the within-term variation shown by the bars.

Figure H.20: Falsification: Past Trade Volumes and Future Yardstick Competition



Notes: These figures test whether past trade growth predicts future political competition. The dependent variables are log changes in trade volumes during 2000–2005: total trade (panel a), motorway trade (panel b), waterway trade (panel c), and export value (panel d). The independent variable is the average Politician Age Differential during the subsequent 2006–2010 period. A significant coefficient would suggest pre-existing trends, casting doubt on the causal interpretation. The insignificant coefficients indicate no evidence of such pre-trends. All specifications include province fixed effects and baseline city controls from 2000. Standard errors are bootstrapped at the province level by 500 times.

I Tables

Table I.1: Promotion Statistics of City Party Secretaries (2004-2014)

Politician Category	Promotion Metric	Value
Promoted City Party Secretaries	To central gov. or other provinces	2%
	Within-province promotion	98%
	Overall promotion rate	18%
City Party Secretaries' Career Path	Previously served as city mayor	53%
	Of these, mayor post held before secretary post	96%
Promoted City Mayor	To the same city	70%

Notes: This table shows statistics of promoted city party secretaries and mayors from 2004 to 2014. The 98% within-province promotion of party secretary suggests a strong within-province competition. The career-path rows document the mobility between the two positions: a majority of secretaries previously served as city mayor, and among those who held both posts, the mayor position almost always preceded the secretary position. The mayor-to-secretary ladder, most often within the same city, means mayors' promotion prospects do not run through the within-province secretary tournament, consistent with the null results for mayors in the placebo and horse-race specifications.

Table I.2: Correlation between Trade Volumes in City i and Economic Performance Indicators in Other Cities j (Same Province)

VARIABLES	Log(Nighttime Light Intensity in City $_{j,t}$) (1)	Log(Population Size in City $_{j,t}$) (2)	Log(GDP in City $_{j,t}$) (3)
Trade Volume in City $_{i,t-1}$	0.053*** (0.008)	0.015*** (0.005)	0.050** (0.022)
Observations	41,554	41,554	41,554
R-squared	0.980	0.989	0.520
City Pair FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table examines the spillover effects of trade activity in one city on the economic performance of other cities within the same province. The analysis uses city-pair panel data from 1996 to 2018. The key independent variable 'Trade Volume in City $_i$ ' measures the total trade volume in the focal city, while the dependent variables measure different economic outcomes in connected cities (City $_j$ in the same province): nighttime light intensity, population size, and total GDP. All specifications include city-pair fixed effects and year fixed effects. Standard errors (in parentheses) are clustered at the city $_j$ level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.3: Summary Statistics

VARIABLES	N (1)	mean (2)	sd (3)	min (4)	max (5)
Panel A: Trade and Transportation Variables					
Log(Total Trade Volume)	5,281	8.609	1.149	-5.449	13.23
Log(Motorway Trade Volume)	5,157	8.450	0.951	4.078	13.23
Log(Waterway Trade Volume)	2,924	5.974	2.040	0	11.82
Log(Export Value (USD))	3,988	6.444	2.110	-4.605	12.63
Panel B: Politician Variables					
Politician Age Differential	5,281	2.934	2.208	0	10
Starting Age	5,281	50.50	3.741	36	60
Peer Average Age	5,281	50.52	1.771	40	56
Panel C: Firm Characteristics					
Log(Firm I Intermediate Goods)	26,882	9.867	1.156	2.303	16.68
Log(Firm I Revenue)	26,888	10.21	1.109	5.308	17.06
Log(Firm I Revenue per Worker)	26,883	5.012	0.972	0.408	9.249
Log(Firm I Profit)	26,889	6.498	1.913	0	14.83
Log(Firm J Intermediate Goods)	118,940	10.000	1.087	0	16.32
Log(Firm J Revenue)	118,957	10.34	1.036	5.011	16.52
Log(Firm J Revenue per Worker)	118,943	5.048	0.922	-0.0275	9.232
Log(Firm J Profit)	118,957	6.691	1.882	0	14.03
Panel D: City-sector Characteristics					
Log(City-sector Subsidies)	9,739	14.24	1.660	5.704	24.11
Log(City-sector Firm #)	9,739	0.401	0.609	0	4.078
Log(City-sector Revenue)	9,739	21.54	1.364	16.17	28.57
Log(City-sector Average Wage)	9,739	15.09	0.706	11.41	18.39
Panel E: County-sector Firm Registration					
Log(# Firm Entry)	5,765,780	2.102	1.266	0.693	9.414
1(Tradable Firms)	5,765,780	0.467	0.499	0	1
1(Real Estate Firms)	5,765,780	0.059	0.236	0	1

Notes: This table presents summary statistics for key variables used in the analysis. Panel A shows statistics for different measures of trade and transportation volume. Panel B presents statistics related to politician characteristics, where 'Politician Age Differential' measures the absolute difference between a politician's age and the average age within the region. Panel C shows statistics for firm characteristics. Panel D presents statistics related to city-sector characteristics. Panel E displays the characteristics of county-sector firm registration.

Table I.4: Correlation Between Signed Age Differential and Promotion Outcomes

VARIABLES	1(Promotion)			
	(1)	(2)	(3)	(4)
Politician's Start Age	-0.022*** (0.004)	-0.008 (0.008)	0.007 (0.013)	0.014 (0.014)
Signed Age Differential		-0.016** (0.007)	-0.031** (0.014)	-0.033** (0.014)
Signed Age Differential $\times$ Deputy City Level				0.038* (0.020)
Log(Education Years)				0.018** (0.009)
Average Light Density				0.051 (0.126)
Log(Jurisdictional Population)				0.038 (0.303)
Colleague Ties (Provincial Secretary)				-0.029 (0.060)
Colleague Ties (Governor)				0.005 (0.051)
Hometown Ties (Provincial Secretary)				-0.248 (0.240)
Hometown Ties (Governor)				0.044 (0.113)
Observations	673	673	672	659
R-squared	0.050	0.056	0.408	0.424
City Characteristics	–	–	–	Yes
Politician Characteristics	–	–	–	Yes
Connections to Provincial Politicians	–	–	–	Yes
City FE	–	–	Yes	Yes
Tenure Start Year FE	–	–	Yes	Yes

Notes: This table examines the relationship between signed age differential and promotion outcomes for city party secretaries. The analysis uses politician-level data from 1996 to 2014. The dependent variable is a binary indicator for whether a politician was promoted to a higher position. The key independent variable, Signed Age Differential, is calculated as the politician's age minus the average age of party secretaries in other cities within the same province; negative values indicate younger than peers, positive values indicate older. This signed measure is used here to separately examine career incentives for younger versus older politicians; the main trade regressions use the absolute value of this gap. Control variables include politician characteristics (age, education years), jurisdictional characteristics (GDP and population), and provincial connection measures. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.5: Correlation Between Signed Age Differential and Transfer/Retirement Outcomes

VARIABLES	1(Transfer/Retirement)			
	(1)	(2)	(3)	(4)
Politician's Start Age	0.020*** (0.004)	0.007 (0.008)	-0.014 (0.014)	-0.023 (0.015)
Signed Age Differential		0.015** (0.007)	0.035** (0.014)	0.039*** (0.015)
Signed Age Differential $\times$ Deputy City Level				-0.047** (0.019)
Log(Education Years)				-0.013 (0.009)
Average Light Density				-0.058 (0.122)
Log(Jurisdictional Population)				0.618 (0.419)
Colleague Ties (Provincial Secretary)				0.033 (0.059)
Colleague Ties (Governor)				0.009 (0.053)
Hometown Ties (Provincial Secretary)				0.255 (0.241)
Hometown Ties (Governor)				-0.103 (0.121)
Observations	673	673	672	659
R-squared	0.038	0.043	0.399	0.420
City Characteristics	—	—	—	Yes
Politician Characteristics	—	—	—	Yes
Connections to Provincial Politicians	—	—	—	Yes
City FE	—	—	Yes	Yes
Tenure Start Year FE	—	—	Yes	Yes

Notes: This table examines the relationship between signed age differential and transfer/retirement outcomes for city party secretaries. The analysis uses politician-level data from 1996 to 2014. The dependent variable is a binary indicator for whether the politician was transferred to another position or retired. The key independent variable, Signed Age Differential, is calculated as the politician's age minus the average age of party secretaries in other cities within the same province; negative values indicate younger than peers, positive values indicate older. This signed measure is used here to separately examine career incentives for younger versus older politicians; the main trade regressions use the absolute value of this gap. Control variables include politician characteristics (age, education years), jurisdictional characteristics (GDP and population), and provincial connection measures. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.6: Similar-Age Peer Density and Politician Career Outcomes

VARIABLES	1(Promotion Next Term)				1(Transferred/Retired Next Term)			
	(1) Full Sample	(2) Excl Deputy	(3) Excl Capital	(4) Excl Both	(5) Full sample	(6) Excl deputy	(7) Excl capital	(8) Excl both
# Similar-Age Peers	-0.082* (0.044)	-0.076* (0.046)	-0.080* (0.045)	-0.078* (0.046)	0.098** (0.047)	0.089* (0.049)	0.109** (0.049)	0.102** (0.050)
Observations	672	642	611	597	672	642	611	597
R-squared	0.382	0.372	0.365	0.357	0.384	0.367	0.366	0.350
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Start-year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The dependent variable in Columns (1)–(4) is an indicator equal to one if the focal city party secretary was promoted at the end of the term; in Columns (5)–(8), it is an indicator equal to one if the secretary was transferred to a non-promotion position or retired. The key regressor is the term-average count of provincial peer secretaries whose start age is within ± 3 years of the focal secretary's start age (leave-one-out). City controls include log population and the total number of cities in the province. Columns within each outcome group vary the sample: full sample; excluding deputy-province cities; excluding provincial-capital cities; excluding both. Standard errors clustered at the city level in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.7: Horse Race Specification: Party Secretary versus City Mayor Age Differential

Variables	(1) Log(Total Trade Volume)	(2) Log(Motorway Trade Volume)	(3) Log(Waterway Trade Volume)	(4) Log(Export Value)
Politician Age Differential (Secretary)	0.011** (0.005)	0.011*** (0.004)	0.036*** (0.009)	0.007 (0.007)
Politician Age Differential (Mayor)	0.007 (0.007)	0.003 (0.004)	-0.001 (0.008)	-0.003 (0.005)
Observations	4,375	4,249	2,444	3,366
R-squared	0.892	0.951	0.975	0.982
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Secretary FE	Yes	Yes	Yes	Yes
City-Mayor FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table presents a horse race between the party secretary's and the city mayor's politician age differential in the same specification. Party Secretary Age Differential measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province, while City Mayor Age Differential is defined analogously for city mayors. Including both variables jointly tests whether the trade effects documented for party secretaries are specific to party secretaries, who face within-province promotion competition of the relevant kind, rather than reflecting a general effect of local leadership age differentials. The sample is restricted to city-years with non-missing age differential measures for both the party secretary and the mayor. Column (1) reports effects on total trade volume, column (2) on motorway trade, column (3) on waterway trade, and column (4) on the value of international exports (USD). All trade volumes are logged and measured at 10,000 tons. I control for politician characteristics, city-secretary fixed effects, city-mayor fixed effects, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.8: Instrumental Variable Estimates: Effect of Politician Age Differential on City Trade Volume

VARIABLES	Log(Total Trade Volume) (1)	Log(Motorway Trade Volume) (2)	Log(Waterway Trade Volume) (3)	Export Value (4)
Politician Age Differential	0.021** (0.010)	0.022** (0.010)	0.040* (0.021)	-0.002 (0.019)
Observations	4,907	4,885	2,779	3,815
First-Stage F-stat	363	362	277	304
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes

Notes: This table presents instrumental variable (IV) regression results estimating the causal effect of politician age differential on city trade volumes. The sample covers city-level panel data from 1996 to 2018. Politician Age Differential is instrumented using its one-year lagged value. The dependent variables are log-transformed trade volumes by transportation mode and trade type. All specifications include comprehensive controls for city and politician characteristics, along with city, politician, year, and city-politician fixed effects. The Kleibergen-Paap Wald F-statistic tests for weak identification. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.9: Placebo Test: Cross-Provincial Age Differential and Trade Outcomes in Border Cities

VARIABLES	Log(Total Trade Volume) (1)	Log(Road Trade Volume) (2)	Log(Waterway Trade Volume) (3)	Log(Export Value) (4)
Cross-Provincial Age Differential (Relative to Neighboring Provinces)	-0.003 (0.011)	0.003 (0.005)	0.010 (0.011)	-0.005 (0.008)
Observations	3,320	3,198	1,729	2,419
R-squared	0.856	0.930	0.961	0.965
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table presents a placebo test examining whether cross-provincial age differentials affect trade outcomes in border cities. The analysis is restricted to cities that border other provinces to test whether political competition with neighboring provinces (rather than within-province competition) drives trade patterns. The key independent variable, Cross-Provincial Age Differential, measures the absolute difference between a border city party secretary's age and the average age of party secretaries across all neighboring provinces. For cities bordering multiple provinces, this is calculated as the average across all neighboring province averages. The dependent variables are: total freight trade volume (Column 1), road freight volume (Column 2), waterway freight volume (Column 3), and export value (Column 4), all in logarithmic form. This serves as a placebo test because the theoretical mechanism predicts that within-province yardstick competition "not cross-provincial age differences" should affect local officials' infrastructure investment decisions and resulting trade patterns. Finding null effects supports the interpretation that age-based competition operates primarily within provincial boundaries where officials compete for promotion. The sample includes border city-year observations from 1996 to 2018. City characteristics include logged population, number of firms, GDP, and urban road length. Politician characteristics include lagged secretary and average peer ages. All specifications include city fixed effects, politician fixed effects, city-politician interaction fixed effects, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.10: POLITICIAN AGE DIFFERENTIAL AND CENTRALLY ADMINISTERED TRANSPORT MODES (RAILWAY & AIRWAY)

	City Party Secretary	City Mayor
Variables	(1)	(2)
<i>Panel A: Railway Trade Volume</i>		
Politician Age Differential	0.011 (0.007)	−0.005 (0.006)
Observations	3,700	3,568
R-squared	0.949	0.939
<i>Panel B: Airway Trade Volume</i>		
Politician Age Differential	0.023 (0.033)	0.001 (0.004)
Observations	1,501	1,579
R-squared	0.953	0.932
City Characteristics	Yes	Yes
Politician Characteristics	Yes	Yes
City FE	Yes	Yes
Politician FE	Yes	Yes
City-Politician FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table examines how politician age differentials relate to *long-distance* city-level trade, focusing on railway and airway volumes – modes administered centrally rather than by city governments – as comparison outcomes over which city governments exercise little discretion. The analysis uses city-level panel data from 1996 to 2018. The key independent variable ‘Politician Age Differential’ measures the absolute difference between a city party secretary’s start age and the average start age of party secretaries in other cities within the same province, lagged one year. Column (1) reports the most demanding specification for city party secretaries, including city characteristics and politician characteristics (the lagged own age and the lagged peer average age) along with city, politician, city-politician, and province-year fixed effects. Column (2) repeats the exercise using the age differential of city *mayors*, estimated with city and politician characteristics and city, politician, city-politician, and year fixed effects. Trade volumes are logged and measured at 10,000 tons. Formal tests of coefficient equality across all four transport modes are reported in Table I.11. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.11: FORMAL COEFFICIENT COMPARISON ACROSS TRANSPORT MODES

Variables	Log(Trade Volume) (1)
Politician Age Differential $\times$ Motorway	0.013*** (0.004)
Politician Age Differential $\times$ Waterway	0.026*** (0.009)
Politician Age Differential $\times$ Railway	0.007 (0.007)
Politician Age Differential $\times$ Airway	−0.002 (0.024)
<i>Cross-mode equality tests: F-statistic [p-value]</i>	
All modes equal	1.05 [0.373]
Motorway = Railway	0.51 [0.474]
Waterway = Railway	2.69 [0.102]
Motorway / Waterway avg. = Railway / Airway avg.	1.52 [0.218]
Observations	13,324
R-squared	0.987
City Characteristics	Yes
Politician Characteristics	Yes
City $\times$ Mode FE	Yes
Politician $\times$ Mode FE	Yes
City-Politician $\times$ Mode FE	Yes
Year $\times$ Mode FE	Yes

Notes: This table compares the effect of the politician age differential across the four transport modes within a single stacked regression. The four city-level mode outcomes (motorway, waterway, railway, and airway trade volumes, logged) are stacked into a city-year-mode panel estimated on a common sample; the key independent variable ‘Politician Age Differential’ is interacted with mode indicators; and every fixed effect of the baseline specification is interacted with mode, so that each mode retains its own intercept structure and cross-mode differences in coefficients can be tested within one variance-covariance matrix. The middle panel reports F-tests of coefficient equality across modes with p-values in brackets. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.12: Robustness: Alternative Time Trend and Clustering

VARIABLES	Log(Trade Volume)			
	(1)	(2)	(3)	(4)
<i>Panel A: Total Trade Volume</i>				
Politician Age Differential	0.011*** (0.004)	0.013*** (0.004)	0.011*** (0.003)	0.011*** (0.004)
Observations	5,281	5,281	5,281	5,281
R-squared	0.893	0.903	0.889	0.889
<i>Panel B: Road Trade Volume</i>				
Politician Age Differential	0.012*** (0.003)	0.014*** (0.004)	0.013*** (0.003)	0.013*** (0.003)
Observations	5,101	5,101	5,101	5,101
R-squared	0.937	0.943	0.936	0.936
<i>Panel C: Waterway Trade Volume</i>				
Politician Age Differential	0.025*** (0.009)	0.021** (0.009)	0.026*** (0.009)	0.026*** (0.008)
Observations	2,858	2,858	2,858	2,858
R-squared	0.969	0.974	0.967	0.967
<i>Panel D: Export Value (USD)</i>				
Politician Age Differential	0.011* (0.006)	0.009 (0.006)	0.013** (0.005)	0.013** (0.006)
Observations	3,904	3,904	3,904	3,904
R-squared	0.978	0.982	0.976	0.976
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Province Linear Trend	Yes	–	–	–
City Linear Trend	–	Yes	–	–
Cluster at Province-Year Level	–	–	Yes	–
Cluster at Politician Level	–	–	–	Yes

Notes: This table examines the robustness of how Politician Age Differential between party secretaries (within the same province) affects city-level trade volumes. The analysis uses city-level panel data from 1996 to 2018. The key independent variable ‘Politician Age Differential’ measures the absolute difference between a city party secretary’s age and the average age of party secretaries in other cities within the same province. Panel A shows effects on total trade volume, Panel B on road trade, Panel C on waterway trade, and Panel D on export trade. All trade volumes are logged and measured at 10,000 tons. Column (1) includes province linear trends, Column (2) includes city linear trends, Column (3) clusters standard errors at the province-year level, and Column (4) clusters at the politician level. All specifications control for city characteristics, politician characteristics, city fixed effects, politician fixed effects, city-politician fixed effects, and year fixed effects. Unless otherwise indicated, standard errors (in parentheses) are clustered at the city level. Significance: * p < 0.10, ** p < 0.05, *** p < 0.01.

Table I.13: Robustness: Controlling for Historical Economic Conditions and Trade Volume

VARIABLES	Log(Total Trade Volume) (1)	Log(Total Trade Volume) (2)	Log(Motorway Trade Volume) (3)	Log(Motorway Trade Volume) (4)	Log(Waterway Trade Volume) (5)	Log(Waterway Trade Volume) (6)	Log(Export Value (USD)) (7)	Log(Export Value (USD)) (8)
Politician Age Differential	0.008** (0.004)	0.013*** (0.004)	0.011*** (0.003)	0.013*** (0.004)	0.020** (0.008)	0.024** (0.010)	0.012** (0.005)	0.009 (0.006)
Log(Lagged Total Trade Volume)	0.325*** (0.062)							
Log(Lagged Nighttime Lighting)	0.149* (0.082)		0.065 (0.068)		0.006 (0.129)		0.127 (0.113)	
Log(3-Year Avg Total Trade Volume)		0.400** (0.183)						
Log(3-Year Avg Nighttime Lighting)		0.102 (0.146)		0.200* (0.108)		-0.176 (0.322)		0.245 (0.229)
Log(Lagged Motorway Trade Volume)			0.263*** (0.054)					
Log(3-Year Avg Motorway Trade Volume)				0.232*** (0.089)				
Log(Lagged Waterway Trade Volume)					0.269*** (0.046)			
Log(3-Year Avg Waterway Trade Volume)						0.249*** (0.068)		
Log(Lagged Export Trade Volume)							0.181** (0.078)	
Log(3-Year Avg Export Trade Volume)								0.097 (0.094)
Observations	5,200	4,516	4,942	4,250	2,667	2,278	3,640	2,998
R-squared	0.894	0.903	0.940	0.938	0.971	0.972	0.978	0.982
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the robustness of the main findings by controlling for historical economic conditions and past trade volumes. The dependent variables in Columns (1) through (8) are the log of Total Trade Volume (Columns 1-2), Motorway Trade Volume (Columns 3-4), Waterway Trade Volume (Columns 5-6), and Export Value (USD) (Columns 7-8), respectively. Odd-numbered Columns use lagged variables, while even-numbered Columns use 3-year averages (L3). The key independent variable is 'Politician Age Differential'. Specifications include controls for Log(Lagged Nighttime Lighting) or Log(3-Year Avg Nighttime Lighting), and the respective lagged or 3-year average Log(Trade Volume) for the dependent variable in each column. All specifications include city characteristics and politician characteristics (lagged own age and lagged peer average age) and city, politician, year, and city-politician fixed effects. Standard errors are in parentheses and are clustered at the city level. Significance: * p < 0.10, ** p < 0.05, *** p < 0.01.

Table I.14: Motorway Underinvestment by Administrative Class

VARIABLES	Log(Provincial Motorway Length) (1)	Log(National Motorway Length) (2)	Log(Urban Motorway Length) (3)
Politician Age Differential	-0.046 (0.039)	0.010 (0.046)	-0.003 (0.050)
Politician Age Differential $\times$ Border Region	0.019 (0.047)	0.077 (0.050)	0.014 (0.045)
Observations	34,295	34,295	34,295
R-squared	0.936	0.935	0.954
Politician Characteristics	Yes	Yes	Yes
County-Politician FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table decomposes the motorway results in Table 6 by administrative class: provincial expressways (S-prefix routes), national expressways (G-prefix routes), and urban motorways (unclassified segments). County-year lengths are computed from the annual motorway network shapefiles, classifying each main carriageway by its route number and intersecting it with county boundaries in an equal-area projection. Specifications and variable definitions follow Table 6. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.15: Placebo Test: Effects of Politician Age Differential on Railway Infrastructure

VARIABLES	Log(Total Railway Length) (7)	Log(Single Railway Length) (8)	Log(Railway Link Count) (9)	Log(Railway Node Count) (10)	Log(Railway Station Count) (11)	Log(Railway Betweenness) (12)
Politician Age Differential	-0.004 (0.003)	-0.005 (0.003)	-0.003 (0.003)	-0.002 (0.002)	-0.002 (0.002)	0.001 (0.008)
Observations	5,178	5,149	5,178	5,177	5,177	5,087
R-squared	0.986	0.982	0.988	0.995	0.995	0.978
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents placebo tests examining the effects of politician age differential on railway infrastructure development. Railway infrastructure is used as a placebo because, unlike motorways which are largely under local government control, railway construction is centrally planned by the national government and should not be affected by local politician competition dynamics. The analysis uses panel data with various measures of railway network characteristics from 1995 to 2018. The key independent variable 'Politician Age Differential' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Dependent variables include log-transformed measures of total railway length, single railway length, railway link count, railway node count, railway station count, and railway betweenness centrality. All specifications include politician characteristics, year fixed effects, and city-politician fixed effects. Standard errors (in parentheses) are clustered at the city level. The insignificant coefficients support the validity of our identification strategy. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.16: Effects of Similar-Age Peer Count on City Trade

	Total Trade Volume	Motorway Trade Volume	Waterway Trade Volume	Export Value (USD)
VARIABLES	(1)	(2)	(3)	(4)
# Similar Age Peers	-0.007*** (0.003)	-0.005** (0.002)	-0.012** (0.005)	-0.003 (0.003)
Observations	5,297	5,113	2,858	3,913
R-squared	0.890	0.935	0.967	0.976
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table examines how the number of similar-age peer secretaries within the same province affects city-level trade. The analysis uses city-level panel data from 1996 to 2018. # Similar Age Peers is the lagged count of party secretaries in other cities of the same province whose current age is within three years of the focal city secretary's start age. Columns (1) to (4) report effects on total trade volume, motorway trade, waterway trade, and the value of international exports (USD), respectively. All trade volumes are logged and measured at 10,000 tons. All specifications include city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.17: Signed Decomposition: Politician Age Differential on City Trade

VARIABLES	Total Trade Volume	Motorway Trade Volume	Waterway Trade Volume	Export Value (USD)
	(1)	(2)	(3)	(4)
Years Younger than Peer Average	0.016*** (0.005)	0.015*** (0.004)	0.023** (0.011)	0.012* (0.007)
Years Older than Peer Average	0.006 (0.007)	0.010** (0.004)	0.030*** (0.009)	0.013* (0.008)
Younger = Older	0.240	0.212	0.491	0.869
Observations	5,297	5,113	2,858	3,913
R-squared	0.890	0.936	0.967	0.976
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table reports the signed decomposition of the absolute age differential, addressing reviewer concerns about the symmetry of the absolute-value specification. The two key independent variables decompose the absolute age differential into non-negative components: ‘Years Younger than Peer Average’ equals $\max(\text{peer average age} - \text{focal age}, 0)$, measuring how many years the focal city party secretary is below the provincial peer average (zero if focal is older); ‘Years Older than Peer Average’ equals $\max(\text{focal age} - \text{peer average age}, 0)$, measuring how many years the focal is above (zero if younger). By construction, their sum equals the absolute age differential used in the baseline. Both regressors are lagged one year. The third row reports the p-value of an F-test for the symmetry restriction (coefficients equal across the two variables) implicit in the absolute-value specification. Dependent variables are logged: total trade volume in column (1), motorway in (2), waterway in (3), and the value of international exports (USD) in (4). All specifications include city characteristics, politician characteristics, and city, politician, city-politician, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.18: Robustness to Focal-City Secretary Turnover

VARIABLES	(1) Full Sample	Sample: No Focal-City Secretary Turnover
		(2) Drop Focal Turnover Year (<i>t</i>)
<i>Panel A: Total Trade Volume</i>		
Politician Age Differential	0.011*** (0.004)	0.034*** (0.012)
Observations	5,281	3,710
R-squared	0.889	0.917
<i>Panel B: Motorway Trade Volume</i>		
Politician Age Differential	0.013*** (0.004)	0.019* (0.011)
Observations	5,101	3,561
R-squared	0.936	0.939
<i>Panel C: Waterway Trade Volume</i>		
Politician Age Differential	0.026*** (0.009)	0.028 (0.028)
Observations	2,858	1,962
R-squared	0.967	0.972
<i>Panel D: Export Value (USD)</i>		
Politician Age Differential	0.013** (0.006)	0.016 (0.020)
Observations	3,904	2,719
R-squared	0.976	0.977
City Characteristics	Yes	Yes
Politician Characteristics	Yes	Yes
City FE	Yes	Yes
Politician FE	Yes	Yes
City-Politician FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table tests the robustness of the Politician Age Differential effect on city trade by dropping city-years close to the focal city's own secretary turnover. Column (1) uses the full sample. Column (2) drops city-years in which the focal city had its own secretary turnover in year *t*. Across both samples, the identifying variation in 'Politician Age Differential' comes from peer turnovers in other cities of the same province. Each panel reports a different trade outcome: total trade volume in Panel A, motorway in B, waterway in C, and the value of international exports (USD) in D. All trade volumes are logged and measured at 10,000 tons. All specifications include city characteristics and politician characteristics (lagged politician start age and lagged peer average age) and city, politician, city-politician, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.19: Competition Mechanism: Effect of Politician Age Differential on Short-Term Nighttime Lights Among Competitors

VARIABLES	Light Gap Relative to Provincial Average (DN)		Light as % Deviation From Provincial Average		Next-Year Lights (DN)	
	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	-0.019 (0.025)		0.580 (0.408)		-0.035 (0.028)	
# Similar Age Peers		0.026** (0.011)		0.317 (0.304)		0.024 (0.015)
Observations	5,076	5,076	5,076	5,076	5,076	5,076
R-squared	0.989	0.989	0.985	0.984	0.991	0.991
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines the relationship between within-province politician competition and short-term nighttime light performance relative to provincial peers. The analysis uses city-level panel data from 1996 to 2018. Politician Age Differential measures the lagged absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. # Similar Age Peers is the count of within-province peer secretaries whose age is similar to the focal secretary. Columns (1) and (2) measure the nighttime light gap in digital number (DN) units between a city and the provincial average. Columns (3) and (4) express this gap as a percentage deviation from the provincial mean. Columns (5) and (6) use next-year nighttime lights (DN) as the outcome. All specifications include city characteristics, politician characteristics, and city, politician, and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.20: Effect of Motorway Development on Motorway Trade Volume

VARIABLES	Log(Motorway Trade Volume)				
	(1)	(2)	(3)	(4)	(5)
Log(Motorway Length)	0.041** (0.019)				
Log(Motorway Entry/Exit)		0.093*** (0.021)			
Log(Motorway Link)			0.077*** (0.021)		
Log(Motorway Node)				0.093*** (0.021)	
Log(Motorway Betweenness)					0.004 (0.006)
Observations	4,694	4,646	4,694	4,646	4,469
R-squared	0.803	0.806	0.803	0.806	0.804
City FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how different measures of motorway development affect city-level trade volumes. The analysis uses city-level panel data from 1996 to 2018. The key independent variables are different measures of motorway development, including motorway length (Column 1), the number of motorway entry and exit points (Column 2), the number of motorway links connecting cities (Column 3), the number of motorway nodes or junctions (Column 4), and motorway betweenness centrality measuring a city's position in the motorway network (Column 5). The dependent variable is the logarithm of trade volume via motorway transport. All specifications include city and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.21: Politician Age Differential and Number of Firm Entries

VARIABLES	Log(# Firm New Entry + 1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Politician Age Differential	0.0075***	0.0078***	0.0077***			
× Tradable Sector	(0.0024)	(0.0024)	(0.0024)			
Politician Age Differential				-0.0114***	-0.0105***	-0.0094***
× Real Estate Sector				(0.0034)	(0.0033)	(0.0033)
Observations	5,765,425	5,765,425	5,765,207	5,765,425	5,765,425	5,765,207
R-squared	0.548	0.558	0.574	0.548	0.558	0.574
County-Sector FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	–	Yes	Yes	–
Politician FE	–	Yes	Yes	–	Yes	Yes
County-Year FE	–	–	Yes	–	–	Yes

Notes: This table examines the impact of Politician Age Differential on the number of new firm entries, distinguished by sector type (tradable versus real estate). The analysis utilizes county-industry-year panel data from 1996 to 2018, with the dependent variable being the logarithm of the number of new firm entries plus one. The key independent variables are the interaction terms between Politician Age Differential and indicators for tradable sectors and real estate sectors, respectively. All specifications control for county-sector fixed effects, politician fixed effects, and county-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * p < 0.10, ** p < 0.05, *** p < 0.01.

Table I.22: Example Matched Firm Pair

Attribute	Firm I Details	Firm J Details
Similarity Score from LLM (96.81%)		
Firm ID	13346	1267002
Name	Shanghai Gaodong Paint Factory	Shanghai Rongsheng Coating Co., Ltd.
Industry Subcategory	Organic Chemical Products Manufacturing	Organic Chemical Products Manufacturing
Activity 1	Paint and Coating	Paint Spraying Processing
City	Pudong New District	Qingpu District

Notes: This table shows an example of firms connected via business networks based on main business activity and sectors. The similarity score also considers the financial status of firms.

Table I.23: Robustness: Effect of Politician Age Differential on Firms with High-Connectivity Input-Output Relationships (1997)

VARIABLES	Firm _j Connected to Firm _i in Provincial Input-Output Table (Above Median Connectivity) in 1997			
	Log(Intermediate Good Inputs) (1)	Log(Revenue) (2)	Log(Revenue per Worker) (3)	Log(Profits) (4)
Politician Age Differential in Firm _i City	0.011*** (0.004)	0.013*** (0.003)	0.009** (0.004)	0.001 (0.003)
Observations	1,146,985	1,146,940	1,146,940	1,147,128
R-squared	0.894	0.922	0.817	0.798
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes

Notes: This table presents robustness checks examining how political age differentials affect firm intermediate good inputs and other firm outcomes through province-specific input-output linkages. The analysis focuses on firm pairs connected via provincial input-output relationships with above-median connectivity levels based on the 1997 input-output table. The key independent variable 'Politician Age Differential in Firm_i City' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. All dependent variables are in logarithmic form. The sample includes manufacturing firms from the Annual Survey of Industrial Firms (ASIF) database. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.24: Effect of Politician Age Differential on Firm Performance: Heterogeneity by State Ownership

VARIABLES	Firm _i Located in City _i Direct Effect on Focal Firms		Firm _j Connected to Firm _i in Business Network Indirect/Spillover Effect on Connected Firms	
	Log(Revenue) (1)	Log(Revenue per Worker) (2)	Log(Revenue) (3)	Log(Revenue per Worker) (4)
Politician Age Differential	0.023*** (0.007)	0.022*** (0.008)	0.007*** (0.002)	0.009*** (0.003)
Politician Age Differential × State-Owned Firms (SOE)	-0.020*** (0.006)	-0.022*** (0.006)	-0.011** (0.005)	-0.009** (0.004)
Observations	26,887	26,882	118,957	118,943
R-squared	0.909	0.863	0.905	0.861
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector × Year FE	Yes	Yes	Yes	Yes
City _j × Year FE	–	–	Yes	Yes

Notes: This table examines how the politician age differential between party secretaries affects firm performance, and how the effect differs for state-owned enterprises (SOEs). The analysis uses firm-level panel data, where Columns (1) and (2) report outcomes for the focal firm (Firm_i) located in City_i, and Columns (3) and (4) report outcomes for firms connected to the focal firm through business networks (Firm_j). The key independent variable ‘Politician Age Differential in City_i’ measures the absolute difference between a city party secretary’s age and the average age of party secretaries in other cities within the same province. The row ‘× State-Owned (SOE)’ interacts this differential with an indicator for whether the relevant firm is state-owned, so the total effect for an SOE is the sum of the two coefficients; the interaction offsets the main effect almost exactly (e.g., $0.023 - 0.020 \approx 0.003$ for focal-firm revenue and $0.022 - 0.022 \approx 0$ for revenue per worker), leaving a near-zero net effect for state-owned firms. All specifications include firm pair fixed effects, politician fixed effects, and sector-by-year fixed effects; Columns (3) and (4) additionally include connected city-by-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.25: Yardstick Competition and the Sectoral Composition of GDP

VARIABLES	Tradable VA Share of GDP (pp)	Non-Tradable VA Share of GDP (pp)
	(1)	(2)
Politician Age Differential	0.021 (0.032)	-0.010 (0.029)
Observations	4,921	4,925
R-squared	0.960	0.958
City Characteristics	Yes	Yes
Politician Characteristics	Yes	Yes
City-Politician FE	Yes	Yes
Year FE	Yes	Yes

Notes: This table decomposes city GDP into tradable and non-tradable components and estimates the effect of yardstick competition on the composition of value added, implementing the decomposition that underlies the reconciliation of the firm-level and GDP results. The analysis uses city-level panel data from 1996 to 2018. The dependent variables are the tradable and non-tradable (including real estate) shares of city GDP, in percentage points. The key independent variable 'Politician Age Differential' is the lagged absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. A positive coefficient in Column (1) and a negative coefficient in Column (2) indicate that weaker competition tilts the composition of GDP toward tradables, and intense competition toward non-tradables. Both specifications include city characteristics, politician characteristics, and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.26: Placebo Test: Effect of Politician Age Differential on Firm Networks in Bordering Cities

VARIABLES	Firm _j Connected to Firm _i in Business Network (Different Provinces) (Firms Show a 95% Similarity Score)			
	Log(Intermediate Good Inputs) (1)	Log(Revenue) (2)	Log(Revenue per Worker) (3)	Log(Profits) (4)
Politician Age Differential in Firm _i City	-0.007 (0.045)	-0.012 (0.036)	-0.004 (0.034)	0.087 (0.083)
Observations	1,655	1,655	1,655	1,655
R-squared	0.939	0.960	0.926	0.843
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes

Notes: This table presents a falsification test examining whether age differentials between party secretaries affect firms in bordering cities across different provinces. The analysis uses firm-pair panel data where firms are connected through business networks but located in different provinces. The key independent variable 'Politician Age Differential in Firm_i City' measures the absolute difference between a city party secretary's age and the average age of party secretaries in other cities within the same province. Dependent variables measure outcomes for connected firms (Firm_j in different provinces): intermediate good inputs, revenue, revenue per worker, and profits (all log-transformed). All specifications include firm pair fixed effects, politician fixed effects, and sector-year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.27: Politician Age Differential (1 SD Decrease) and Firm TFP

Sector	Coefficient	Input Loss (%)	TFP Loss (%)
Full Sample	0.023	5.04	1.21
Tradable	0.026	5.70	1.37
Manufacturing	0.019	4.17	1.00
Processing	0.042	9.21	2.21
Natural Resources	0.019	4.17	1.00
Products	0.028	6.14	1.47
Textiles	0.026	5.70	1.37

Notes: This table presents the estimated percentage loss in intermediate inputs and Total Factor Productivity (TFP) for firms across various sectors following a one standard deviation decrease in Politician Age Differential. Input Loss is calculated as Coefficient $\times$ 2.192. TFP loss is calculated based on the reduction in intermediate input availability using the elasticity from [Halpern et al. \(2015\)](#), where a 1% increase in intermediate inputs yields a 0.24% increase in productivity. Data is derived from firm-level analysis using the Annual Survey of Industrial Firms (ASIF) from 2000–2007.

Table I.28: Calculation of Tradable-Sector GDP Impact (1 SD Decrease in Politician Age Differential)

Sector	TFP Loss (%)	GDP Share (%)	GDP Impact (%)
Processing	2.21	10.23	0.226
Products	1.47	14.18	0.208
Textiles	1.37	4.08	0.056
Natural Resources	1.00	5.03	0.050
Total			0.540

Notes: This table calculates the estimated impact on tradable-sector GDP resulting from TFP losses induced by a one standard deviation decrease in Politician Age Differential. The GDP Impact (%) for each sector is calculated by multiplying the sector's TFP Loss (%) by its corresponding GDP Share (%) obtained from China's 2000 Input-Output table. The total is the sum across the listed tradable sectors, which jointly account for 33.5% of GDP; it is a tradable-sector productivity cost, not an overall GDP effect, since the non-tradable expansion induced by competition is deliberately excluded from this calculation.

Table I.29: Effects of Provincial Politician Age Differential on Bilateral Trade Volume (Inter-province Trade)

VARIABLES	Log(Bilateral Provincial Trade Volume)				
	(1)	(2)	(3)	(4)	(5)
Provincial Politician Age Differential (Origin Province)	0.021** (0.009)	0.022** (0.009)	0.022** (0.009)		
Provincial Politician Age Differential (Destination Province)	0.011 (0.009)	0.012 (0.009)		0.012 (0.009)	
Absolute Age Difference (Origin vs. Destination)	0.002 (0.003)				
# Similar Age Peers (Origin Province)					-0.013*** (0.003)
# Similar Age Peers (Destination Province)					-0.009*** (0.003)
Observations	16,215	16,215	16,215	16,215	16,215
R-squared	0.943	0.943	0.943	0.943	0.943
Origin-Destination FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Origin Province $\times$ Origin Politician FE	Yes	Yes	Yes	Yes	Yes
Destination Province $\times$ Destination Politician FE	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how politician age differentials affect bilateral trade volumes between Chinese provinces. The analysis uses province-pair panel data from the Railway Statistical Yearbook covering 2000 to 2018. The origin politician age differential measures the absolute difference between the origin province's party secretary age and the national average age of all provincial party secretaries; the destination politician age differential applies the same measure to the destination province. Column (1) includes both differentials and their absolute difference, Column (2) drops the relative-age term, Columns (3) and (4) retain only the origin and only the destination differential respectively, and Column (5) replaces the differentials with the number of similar-age provincial peers, defined as the count of other provincial party secretaries whose start age is within three years of the focal secretary, for the origin and destination provinces. All specifications include origin-destination fixed effects, year fixed effects, and origin-province-by-politician and destination-province-by-politician fixed effects. Standard errors (in parentheses) are clustered at the origin-destination pair level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.30: Politician Age Differential and Firm Suppliers/Competitors

VARIABLES	Firm _j Connected to Firm _i in Business Network Indirect/Spillover Effect on Firms Show a 95% Similarity Score (LLM)			
	Log(Intermediate Good Inputs) (1)	Log(Profits) (2)	Log(Revenue) (3)	Log(Revenue per Worker) (4)
<i>Panel A: Potential Supplier and Partner</i>				
Politician Age Differential	0.008** (0.004)	0.012 (0.009)	0.006*** (0.002)	0.007** (0.004)
Observations	99,057	99,067	99,067	99,054
R-squared	0.869	0.792	0.903	0.861
<i>Panel B: Potential Competitor: Similar Products/Business Activities</i>				
Politician Age Differential	-0.001 (0.008)	0.012 (0.025)	0.000 (0.004)	0.006 (0.005)
Observations	15,395	15,400	15,400	15,399
R-squared	0.945	0.858	0.976	0.944
Firm Pair FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Sector-Year FE	Yes	Yes	Yes	Yes
City-Year FE	Yes	Yes	Yes	Yes

Notes: This table examines the heterogeneous effects of yardstick political competition on connected firms (firm j), differentiating by the type of inter-firm relationship. The analysis uses firm-pair panel data from 2000 to 2007. Panel A focuses on firm pairs classified as potential suppliers or partners, while Panel B focuses on firm pairs classified as potential competitors. The key independent variable is the Politician Age Differential in the city governing firm i . Dependent variables, shown in Columns (1) through (4), are log-transformed intermediate good inputs, profits, revenue, and revenue per worker for firm j . All specifications include firm pair fixed effects, politician fixed effects, sector-by-year fixed effects, and city-by-year fixed effects. Standard errors (in parentheses) are clustered at city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.31: Effect of Politician Age Differential by City Industry Structure

VARIABLES	Log(Total Trade Volume)					
	Eastern Region (1)	Initial Trade Share (2)	City Government Firm Tax Revenue (3)	VA in Manuf Share (4)	2nd Industry Employment (5)	Industrial Firm Number (6)
Politician Age Differential	0.016*** (0.005)	0.033*** (0.009)	0.048* (0.024)	0.043** (0.018)	0.034*** (0.012)	0.056*** (0.020)
Age Diff × Eastern Region	-0.012* (0.007)					
Initial Trade Share		0.336*** (0.089)				
Age Diff × Initial Trade Share		-0.048** (0.020)				
City Firm Tax Revenue			-0.027 (0.053)			
Age Diff × Firm Tax Revenue			-0.003 (0.002)			
Manufacturing Value-Added Share				0.010*** (0.003)		
Age Diff × Manufacturing VA				-0.001* (0.000)		
City Manufacturing Employment					-0.050 (0.058)	
Age Diff × Manufacturing Emp					-0.007* (0.004)	
Number of Industrial Firms						0.172*** (0.063)
Age Diff × Industrial Firms						-0.006** (0.003)
Observations	5,281	4,426	4,151	3,933	4,892	4,103
R-squared	0.889	0.881	0.897	0.916	0.907	0.887
City Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table examines how the effect of politician age differential on trade volume varies across cities with different regional characteristics, initial trade conditions, and industry structures. Column (1) examines whether a city is located in the eastern region, Column (2) examines the city's initial trade share, Column (3) examines city government firm tax revenue, Column (4) examines the manufacturing value-added share of GDP, Column (5) examines manufacturing employment, and Column (6) examines the number of industrial firms. The dependent variable is log total trade volume across all specifications. The main variable of interest is politician age differential and its interactions with city characteristics and industry structure measures. All specifications include city characteristics, politician characteristics, year fixed effects, and city-politician fixed effects. Standard errors are in parentheses and are clustered at the city level. Significance: * p < 0.10, ** p < 0.05, *** p < 0.01.

Table I.32: Competition Effect by Tenure Proximity of Provincial Peers

VARIABLES	Total Trade Volume	Motorway Trade Volume	Waterway Trade Volume	Export Value (USD)
	(1)	(2)	(3)	(4)
Politician Age Differential: Near-Tenure Peers (≤ 2 Yr Gap)	0.008** (0.004)	0.009*** (0.003)	0.030*** (0.008)	0.007 (0.005)
Politician Age Differential: Mid-Tenure Peers (3–5 Yr Gap)	0.003 (0.004)	0.003 (0.002)	0.000 (0.005)	0.005 (0.004)
Politician Age Differential: Far-Tenure Peers (≥ 6 Yr Gap)	0.003 (0.004)	0.006* (0.003)	-0.003 (0.007)	0.009 (0.006)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
City-Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: This table decomposes the Politician Age Differential effect on city trade by the tenure proximity of within-province peer secretaries. For each city-year, peers are split into three disjoint bins by the absolute gap between the focal and peer secretary's tenure year: near-tenure peers (gap of 0 to 2 years), mid-tenure peers (gap of 3 to 5 years), and far-tenure peers (gap of 6 or more years). Each row reports the lagged age differential computed over the corresponding bin, entered jointly in the same regression. Columns (1) to (4) report effects on total trade volume, motorway trade, waterway trade, and the value of international exports (USD), respectively. All trade volumes are logged and measured at 10,000 tons. All specifications include lagged politician characteristics (politician start age and peer average age) and city-politician and year fixed effects. Standard errors (in parentheses) are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.33: Effect of Different Competition Measures

VARIABLES	Log(Total Trade Volume) (1)	Log(Motorway Trade Volume) (2)	Log(Waterway Trade Volume) (3)	Log(Export Value (USD)) (4)
<i>Panel A: Baseline Relative Age Competition</i>				
Politician Age Differential	0.011*** (0.004)	0.013*** (0.004)	0.026*** (0.009)	0.013** (0.006)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
<i>Panel B: Relative Age Competition ($\leq$ Age 57)</i>				
Politician Age Differential $\leq$ Age 57	0.011** (0.004)	0.013*** (0.004)	0.028*** (0.009)	0.012* (0.006)
Observations	4,835	4,677	2,659	3,572
R-squared	0.910	0.937	0.967	0.977
<i>Panel C: Relative Current-Age Competition</i>				
Politician Current-Age Differential	0.005 (0.005)	0.007** (0.003)	0.019** (0.009)	0.010* (0.005)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
<i>Panel D: Relative Tenure Year Competition</i>				
Politician Tenure Year Differential	0.001 (0.007)	-0.000 (0.006)	0.002 (0.013)	0.001 (0.010)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
<i>Panel E: Relative Educational Level Competition</i>				
Politician Education Level Differential	0.006 (0.008)	0.003 (0.004)	-0.002 (0.013)	-0.001 (0.007)
Observations	5,281	5,101	2,858	3,904
R-squared	0.889	0.936	0.967	0.976
<i>Panel F: Relative Experience Year Competition</i>				
Politician Working Years Differential	0.001 (0.002)	0.001 (0.002)	0.014* (0.007)	0.001 (0.003)
Observations	4,761	4,626	2,635	3,611
R-squared	0.911	0.937	0.968	0.976
<i>Panel G: Similar Experience Count</i>				
Similar Working Year (+3/-3) Counts within Province	0.000 (0.004)	-0.003 (0.003)	-0.020*** (0.007)	-0.000 (0.005)
Observations	4,764	4,628	2,635	3,611
R-squared	0.911	0.937	0.968	0.976
Sample	Full	Full	Full	Full
City Characteristics	Yes	Yes	Yes	Yes
Politician Characteristics	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Politician FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
City-Politician FE	Yes	Yes	Yes	Yes

Notes: This table investigates the effect of different yardstick competition measures on trade volumes (log-transformed) and export values. Each panel presents results from separate regressions using different measures of political competition, all lagged one year and constructed as winsorized absolute differentials from the leave-one-out provincial average unless noted otherwise. Panel A shows the baseline relative age competition effect using the start-of-term politician age differential. Panel B restricts the analysis to politicians aged 57 or below to capture promotion incentives before mandatory retirement. Panel C replaces the start-of-term age with the politician's current age in the construction of the age differential. Panel D examines relative tenure competition using politician tenure differential. Panel E examines educational level competition using politician education level differential. Panel F analyzes relative experience year competition using working years differential. Panel G examines similar experience politician counts within provinces; for this count measure the expected sign is negative, since more similar peers imply more intense competition. The dependent variables are Total Trade Volume (Column 1), Motorway Trade Volume (Column 2), Waterway Trade Volume (Column 3), and Export Value in USD (Column 4). All specifications include city characteristics and politician characteristics (lagged own age and lagged peer average age) and city, politician, year, and city-politician fixed effects. Standard errors are in parentheses and are clustered at the city level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.